

87654321

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD
DATE

9

0001102890

ENGINEERING RELEASED

2011-04-07

J40I

MLB

Thu Apr 7 18:48:24 2011

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DOCUMENTS / BOARDS / ASSEMBLIES

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8768	1	SCHEM,MLBI,J40I	SCH		
820-2993	1	PCBF,MLBI,J40I	MLB		
639-1558		PCBA,MLBI,J40I			
085-2372	1	DEV LIST,MLBI,J40I	DEV1		DEVELOPMENT_LIST

DRAWING
TITLE=SHORTSTACK
ABBREV=DRAWING

SCH,MLB,J40

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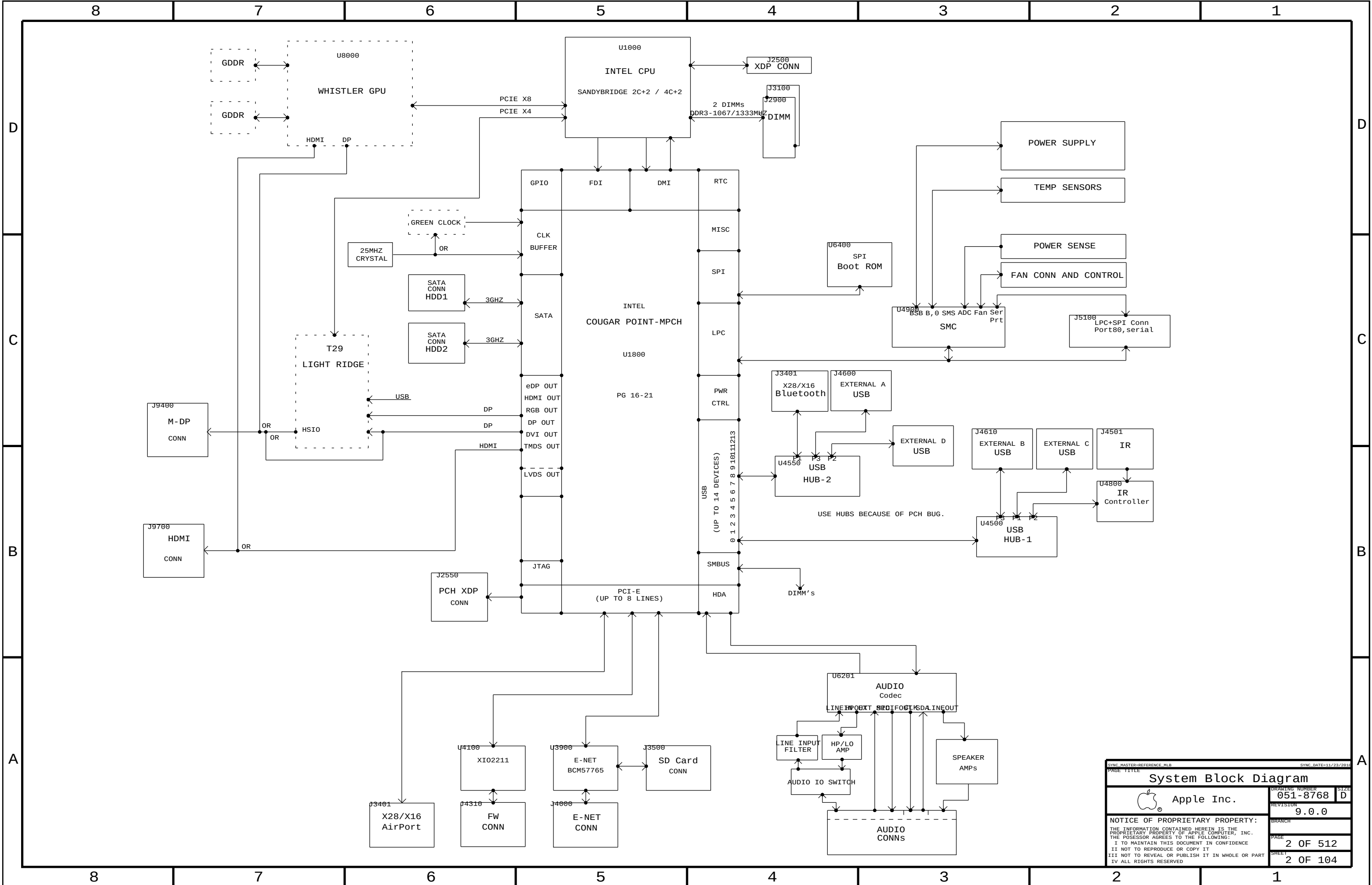
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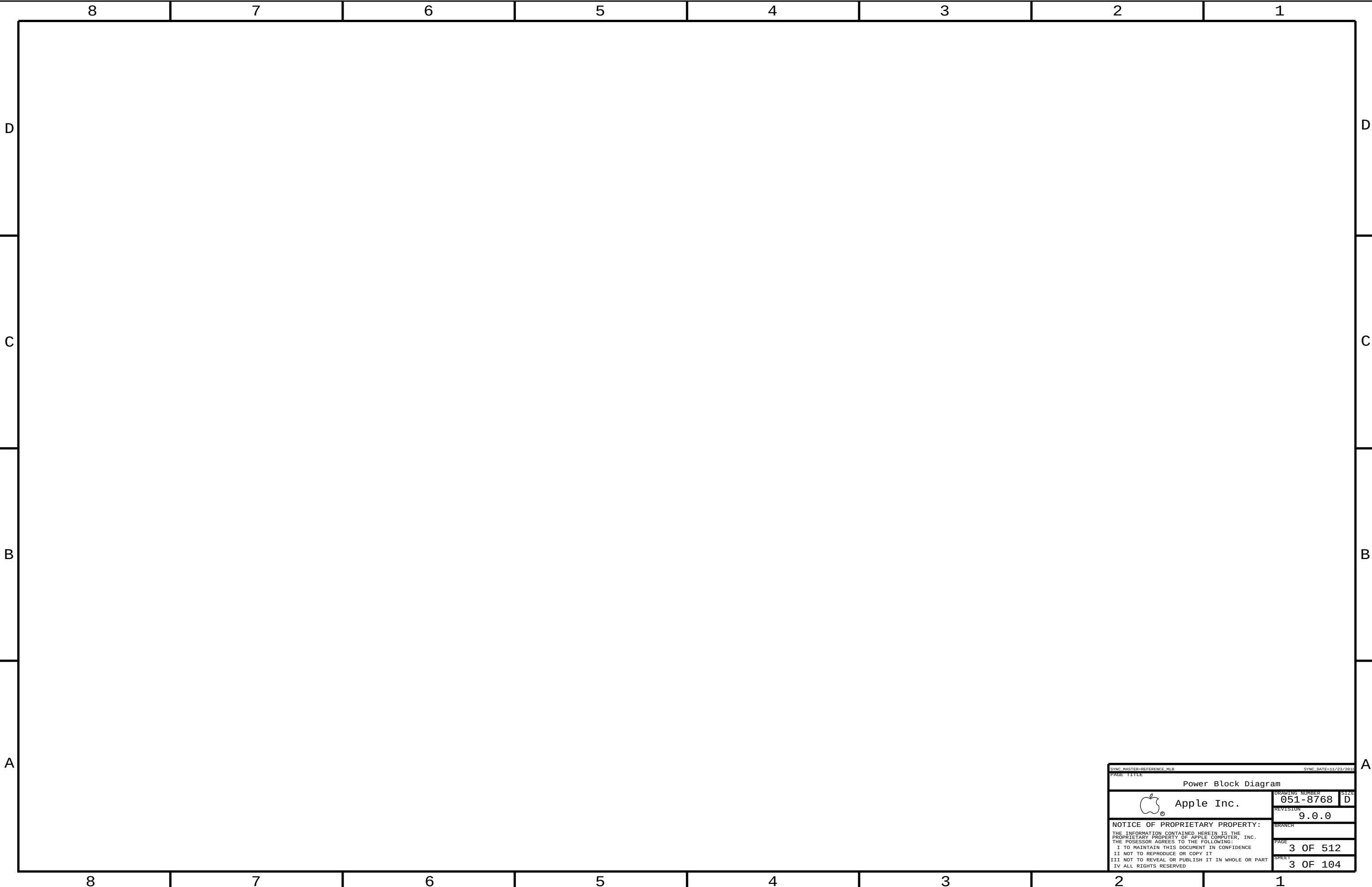


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PLATFORM CONTROL HUB (PCH)

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3980	1	IC, PCH, COUGAR_POINT,QM2Q, SUPER SKU, BGA, ES1/B9	U1800	CRITICAL	COUGAR_POINT:B0
337S4022	1	IC, PCH, COUGAR_POINT,QXXX, SUPER SKU, BGA, Q5/B1	U1800	CRITICAL	COUGAR_POINT:B1
337S4025	1	IC, PCH, COUGAR_POINT,QN3H, SUPER SKU, BGA, Q5/B2	U1800	CRITICAL	COUGAR_POINT:B2
337S4029	1	IC, PCH, COUGAR_POINT,SLH9D, BD82HM65, BGA, PRQ/B2	U1800	CRITICAL	COUGAR_POINT:B2_PRQ_OLD
337S4087	1	IC, PCH, COUGAR_POINT,SLJ4P, BD82HM65, BGA, PRQ/B3	U1800	CRITICAL	COUGAR_POINT:B3_PRQ_OLD

PROGRAMMED PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341T0338	1	IC,EFI BOOTROM, J40	U6400	CRITICAL	
341T0335	1	IC,SMC, J40G	U4900	CRITICAL	SMC:J40G
341T0336	1	IC,SMC, J40I	U4900	CRITICAL	SMC:J40I
341T0337	1	IC,SMC, J40S	U4900	CRITICAL	SMC:J40S
341T0340	1	IC,IR CONTROLLER, J40	U4700	CRITICAL	
341T0341	1	IC,ENET ROM J40	U3990	CRITICAL	

BASE: 335S0770, 335S0769
BASE: 338S0895
BASE: 338S0895
BASE: 338S0895
BASE: 338S0375
BASE: 335S0539, 335S0663, OR 335S0800

NON-SCHEMATIC PARTS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
825-6838	1	LBL,SERIAL NUMBER	LBL		

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0286	128S0248	?	?	39UF CAP FROM SANYO AND CHEMI-CON
132S0099	132S0203	?	?	0.1UF CAP,X7R
138S0676	138S0691		?	22UF CAP FROM MURATA
138S0681	138S0638		?	18UF CAP FROM TAIYO YUDEN
152S0138	152S0694	?	?	INDUCTOR FROM VISHAY
152S1447	152S1269	?	?	INDUCTOR FROM CYNTEC
152S1282	152S1155	?	?	INDUCTOR FROM CYNTEC
152S1366	152S1289	?	?	INDUCTOR FROM TOKO
155S0641	155S0397	?	?	FERRITE FROM LAIRD
155S0457	155S0329		ALL	MAG LAYERS ALT TO MURATA
157S0058	157S0055		ALL	DELTA ALT TO TDK MAGNETICS
353S3196	353S3326	?	?	DIFFERENT TEMP RATING
371S0679	371S0652	?	?	PIN DIODE FROM NXP
376S0914	376S0881		?	SI7121DN AS ALTERNATE
376S0957	376S0999		?	DUAL FET FROM FAIRCHILD
376S0958	376S0953		?	DUAL FET FROM FAIRCHILD
376S1000	376S0960		?	DUAL FET FROM FAIRCHILD
376S0972	376S0612		?	FET FROM ROHM
376S0613	376S0859		?	OLDER DUAL FET FROM TOSHIBA
376S0855	376S0859		?	DUAL FET FROM DDS
376S0977	376S0859		?	DUAL FET FROM DDS
138S0703	138S0648		?	CAP FROM MURATA
197S0339	197S0179		?	OSC. FROM SITIME
376S0926	376S0610		?	DUAL FET FROM FAIRCHILD
376S0572	376S0659		?	FET FROM ALPHA OMEGA
377S0124	377S0057		?	VARIATOR FROM AMOTECH

COMMON PARTS TABLE

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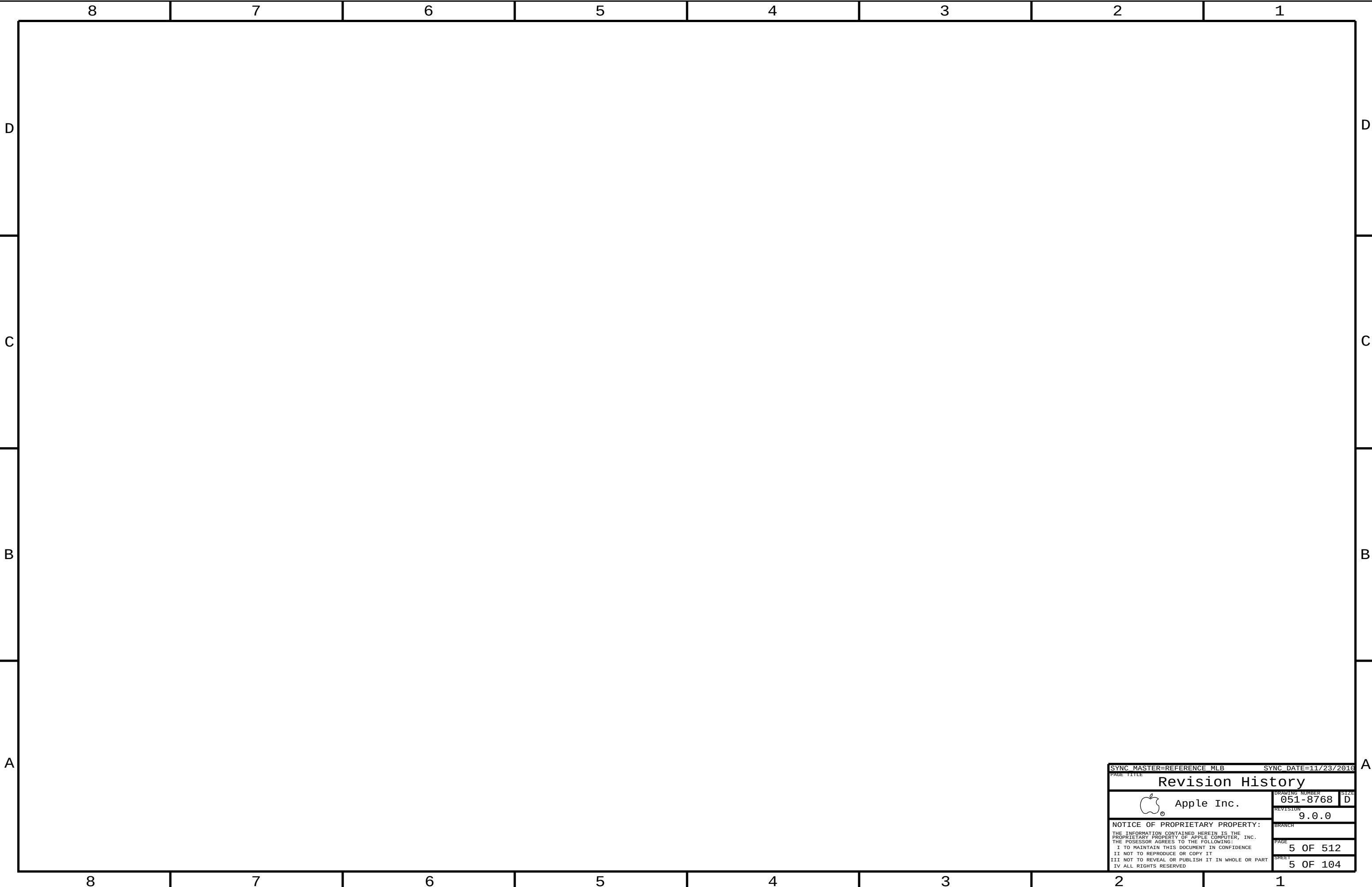
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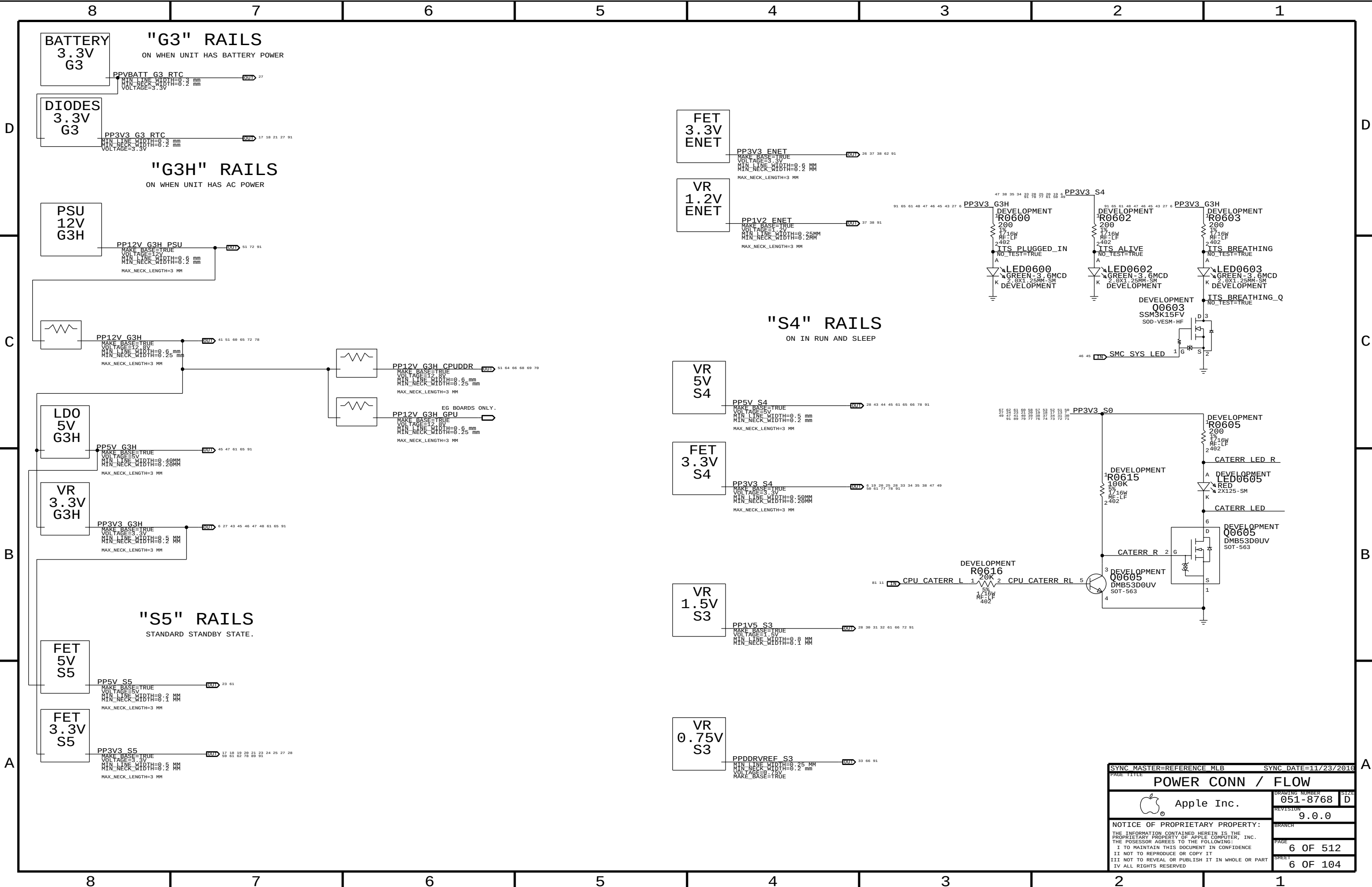
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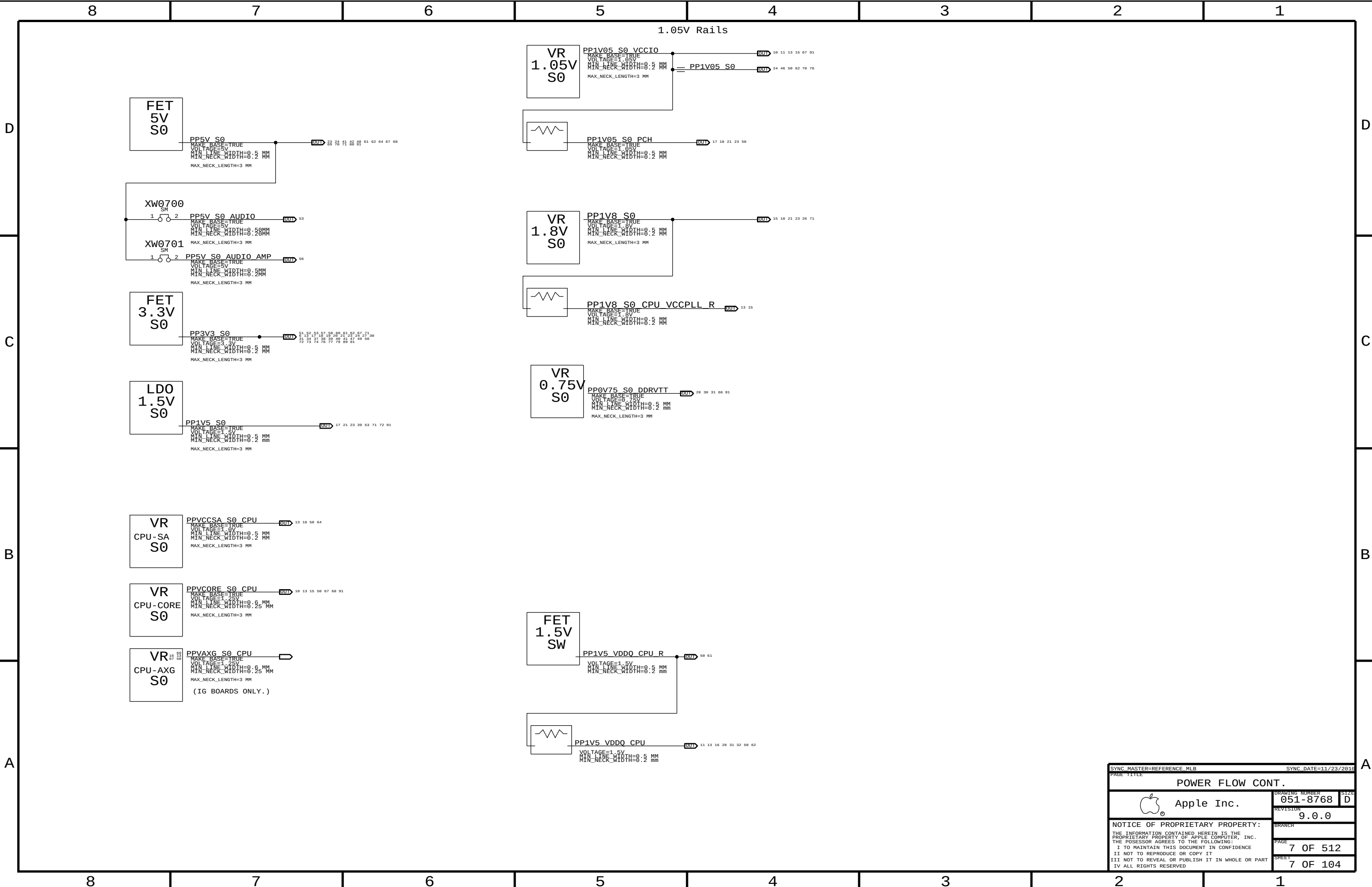
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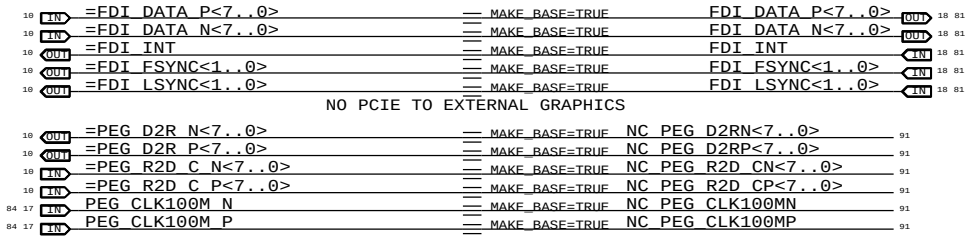


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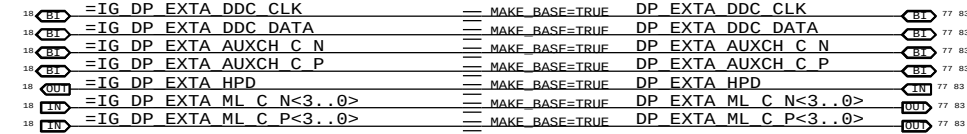


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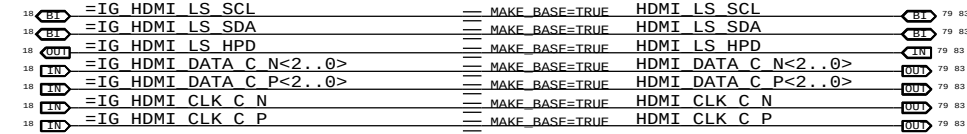
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EXTERNAL DISPLAY PORT

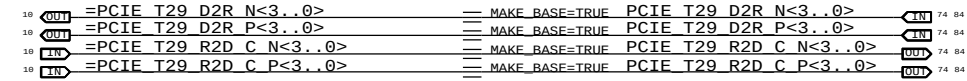


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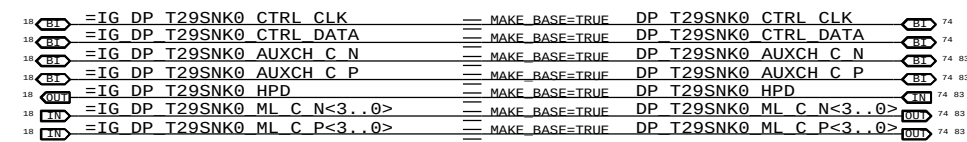


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PCIE



T29 DISPLAY PORT INPUT

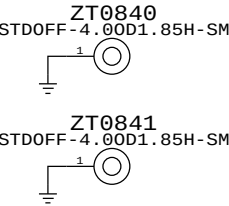


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338S0945	1	IC, T29, SLHAJ, FCBGA, 15X15MM	U9000	CRITICAL	
335S0550	1	IC, EEPROM, SERIAL, SPI, 4KX8, 1.8V, MLP8, LF	U9090	CRITICAL	T29ROM:BLANK
341T0342	1	IC, CONF ROM, T29, J40I, S	U9090	CRITICAL	T29ROM:PROG_IS
341T0380	1	IC, CONF ROM, T29, J40G	U9090	CRITICAL	T29ROM:PROG_G
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341T0339	1	IC, PROGRMD, T29 PORT MCU, HVQFN25	U9330	CRITICAL	T29MCU:PROG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3312	1	IC, CBT104DP081A, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:NXP
353S3312	1	IC, CBT104DP081A, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:BOTH
353S3055	1	IC, PI3VEDP212, 2:1 DP MUX, QFN	U9390	CRITICAL	T29DPMUX:PERICOM

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S3055	353S3312	T29DPMUX:BOTH	?	DP MUX FROM PERICOM

USED FOR T29 IC



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SPECIFIC ALIASES, TABLES

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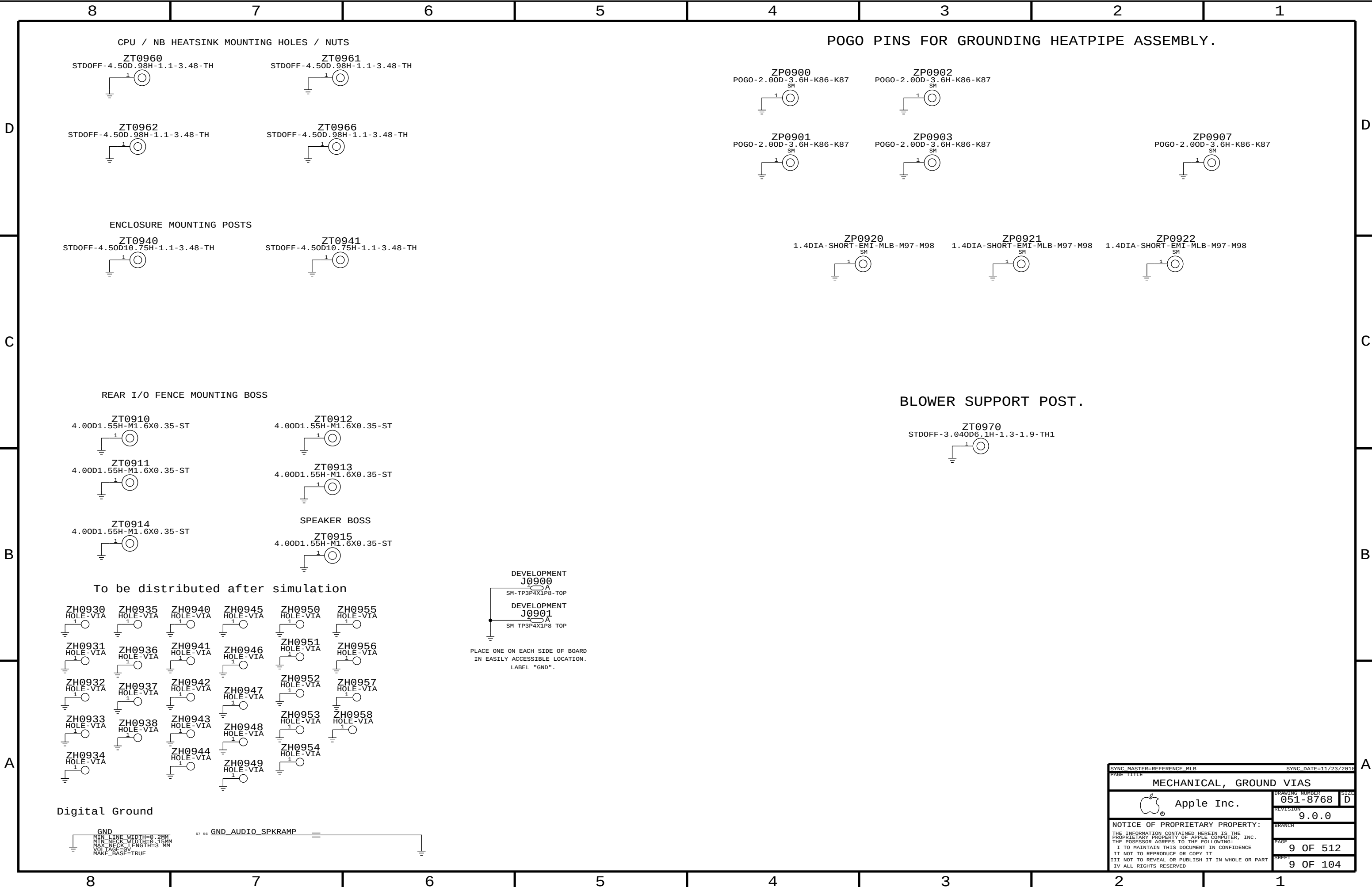
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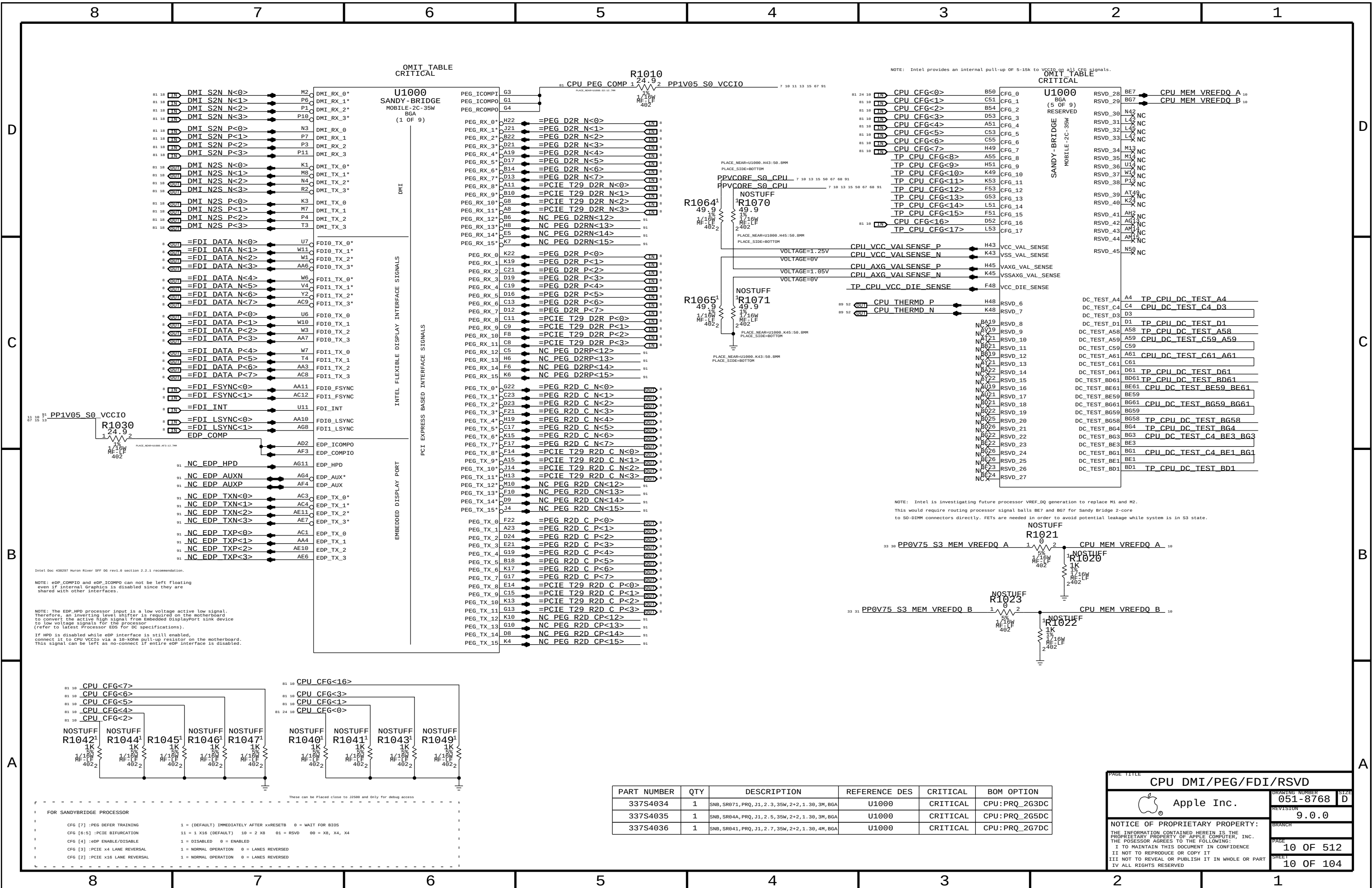
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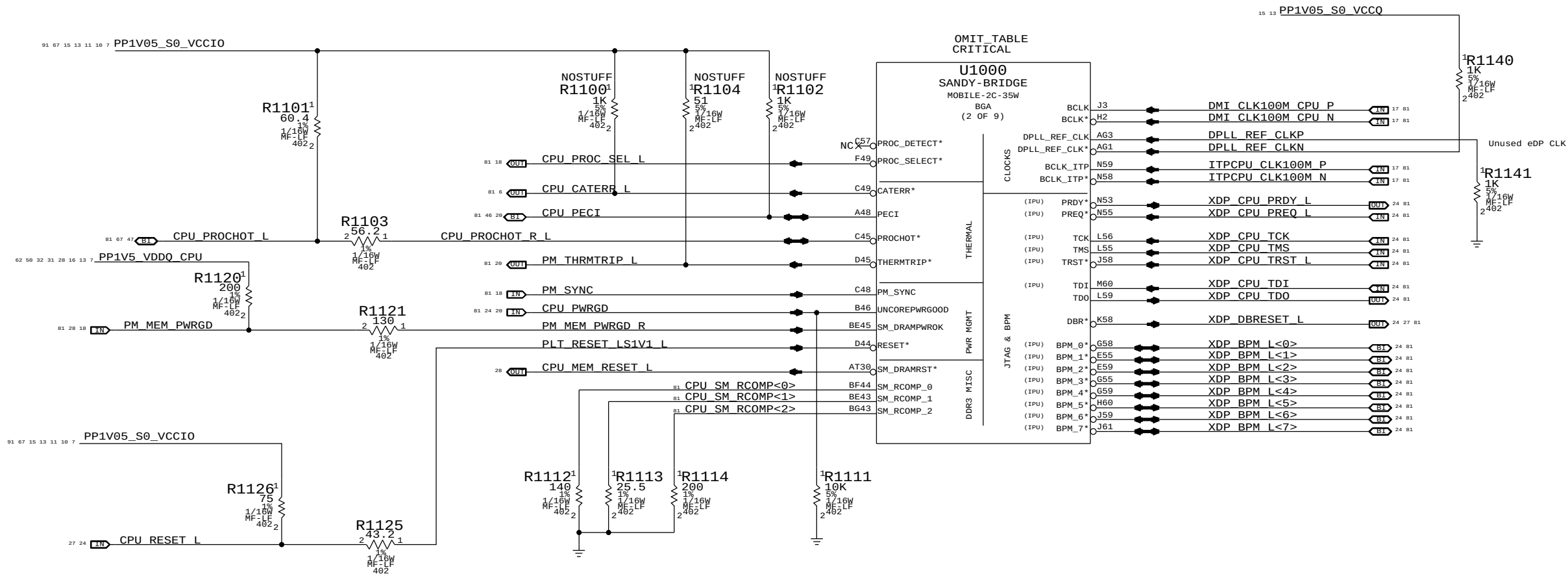
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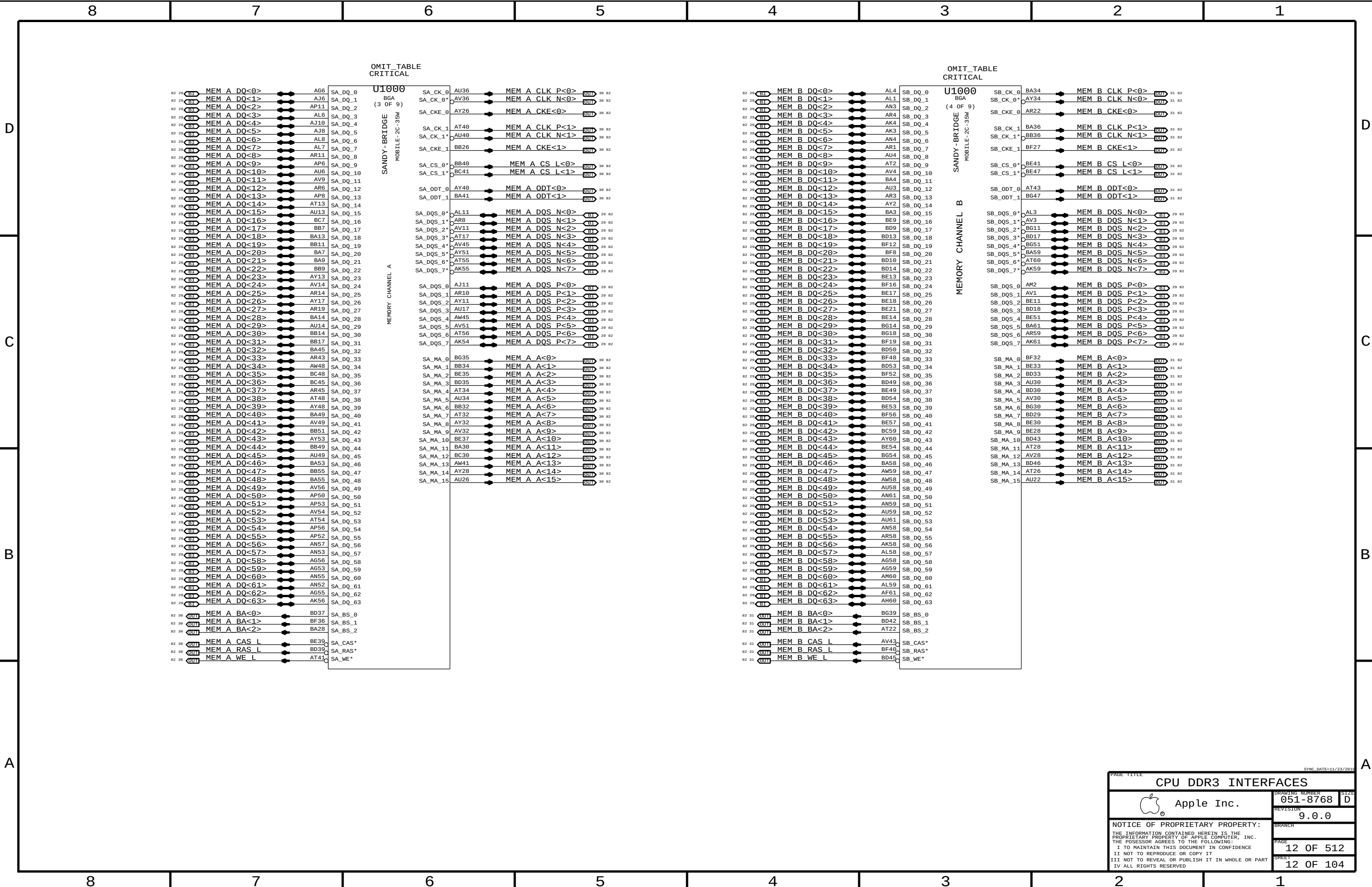
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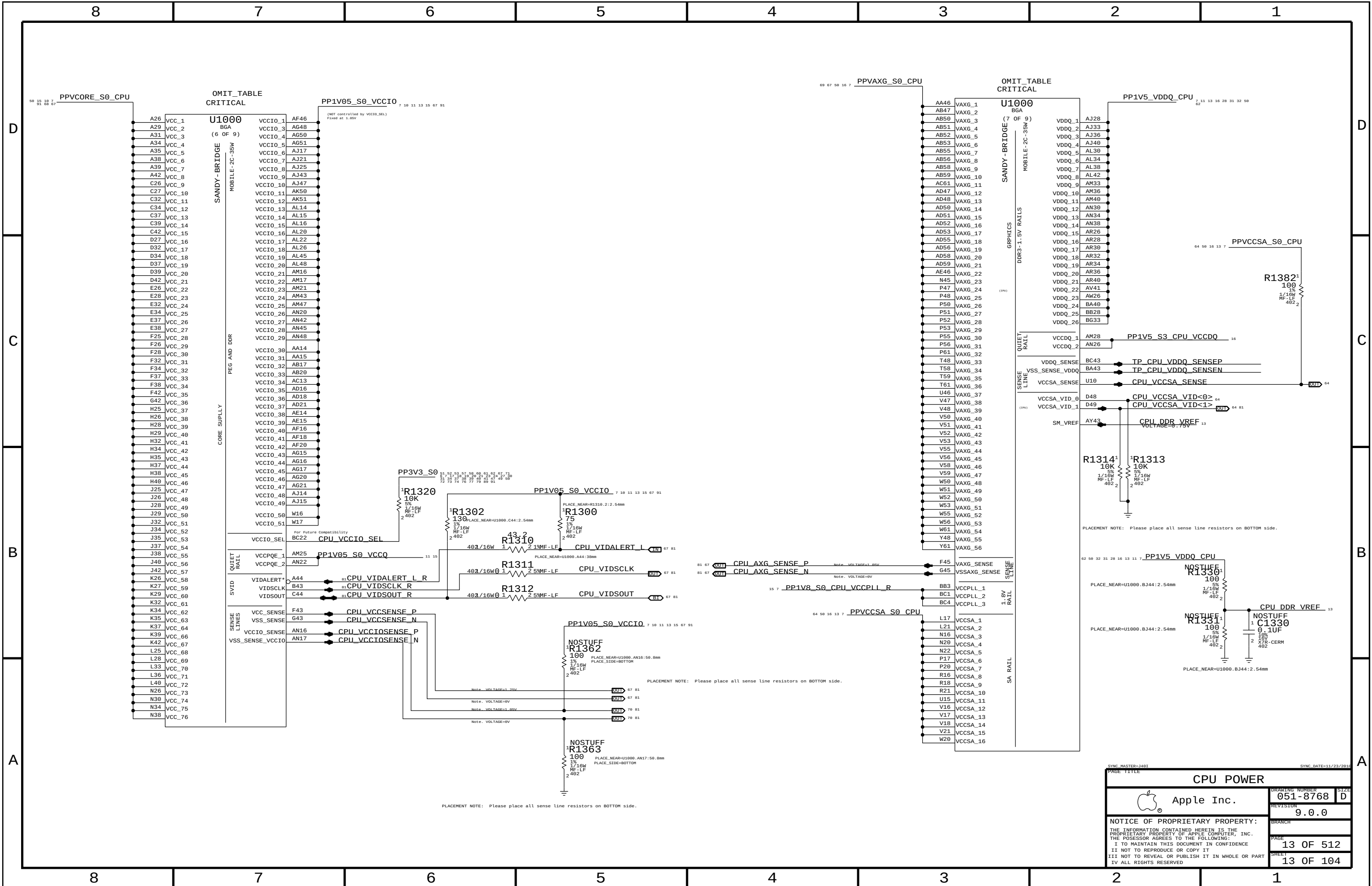


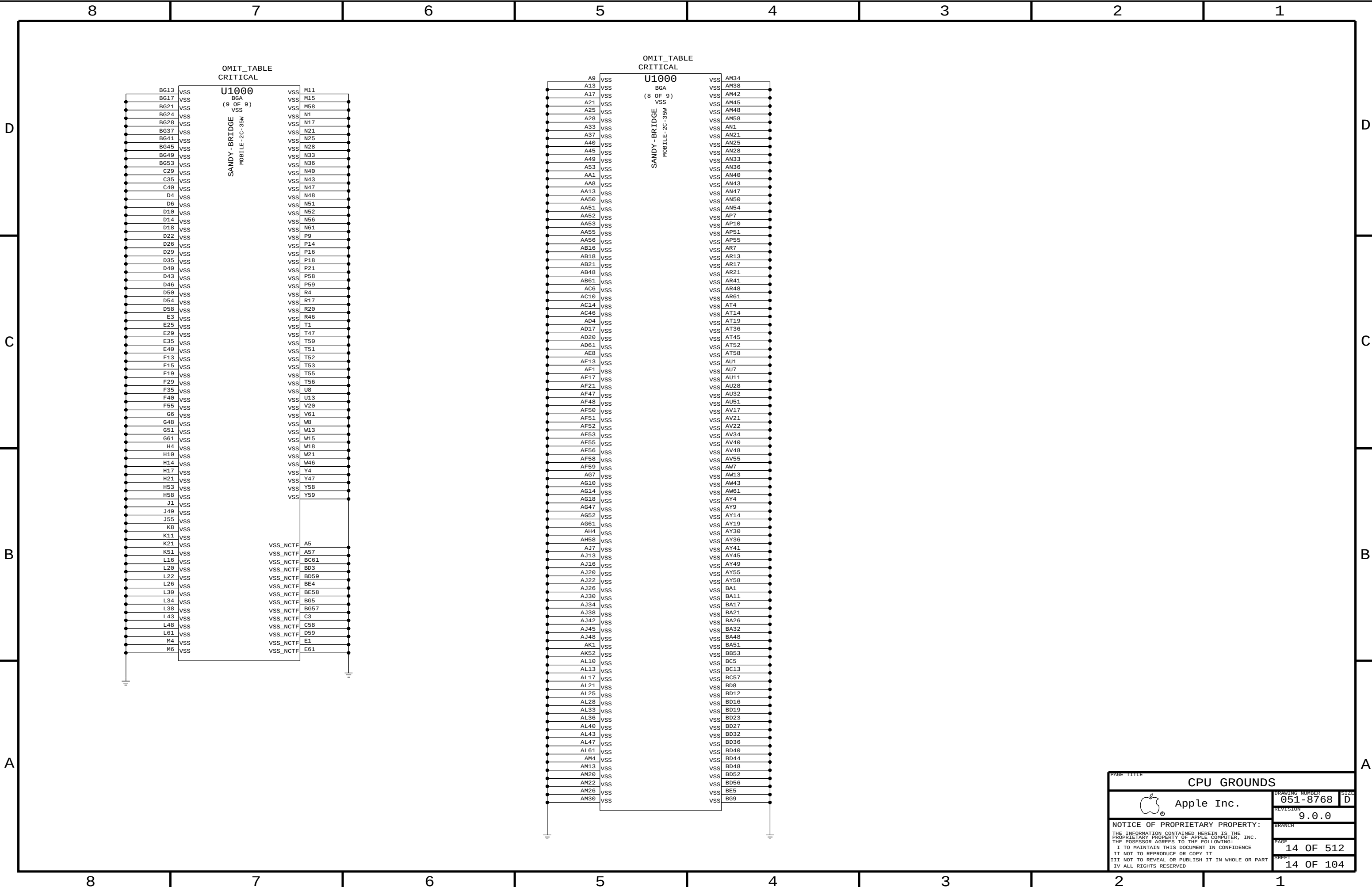
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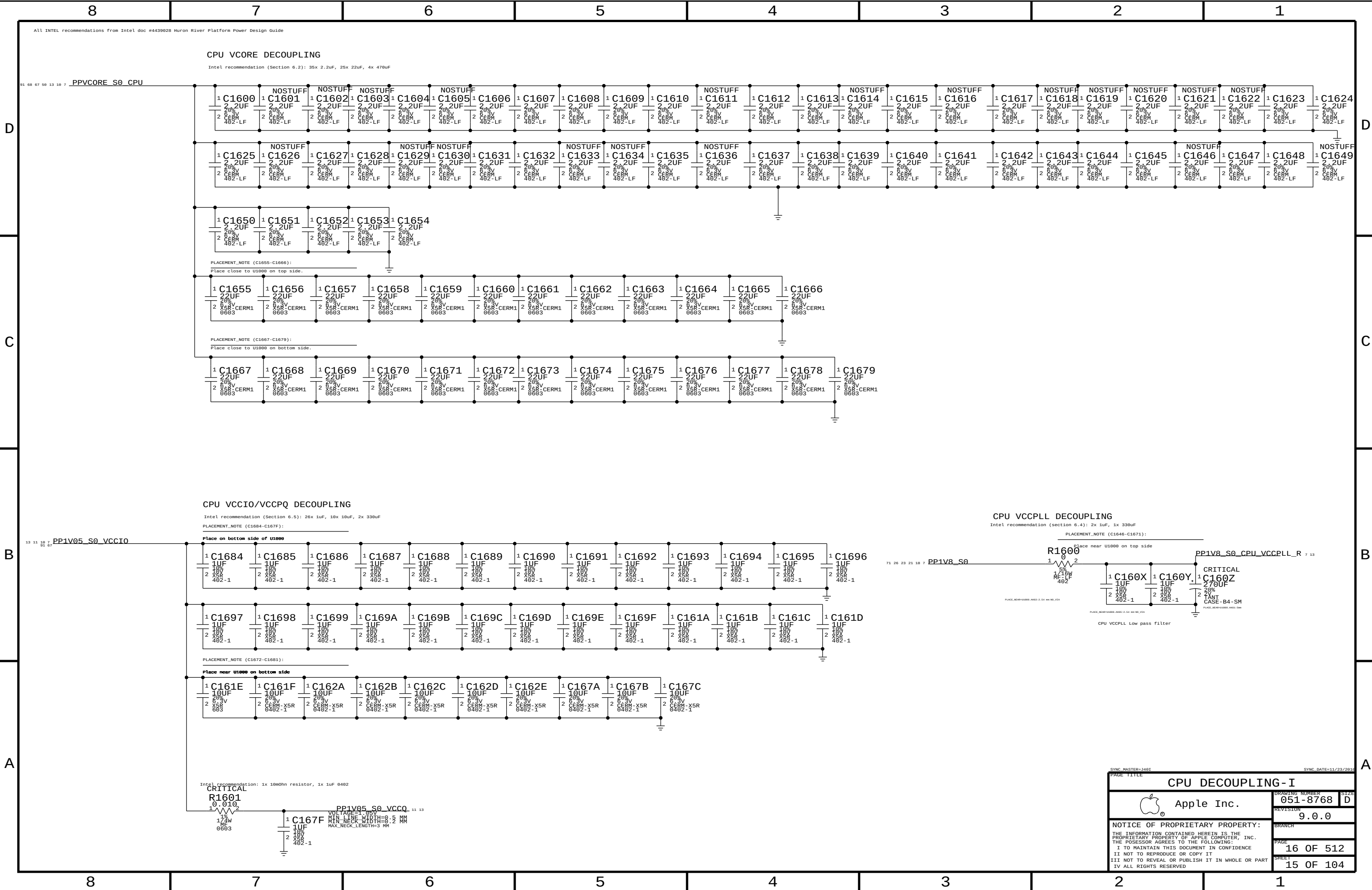


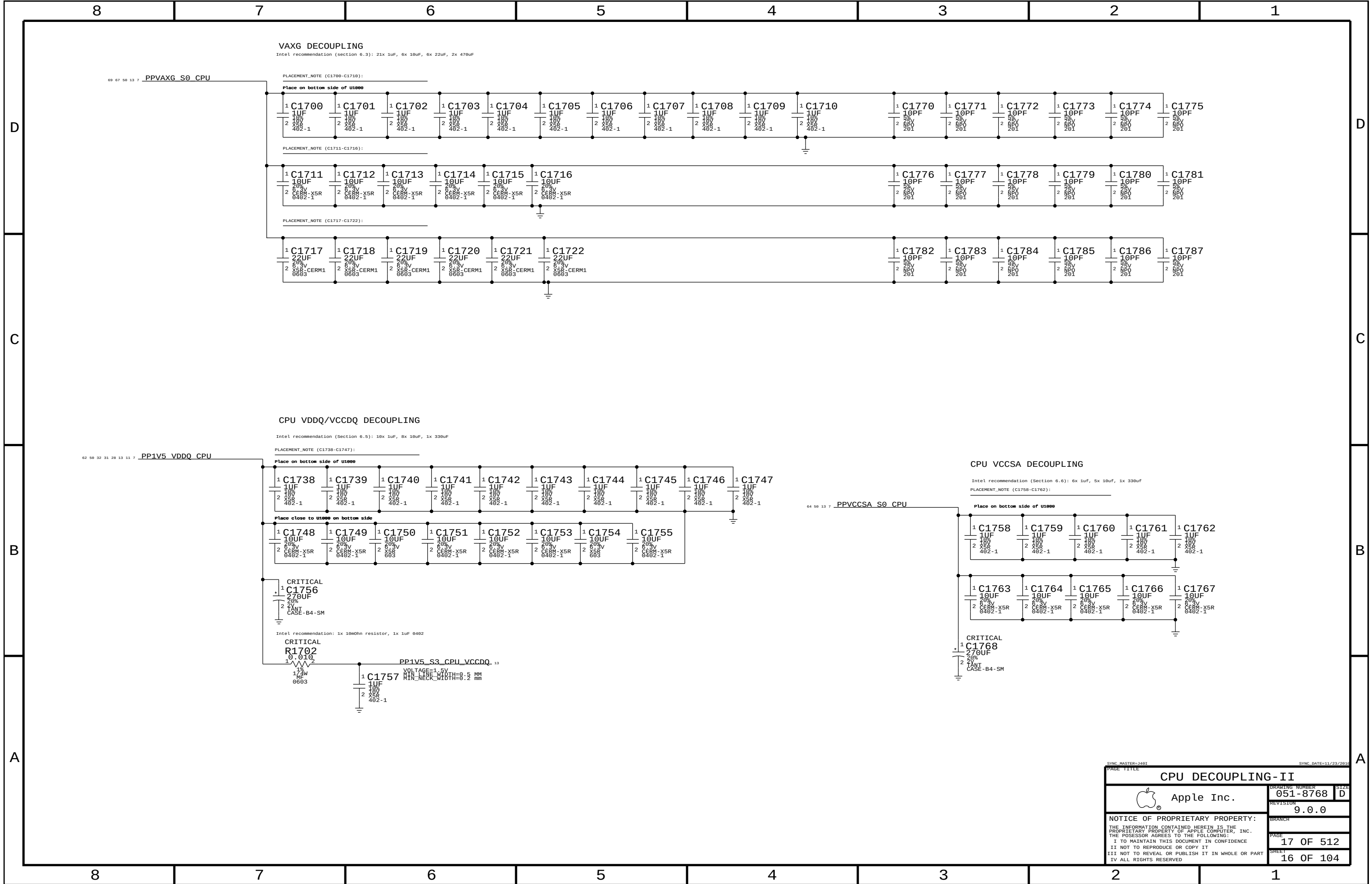


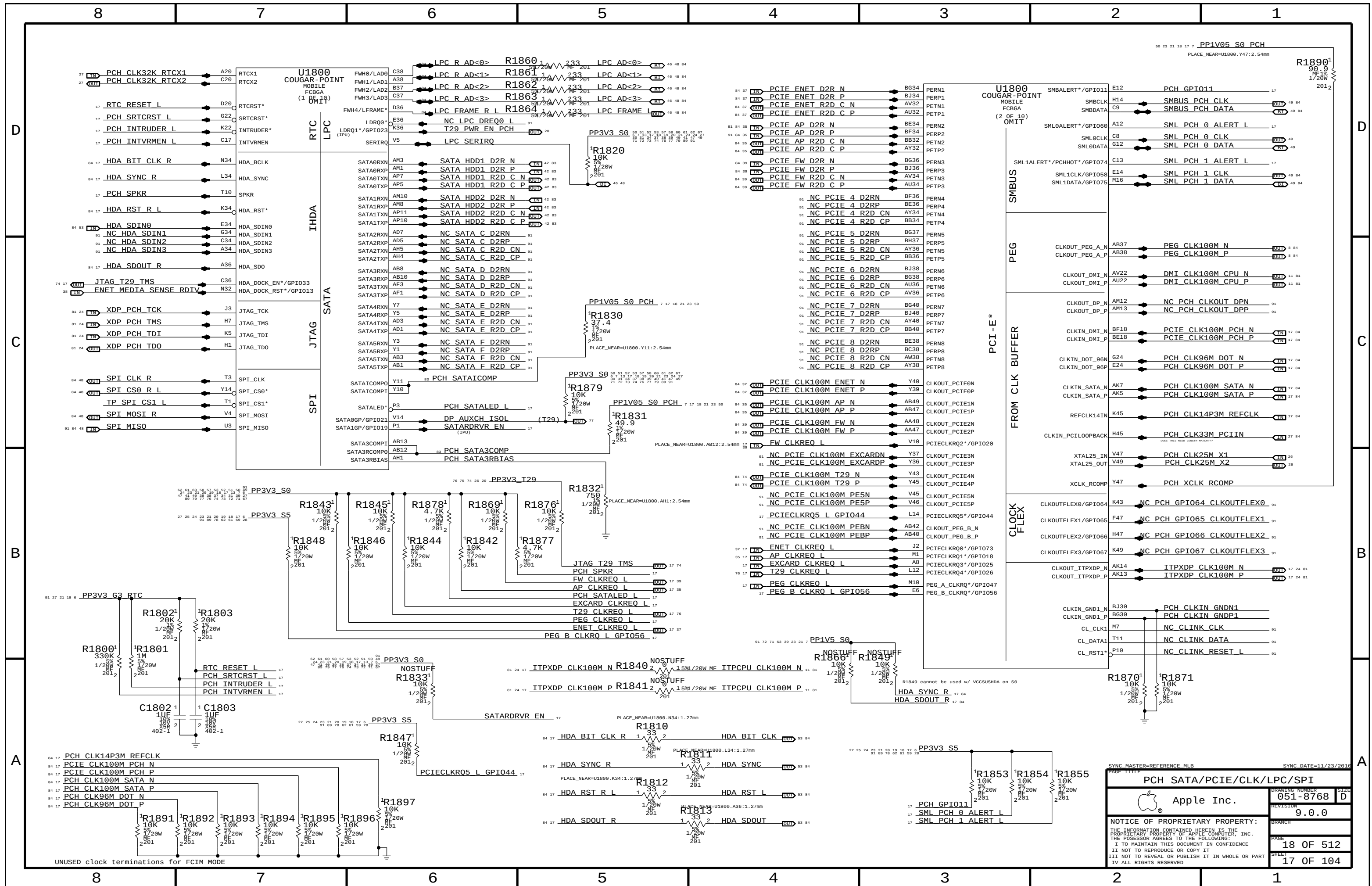


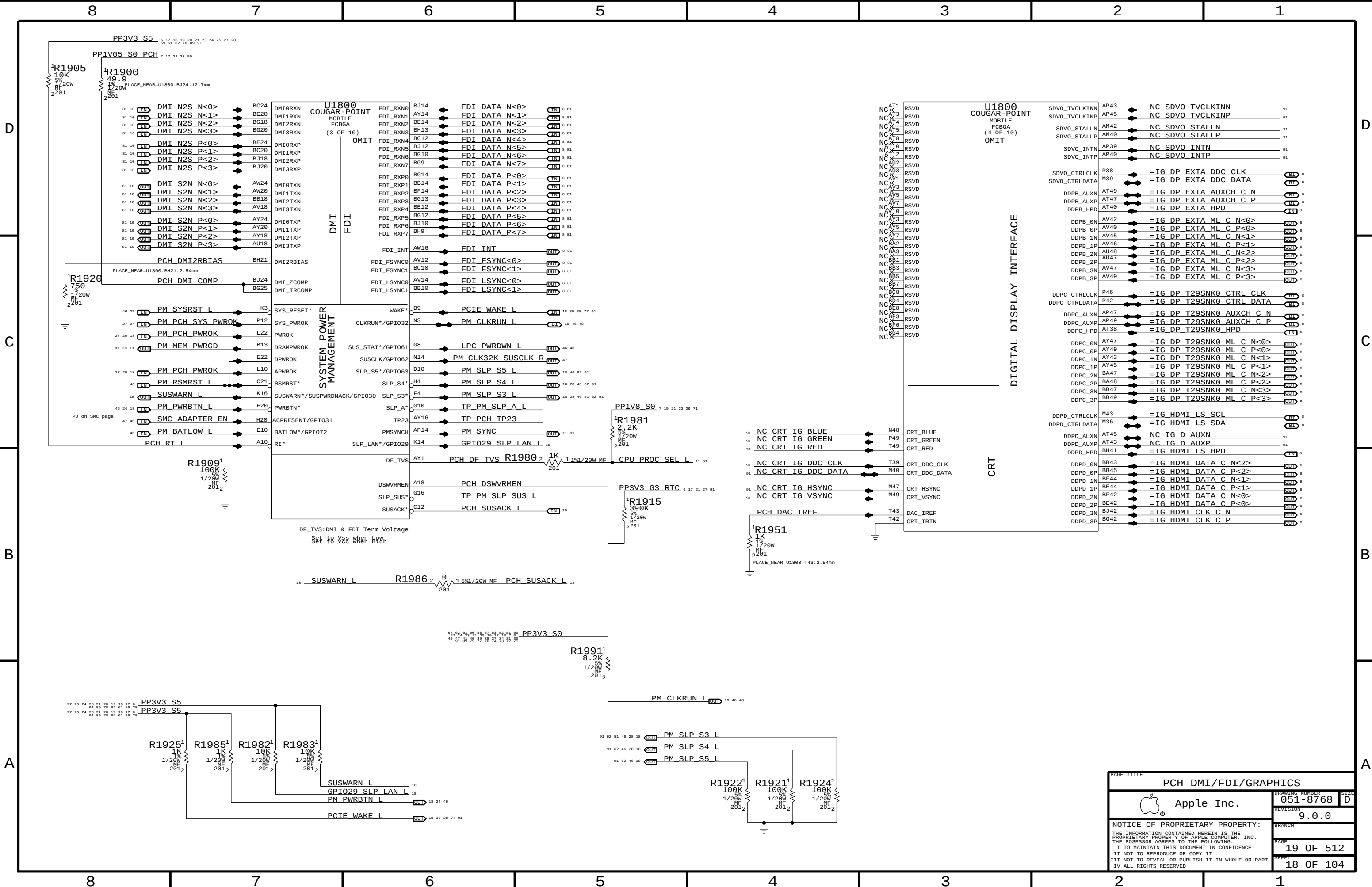


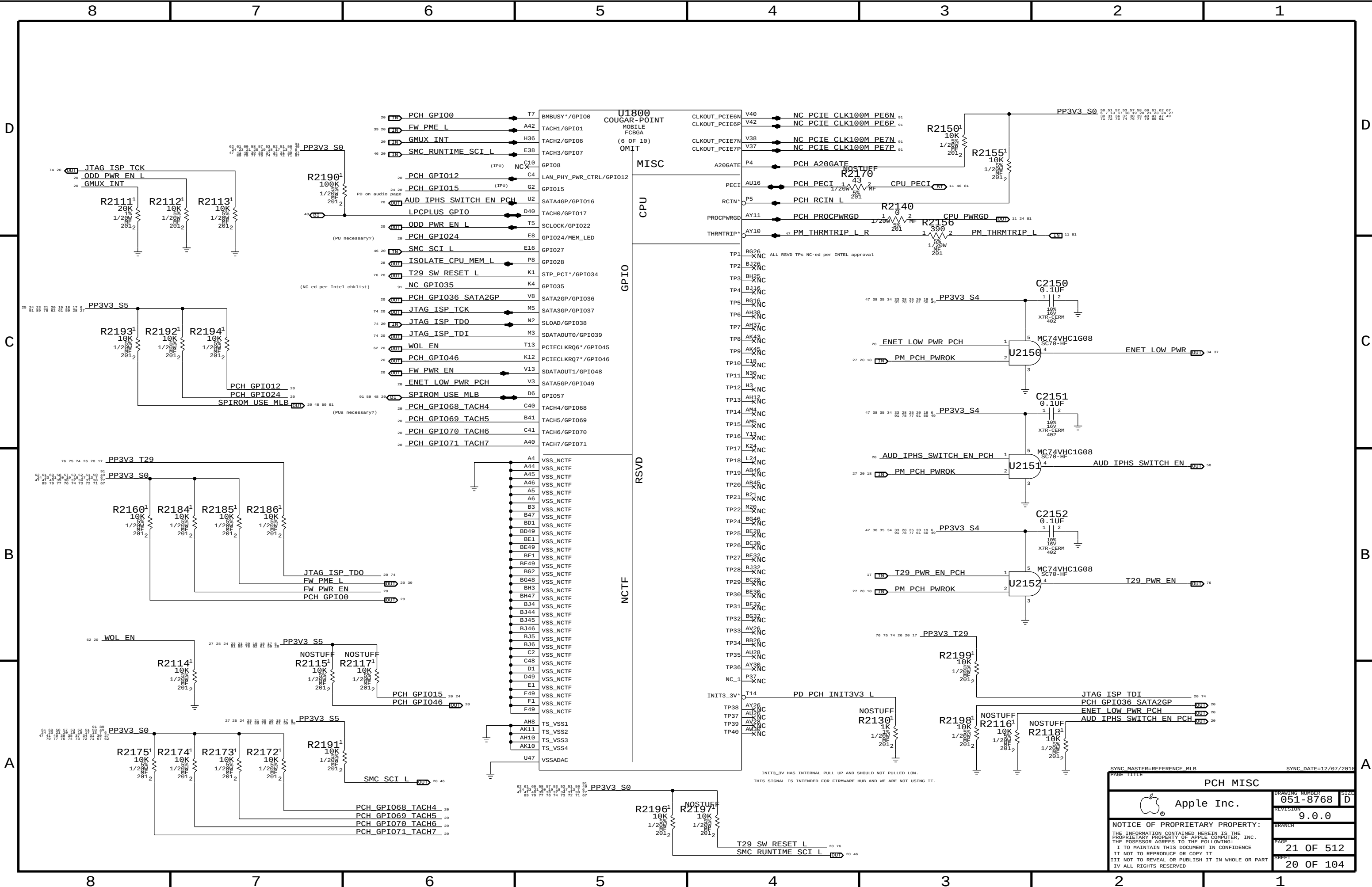


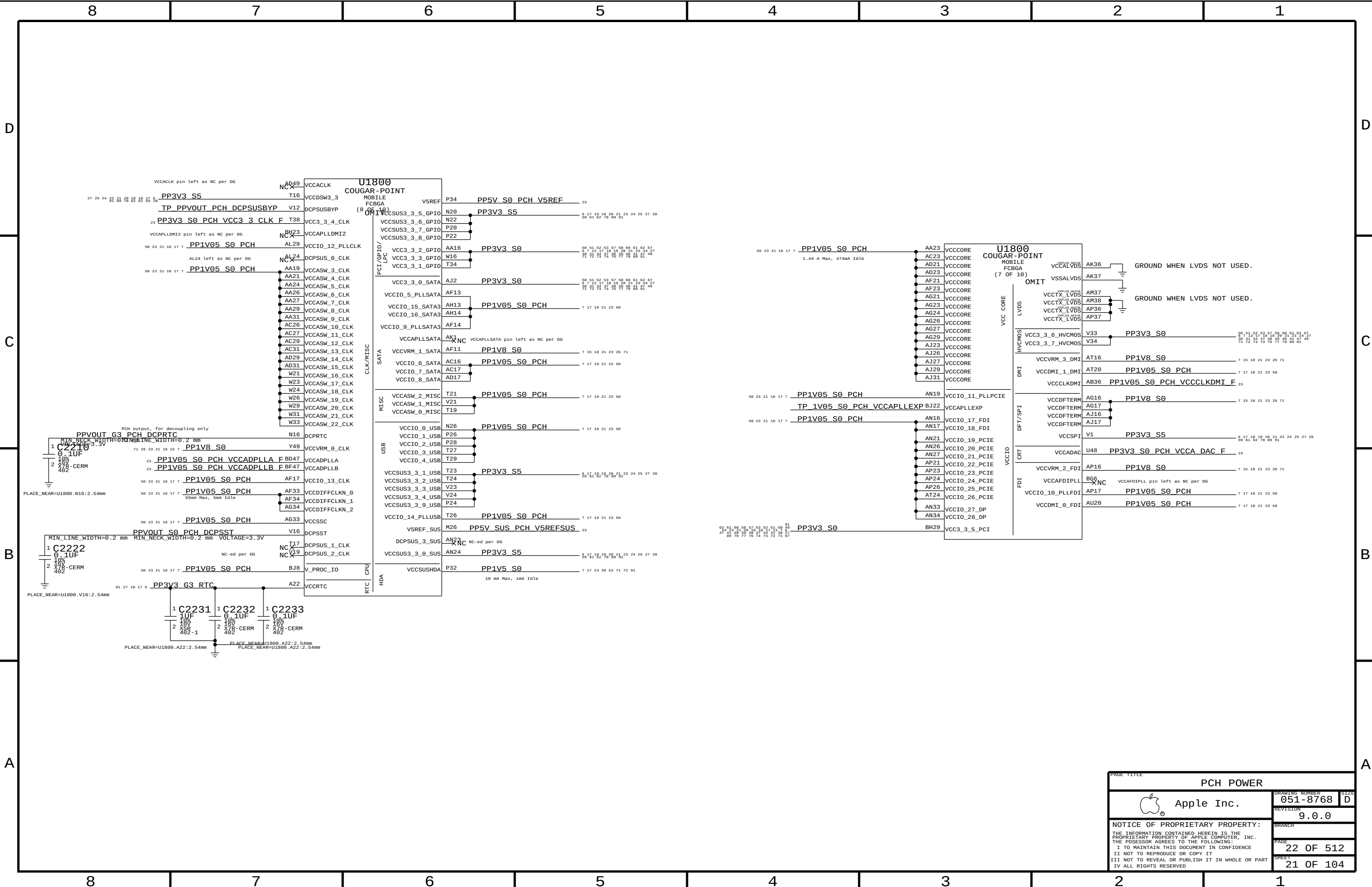


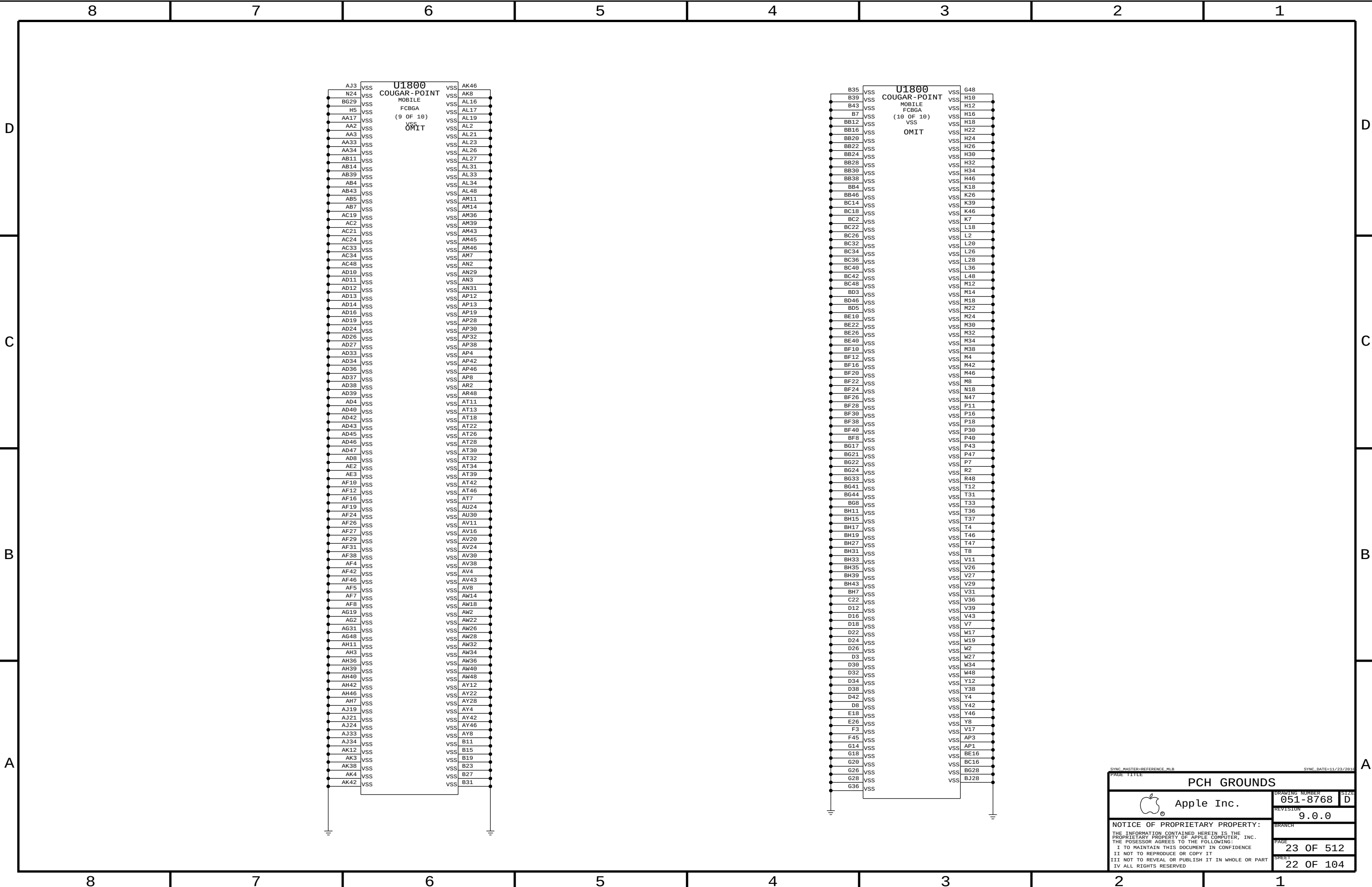




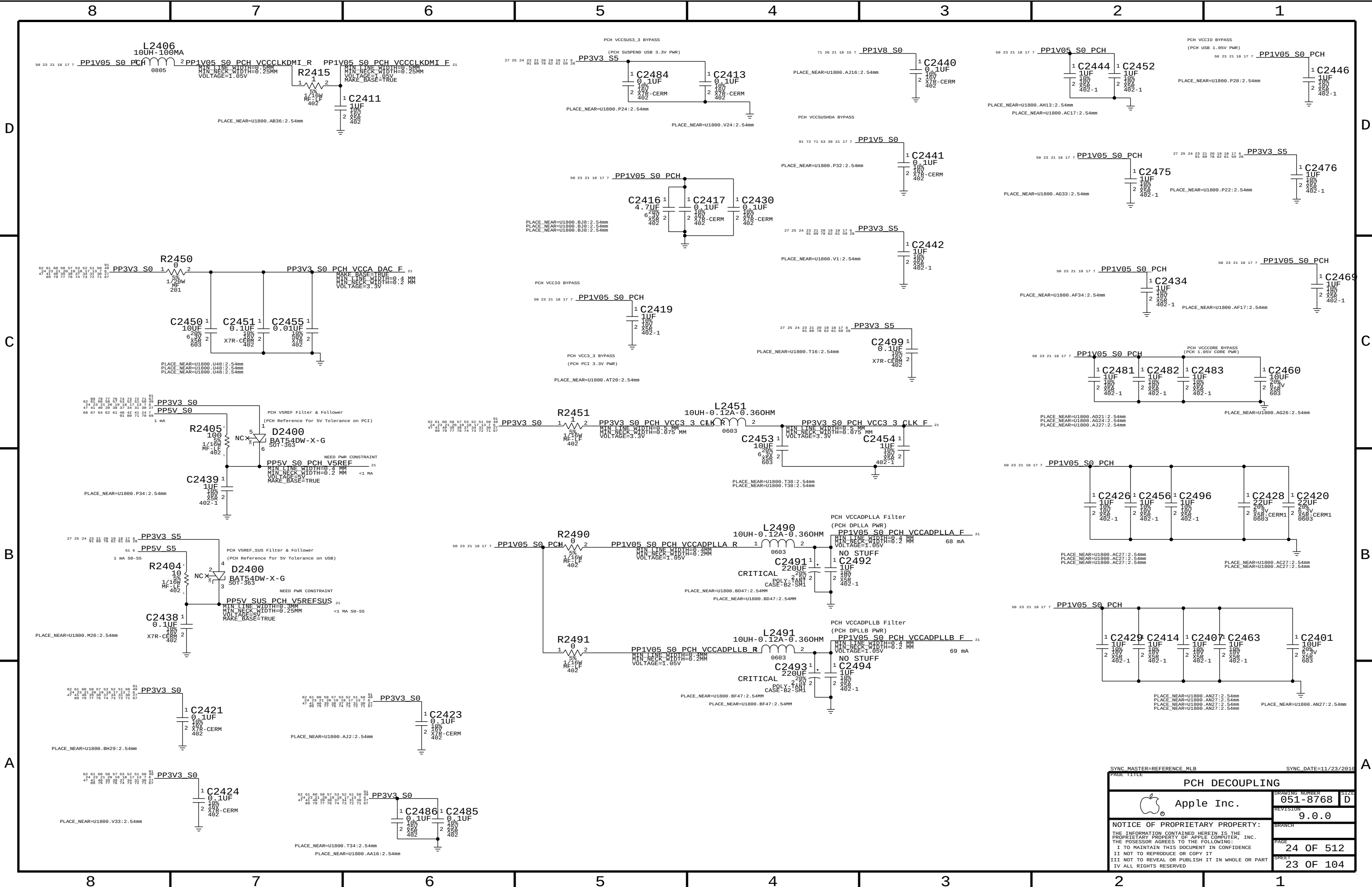




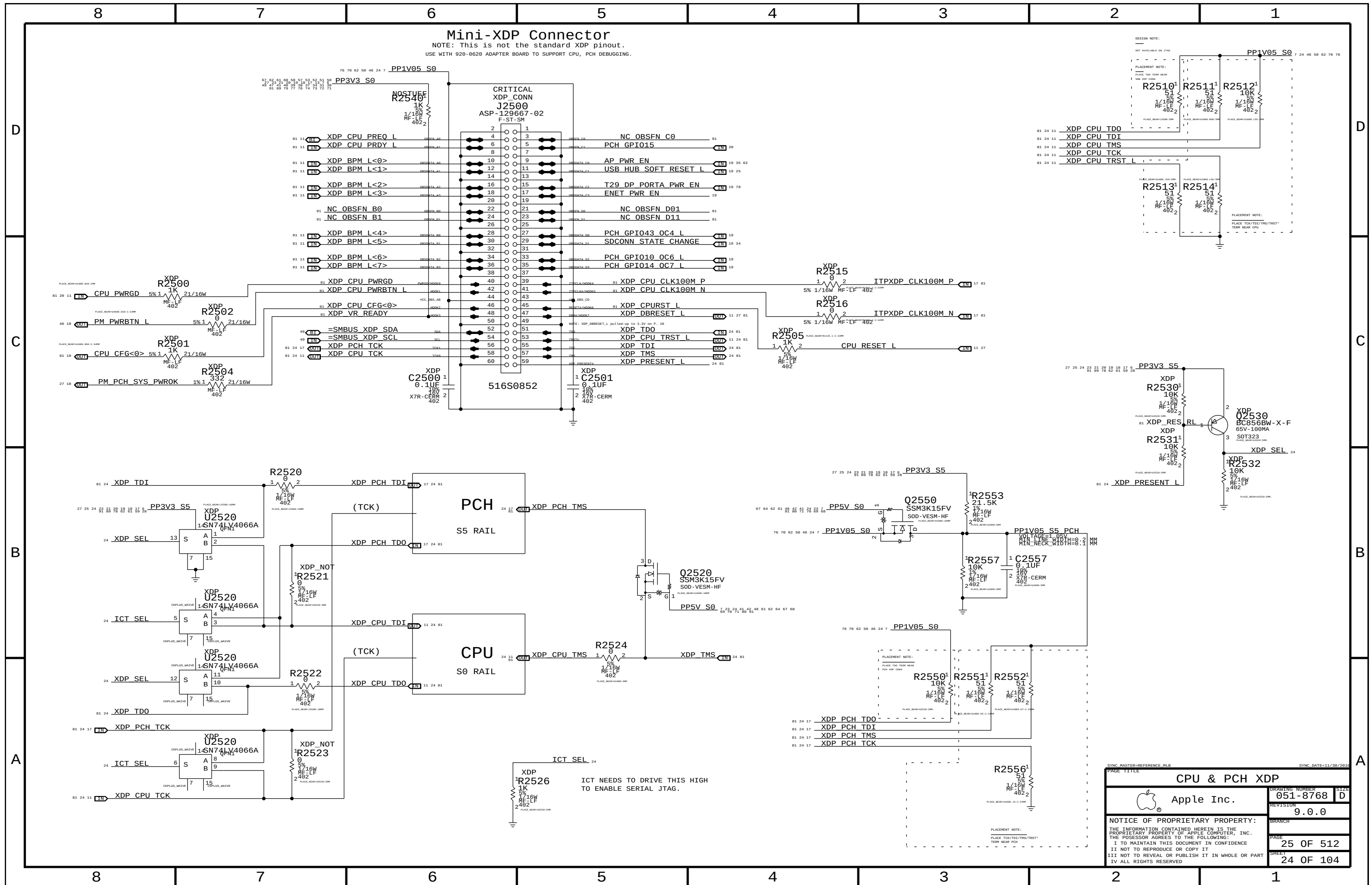


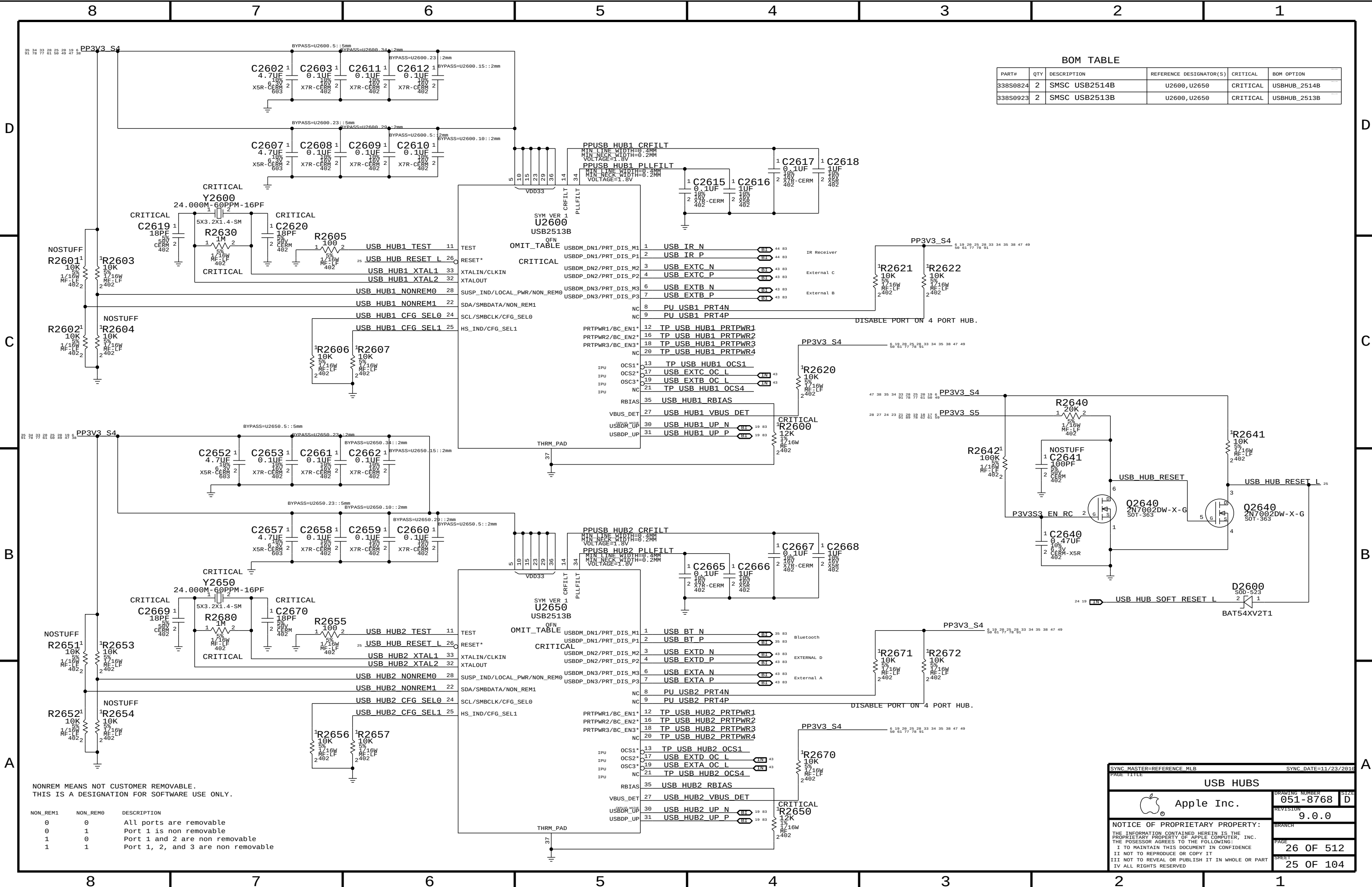


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PCH GROUNDS				DRAWING NUMBER	051-8768
Apple Inc.				REVISION	9.0.0
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PAGE TITLE			
PCH DECOUPLING			
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	BRANCH		
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BOM TABLE					
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0824	2	SMSC USB2514B	U2600,U2650	CRITICAL	USBHUB_2514B
338S0923	2	SMSC USB2513B	U2600,U2650	CRITICAL	USBHUB_2513B

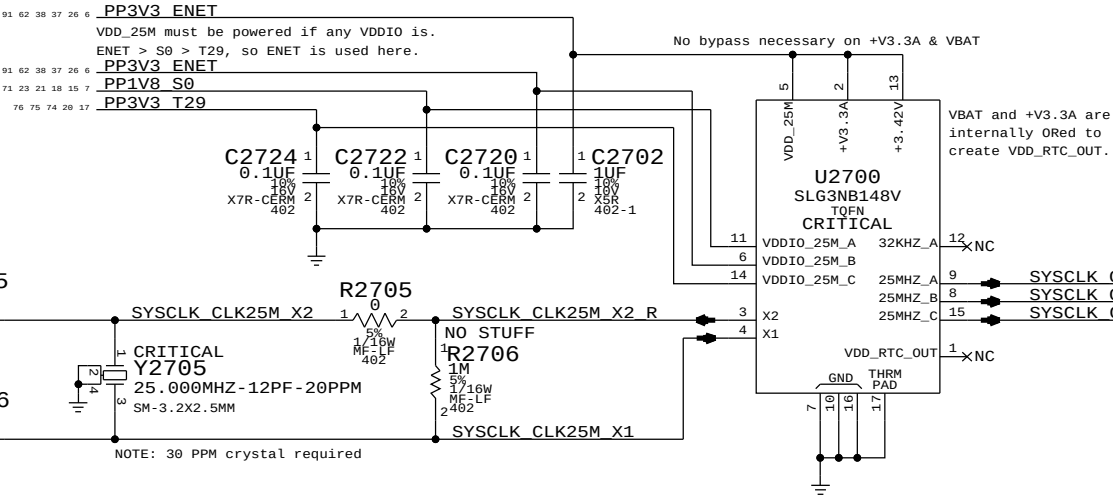
NONREM MEANS NOT CUSTOMER REMOVABLE. THIS IS A DESIGNATION FOR SOFTWARE USE ONLY.		
NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

SYNC MASTER=REFERENCE_MLB		SYNC DATE=11/23/2010	
PAGE TITLE			
USB HUBS		DRAWING NUMBER	SIZE
	Apple Inc.	051-8768	D
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System 25MHz Clock Generator

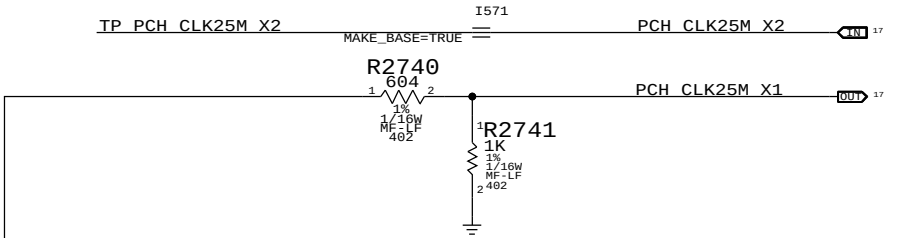
Low-Power 32kHz Source (not used)
Normal-Power 32kHz Source (not used)
GreenClk 25MHz Power

Ethernet XTAL Power
SB XTAL Power
T29 XTAL Power

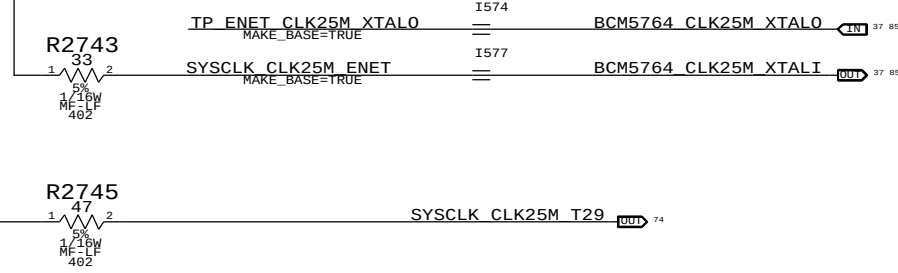


J40 NOTE: WILL NOT BE USING RTC POWER OR CLK FEATURES OF THIS CHIP DUE TO EXTRA COIN CELL LOAD
Working with Silego on a simplified version without this support

COUGAR POINT (PCH) 25MHZ CRYSTAL INPUT

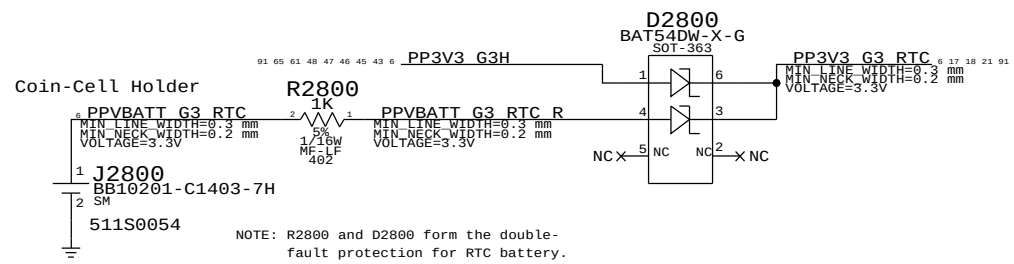


CAESAR IV (ENET) 25MHZ CRYSTAL INPUT



GREEN CLK

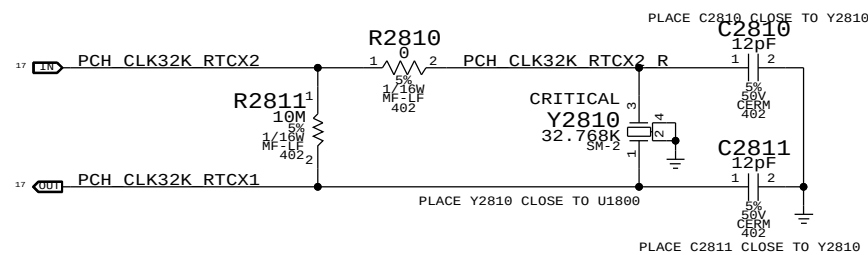
RTC Power Sources



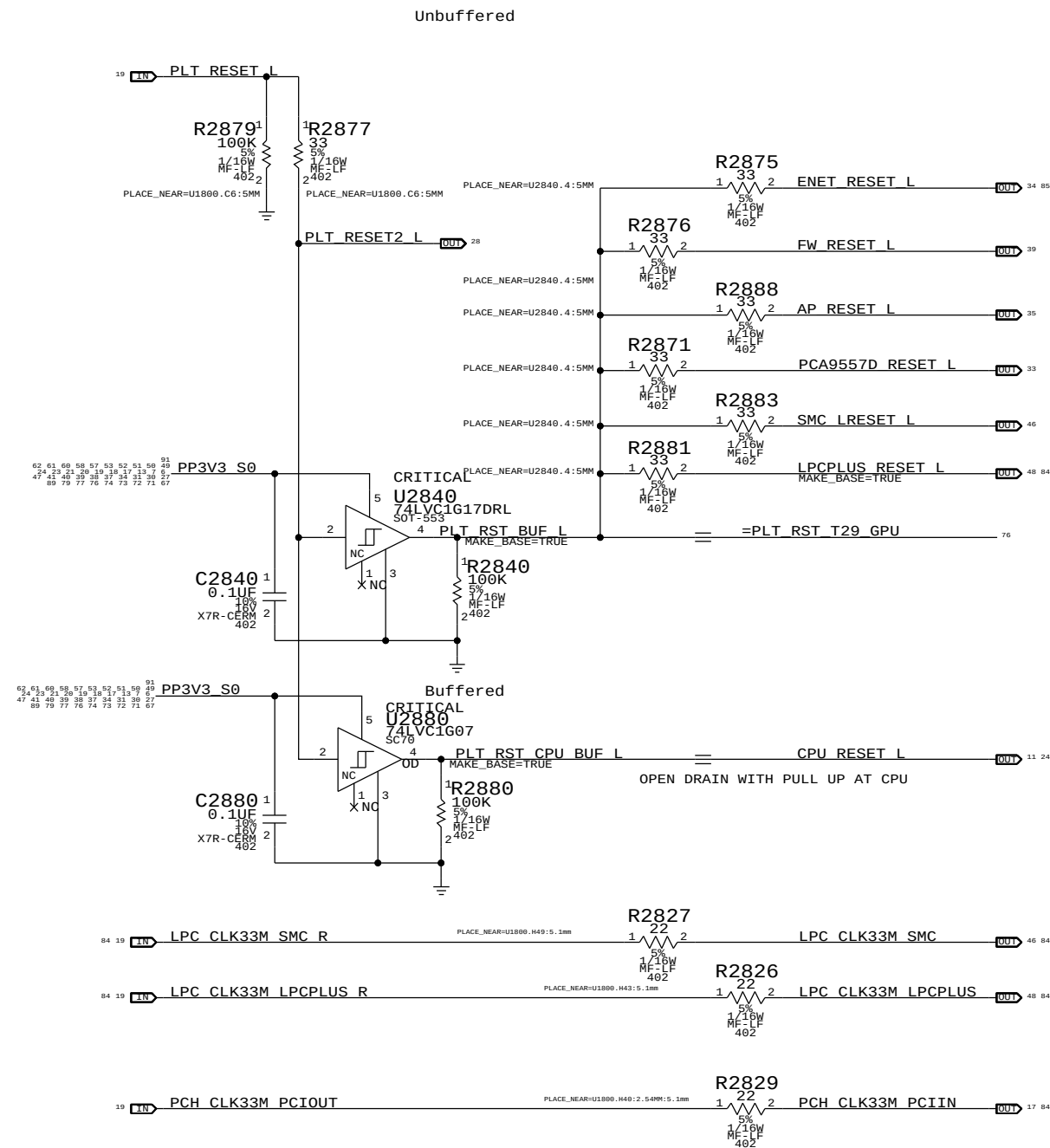
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
742-0062	BAT	COIN, LI/MAG, BR2032, 3V, 190MAH	BAT	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0074	511S0054	?	?	BATTERY HOLDER WITHOUT ALIGNMENT PEGS

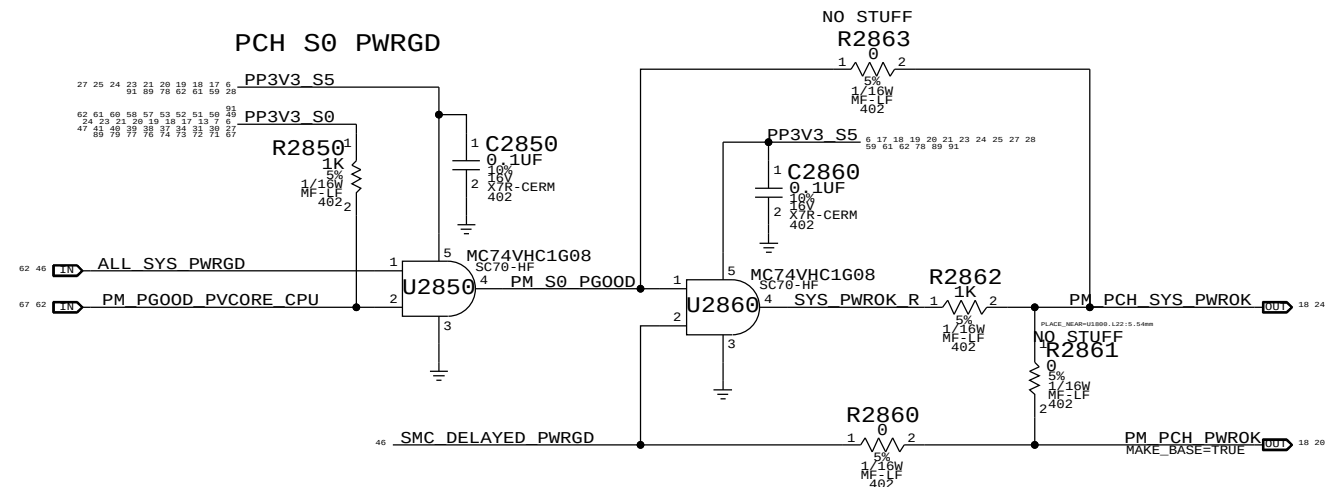
PCH RTC Crystal



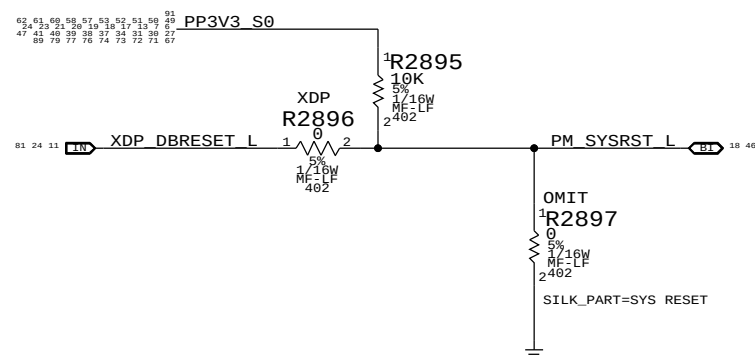
Platform Reset Connections



PCH S0 PWRGD



PCH Reset Button



SYNC MASTER-REFERENCE: HLB		PAGE TITLE		SYNC DATE=11/30/2011	
RTC SUPPORT, RESETS		DRAWING NUMBER		SIZE	
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

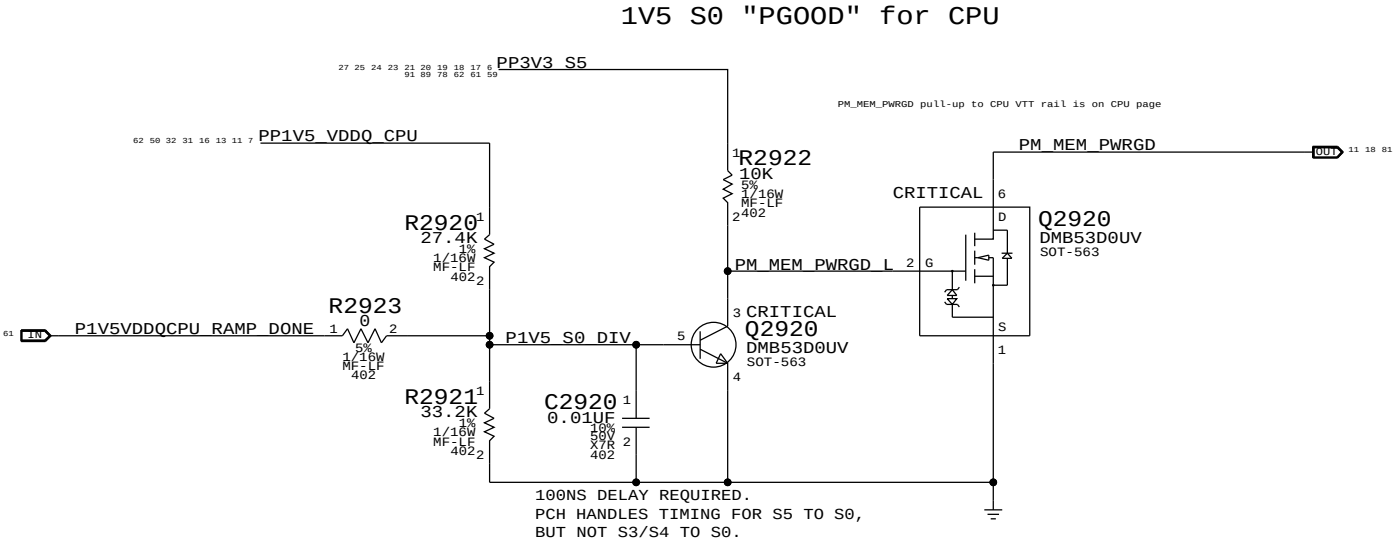
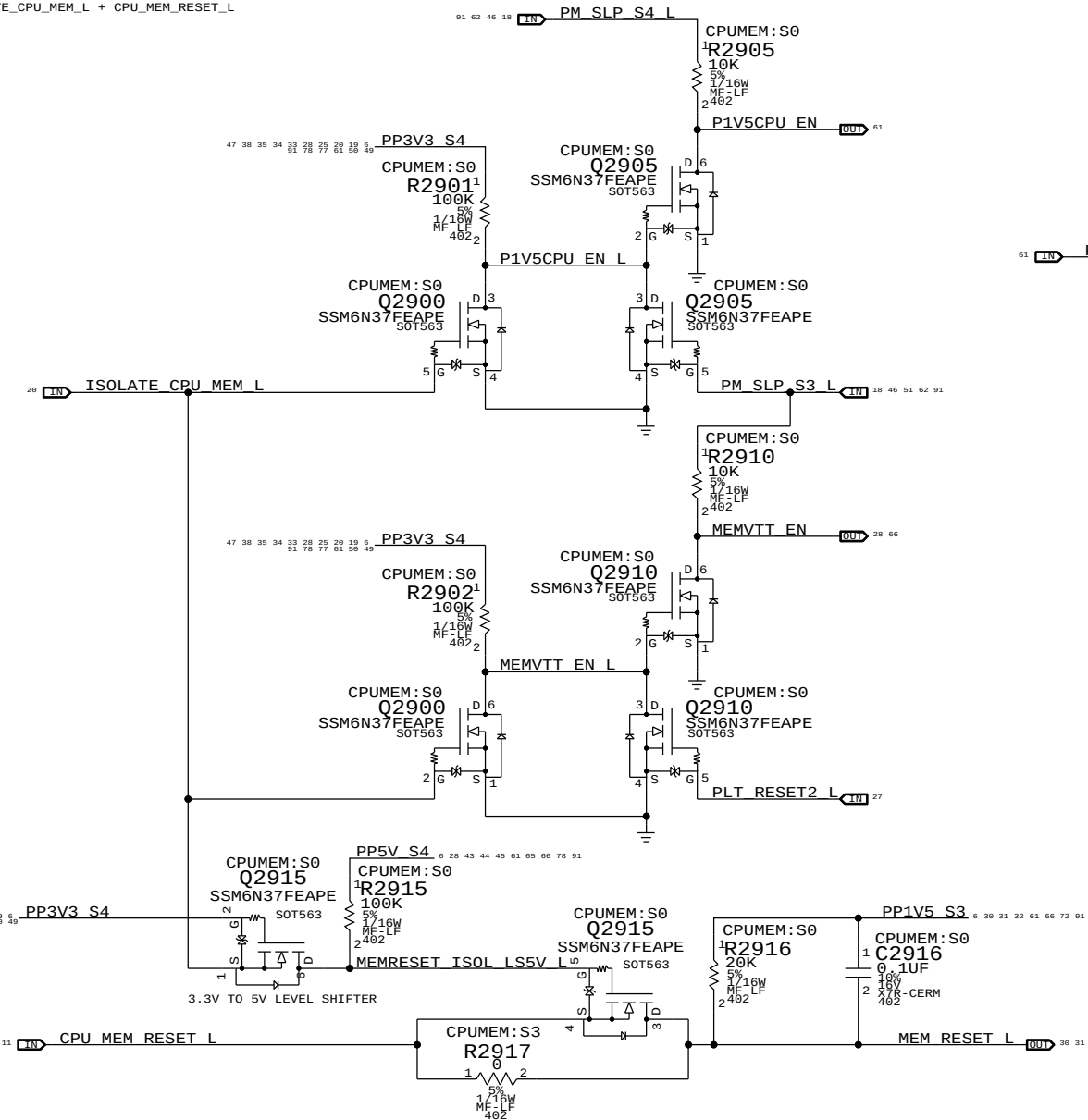
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L

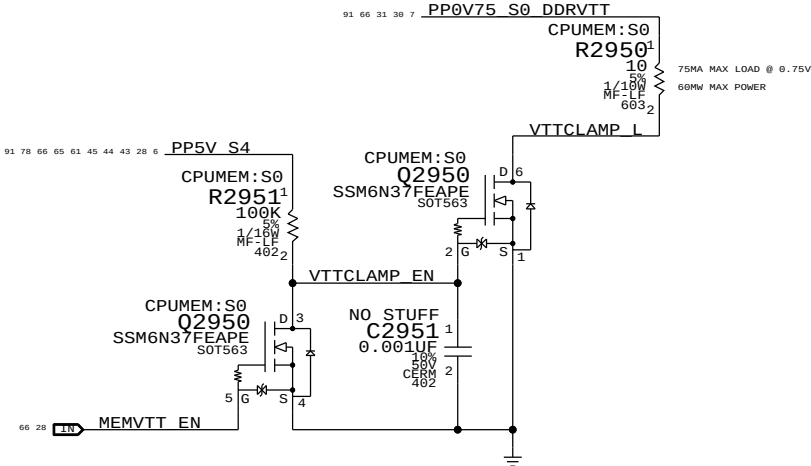
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
	1	0	1	1	1	1	1	1
	2	0	0	1	1	1	0	1
to S3	3	0	0	1	X	1	0	0
	4	0	0	1	1	X	1	1
	5	0	1	1	1	0 (*)	1	1
S0	6	0	1	1	1	1	1	1
	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER-REFERENCE: HLB

SYNC DATE: 11/23/2011

CPU Memory S3 Support

Apple Inc.

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7

6

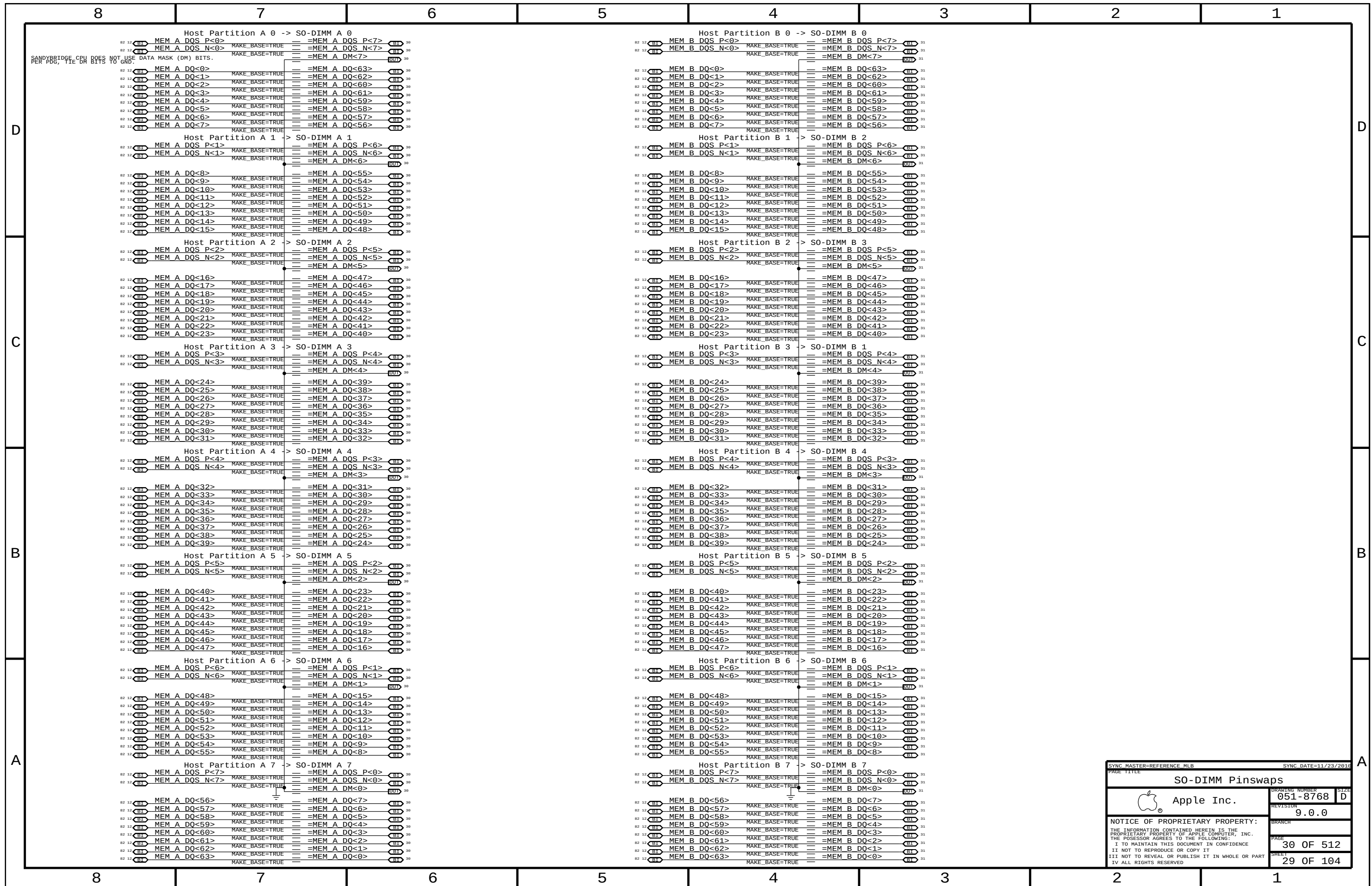
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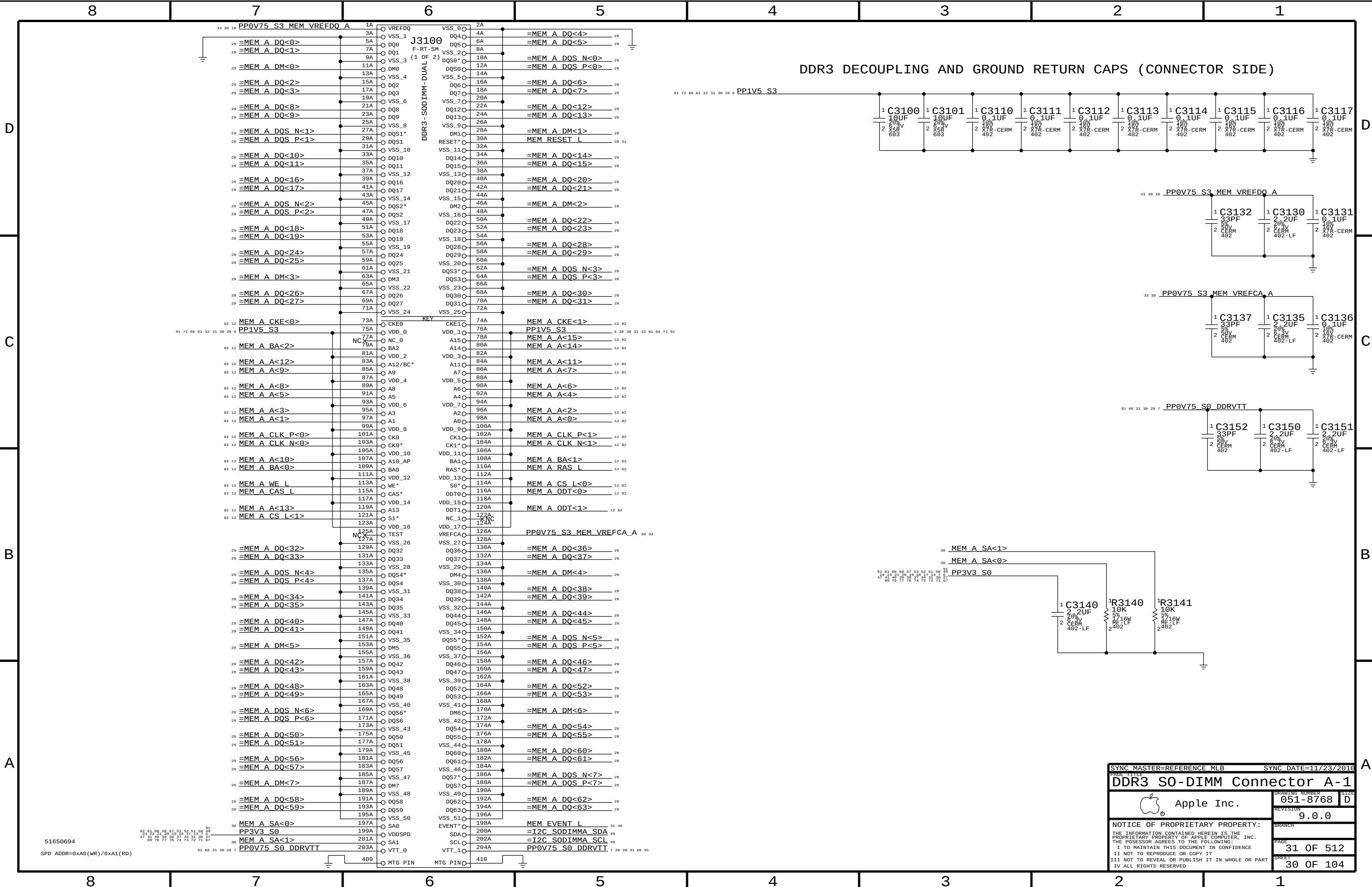
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3

2

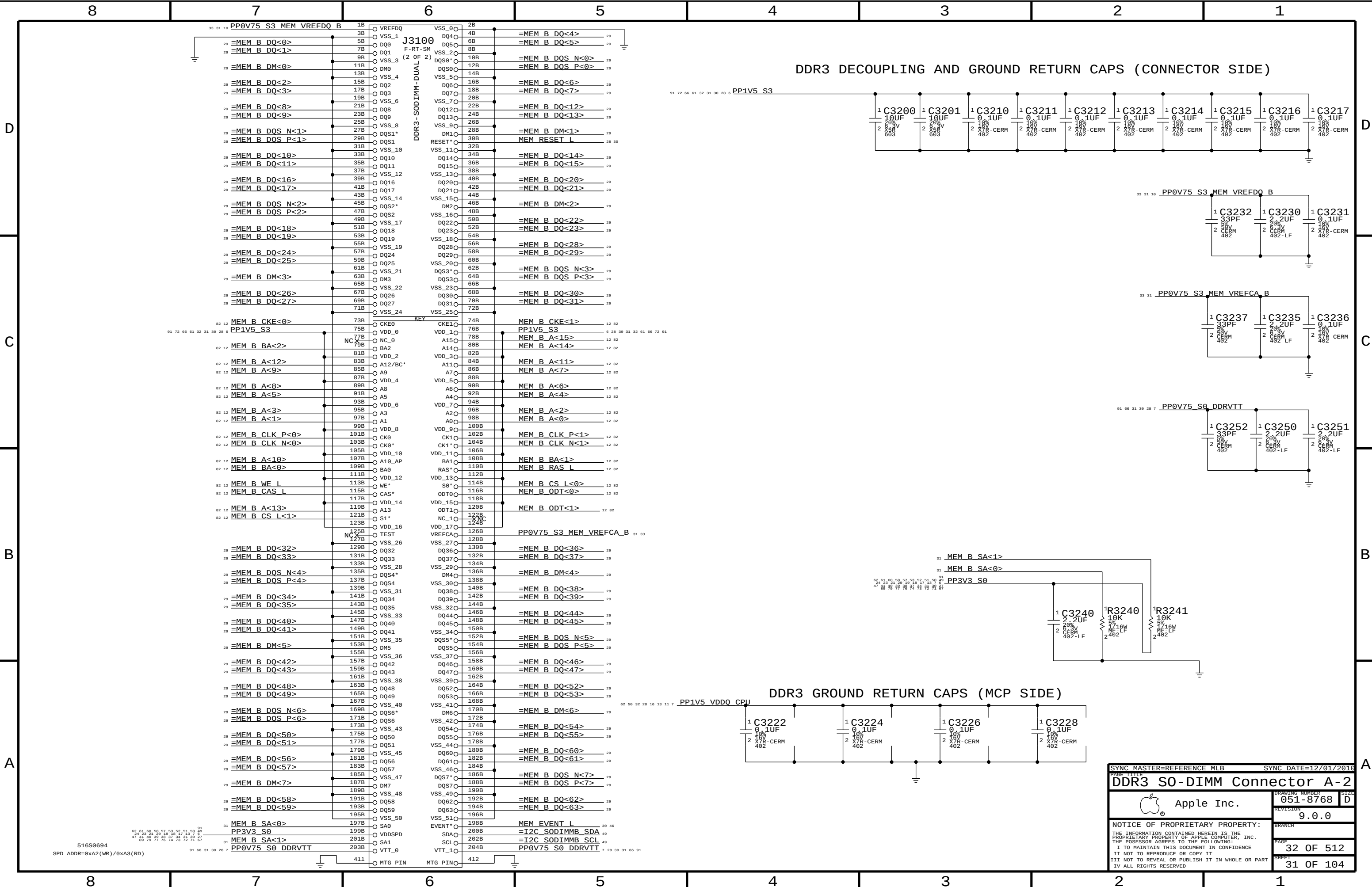
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DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)

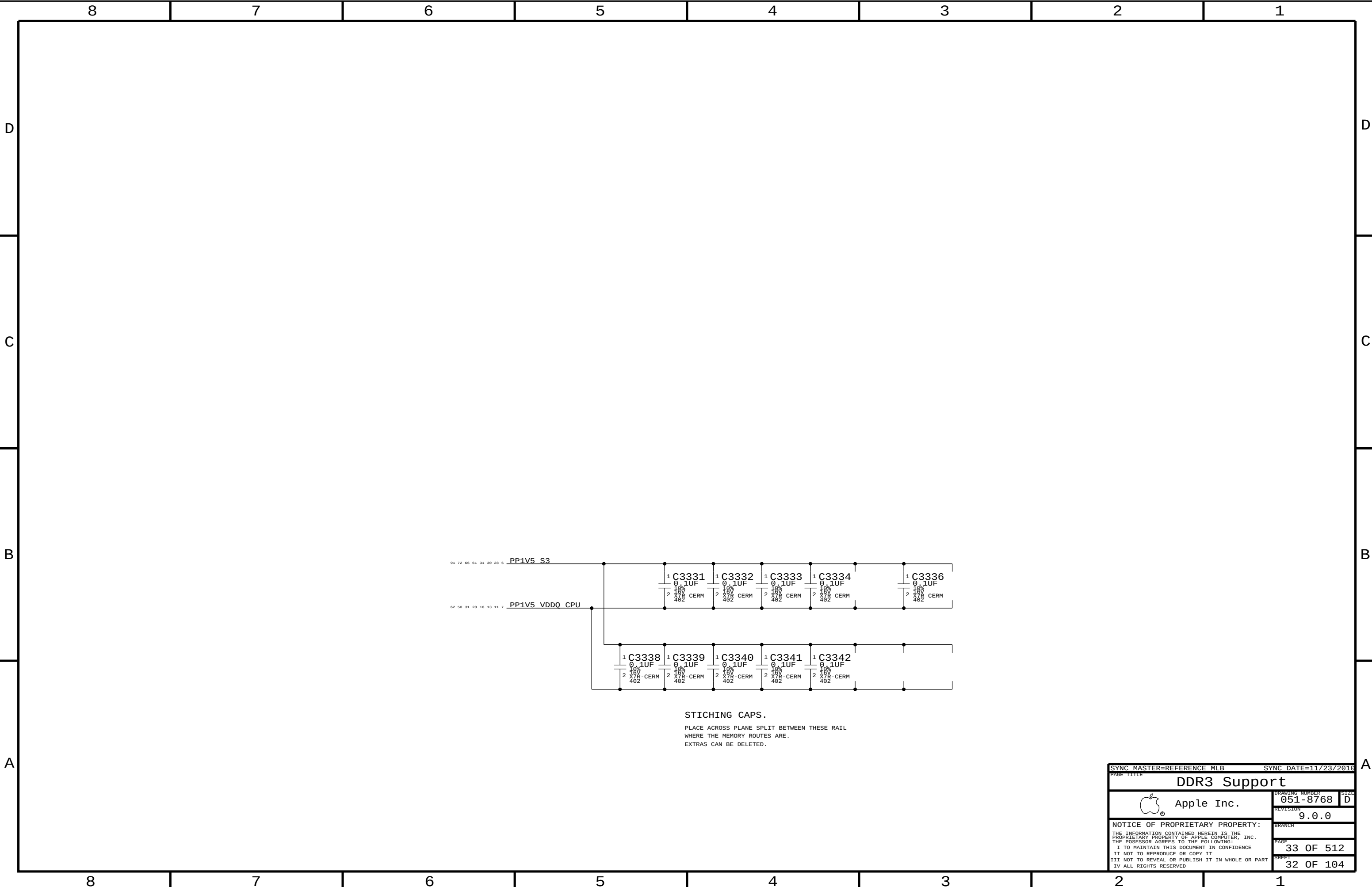
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DDR3 SODIMM Connector A-1		051-8768	
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		SHEET	30 OF 104



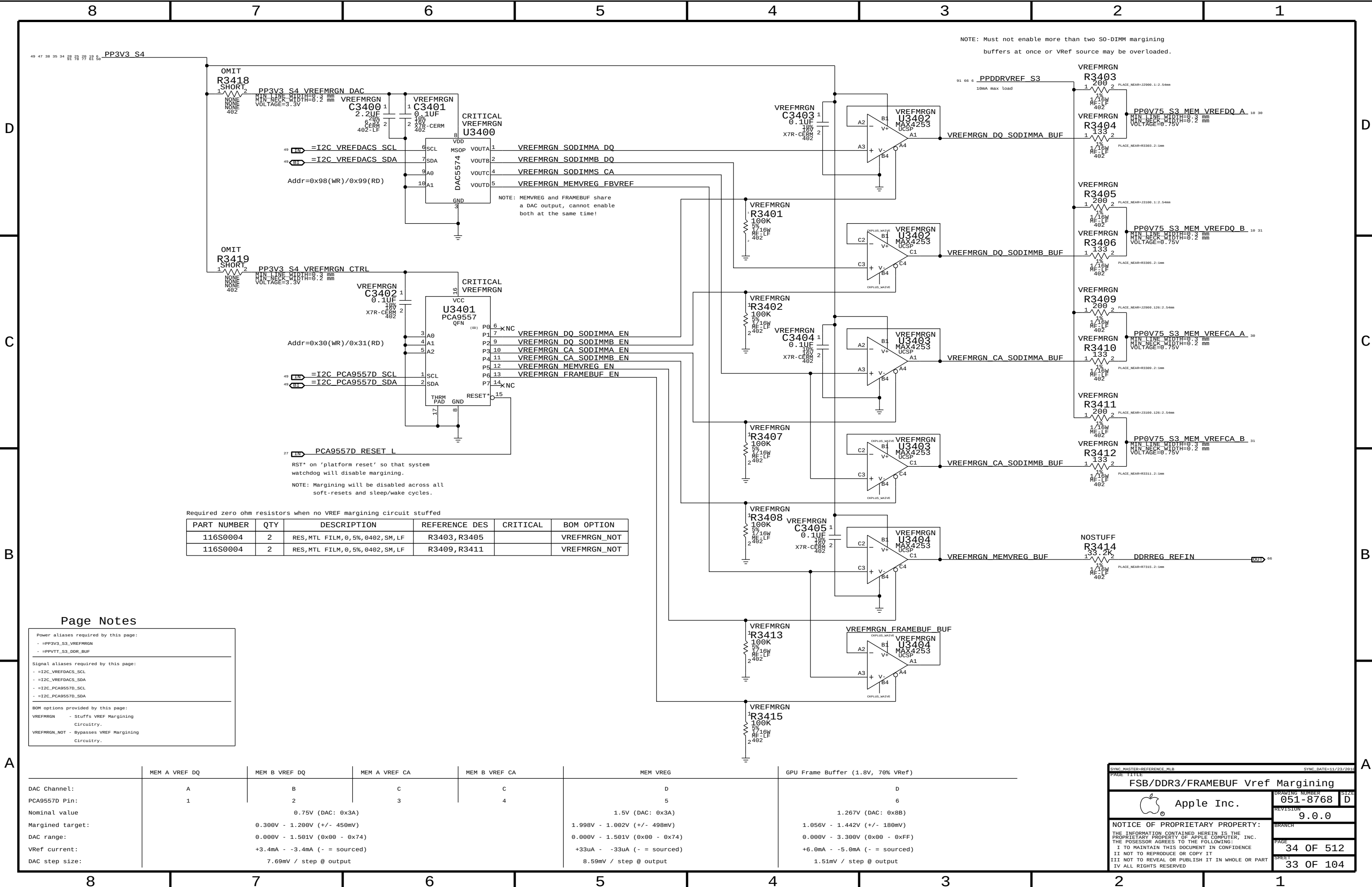
DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)

DDR3 GROUND RETURN CAPS (MCP SIDE)

SYNC MASTER=REFERENCE MLB		SYNC DATE=12/01/2016	
PAGE TITLE			
DDR3 SODIMM Connector A-2		DRAWING NUMBER	051-8768
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DDR3 Support			
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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

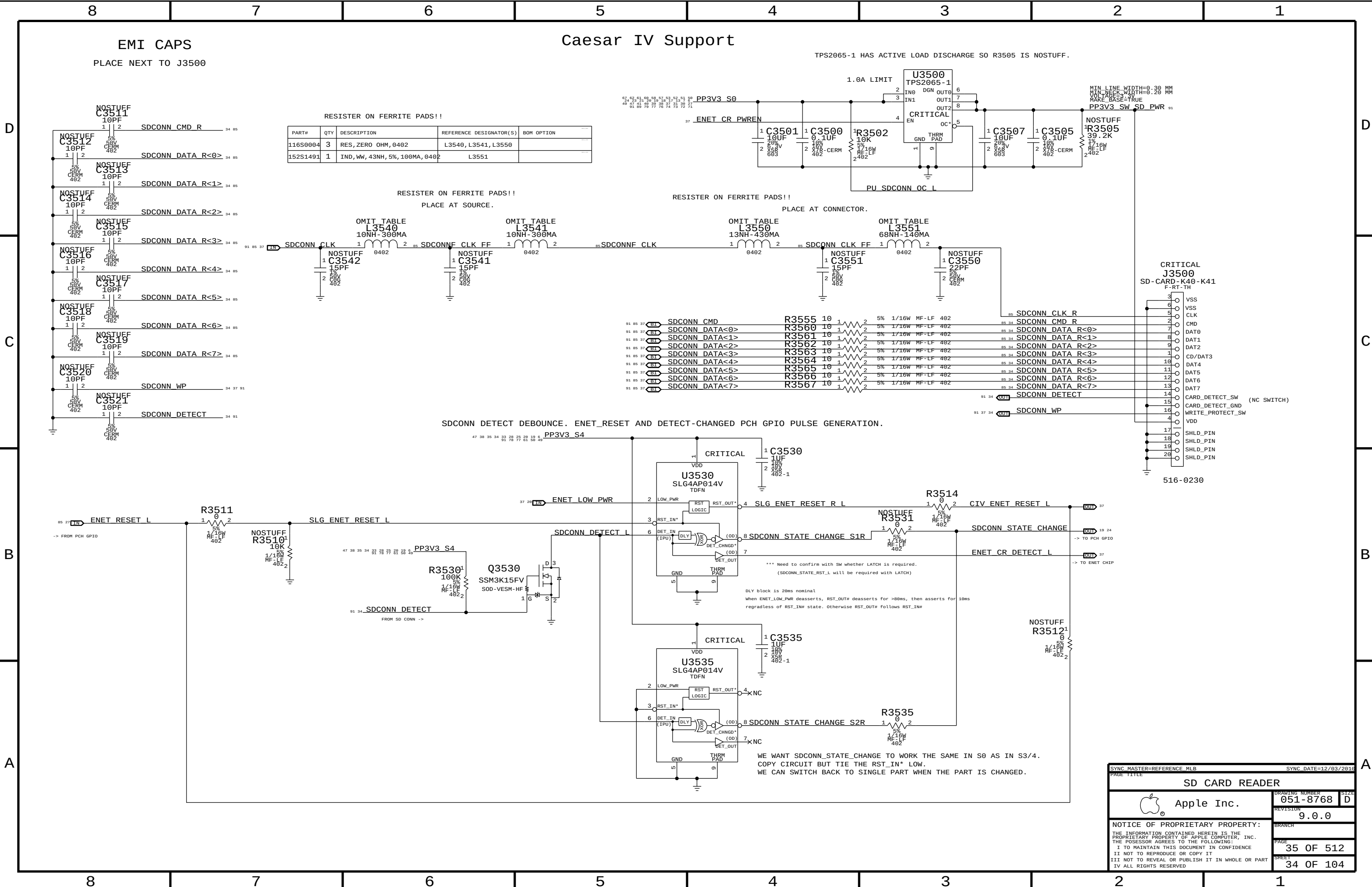
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

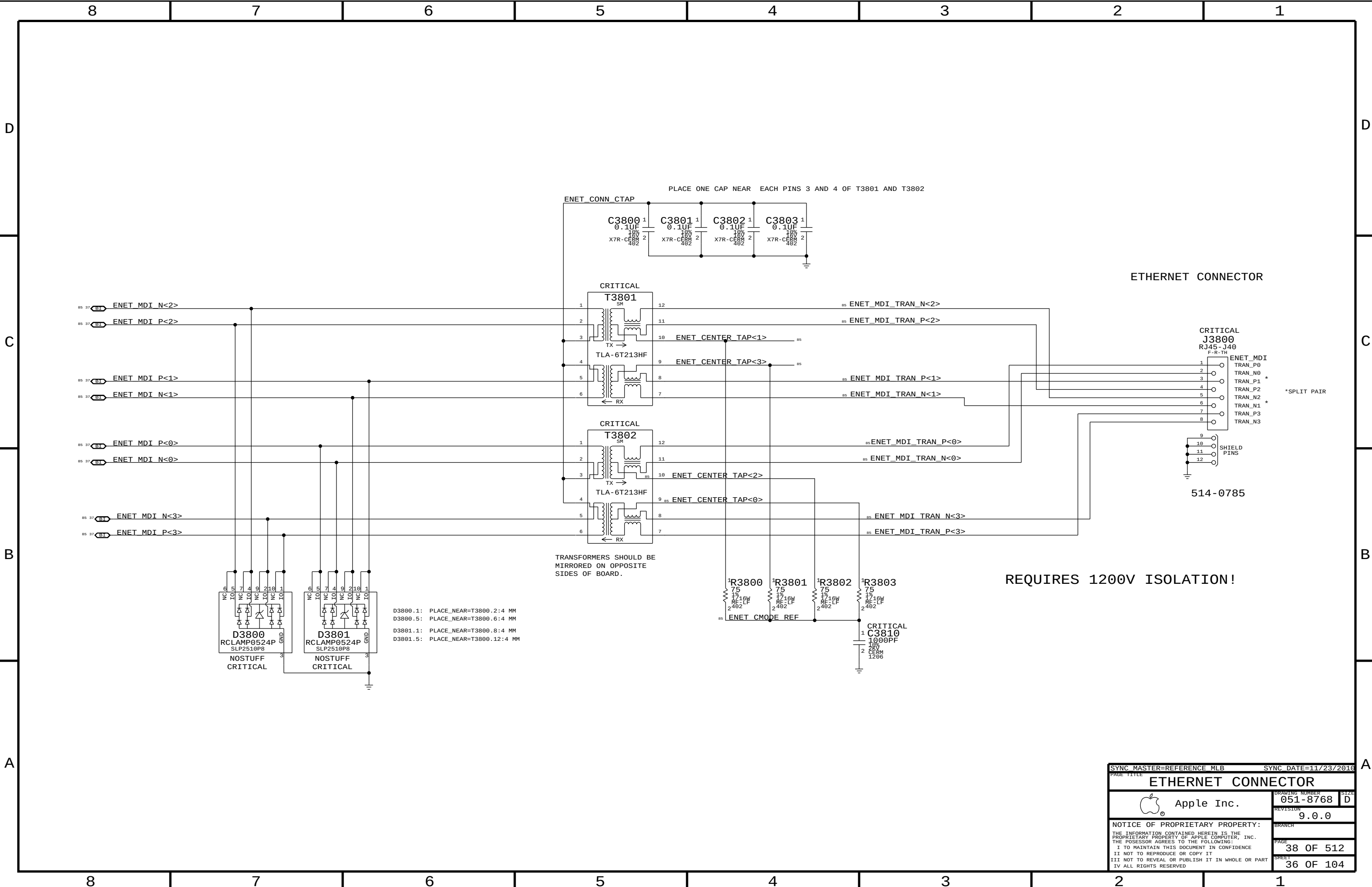
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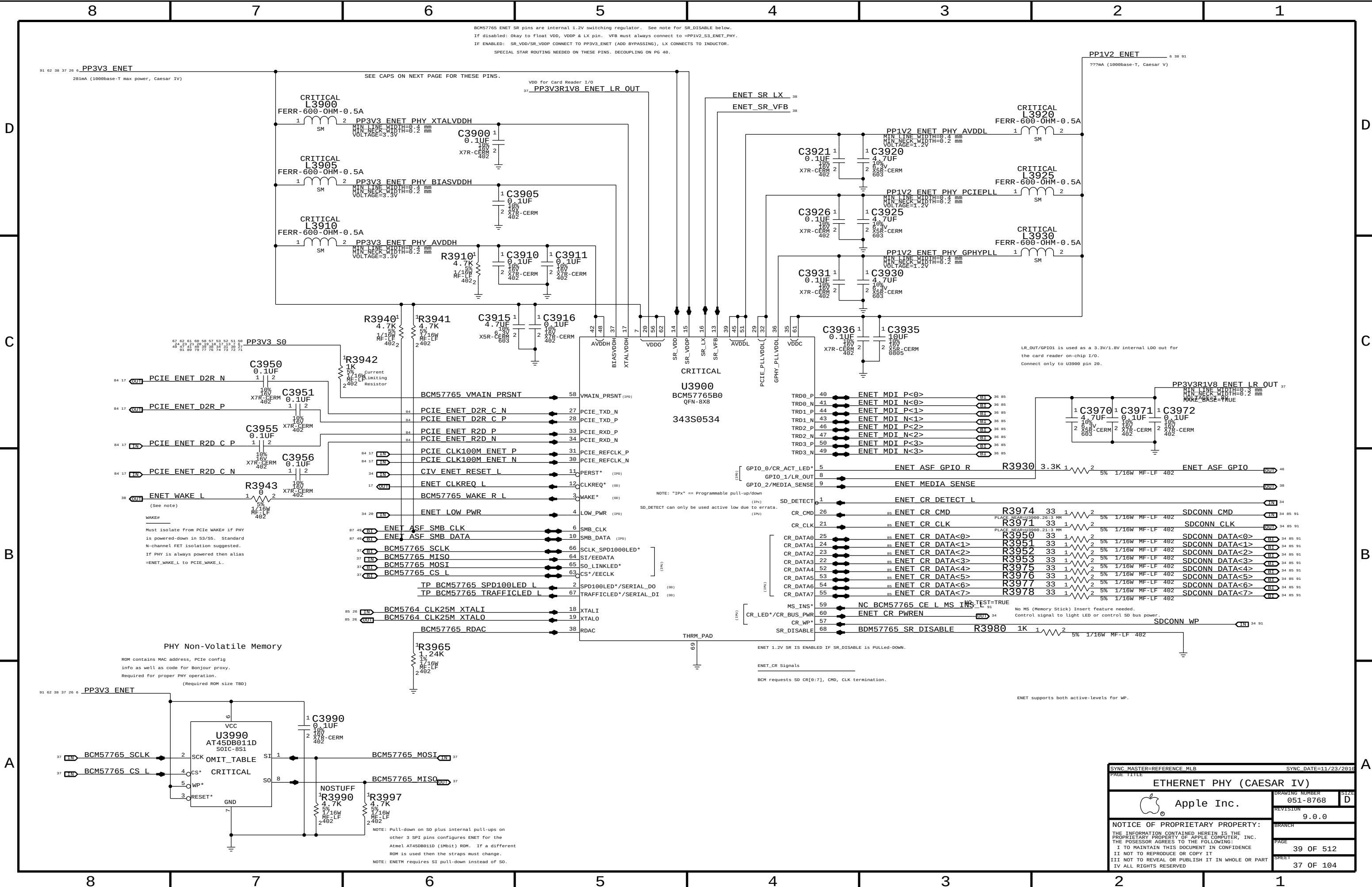
- VREFMRGN - Stuffs VREF Margining Circuitry.
- VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

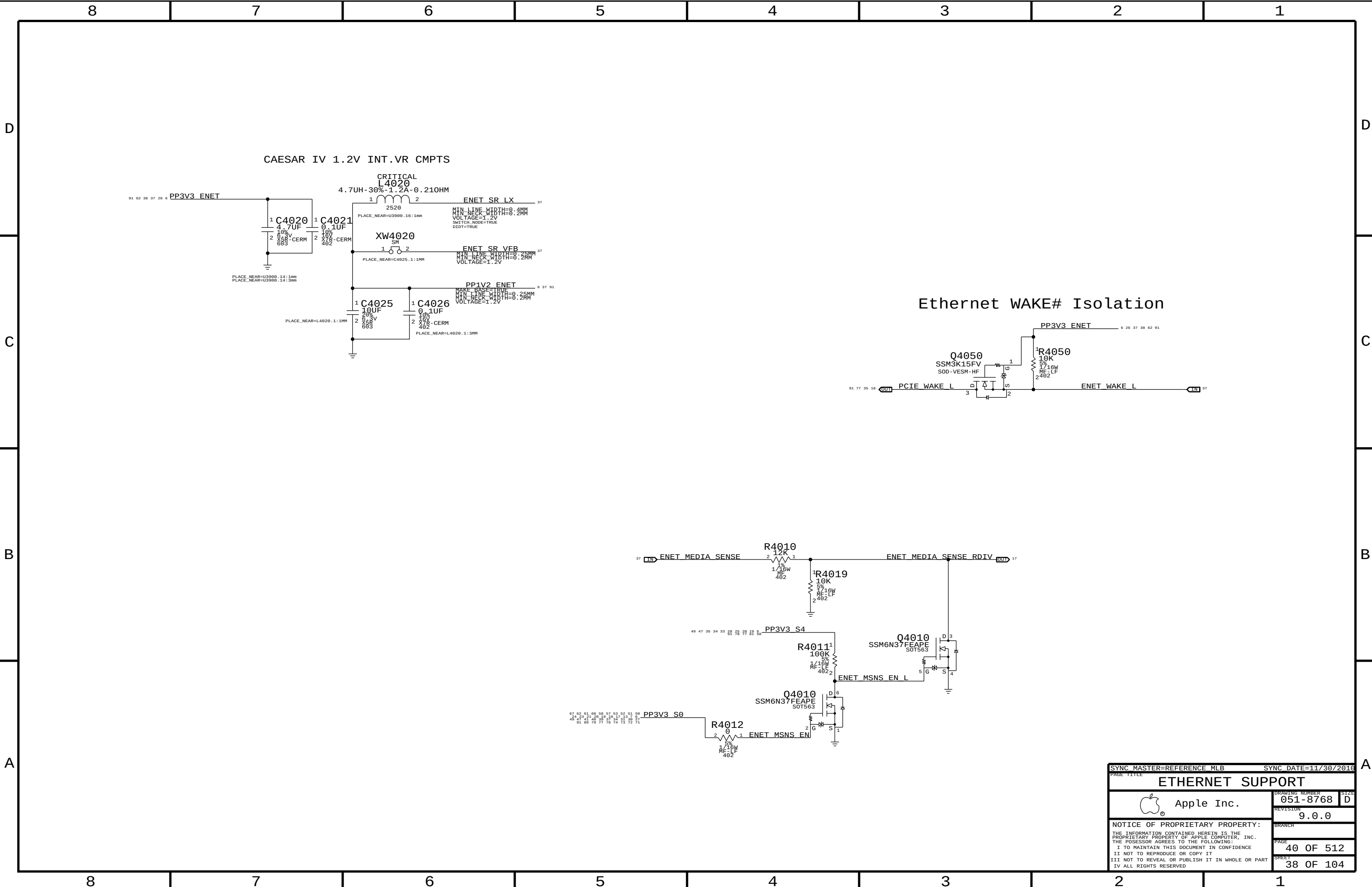
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margin'd target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

PAGE TITLE		DRAWING NUMBER	
FSB/DDR3/FB Vref Margining		051-8768	
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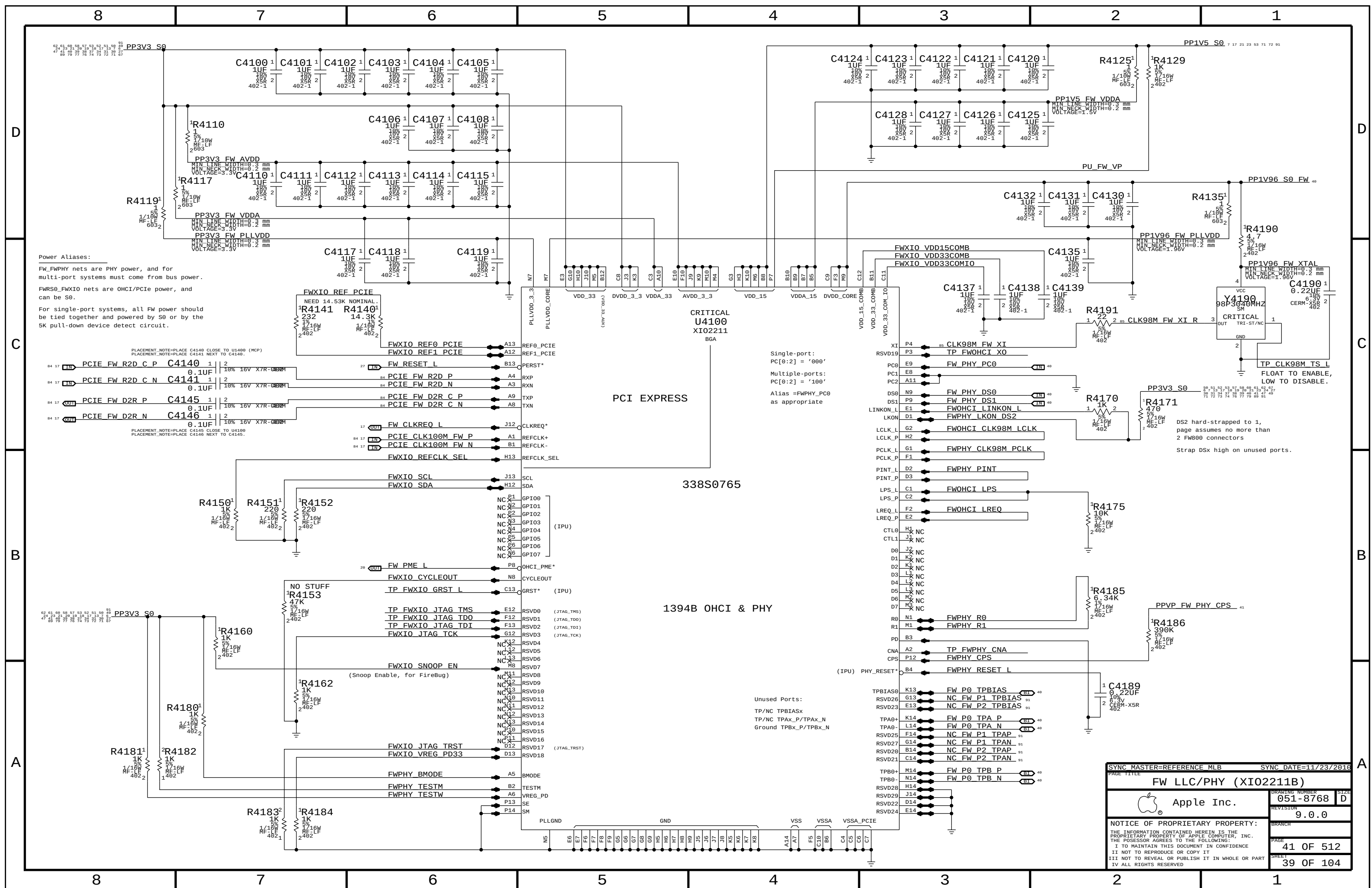


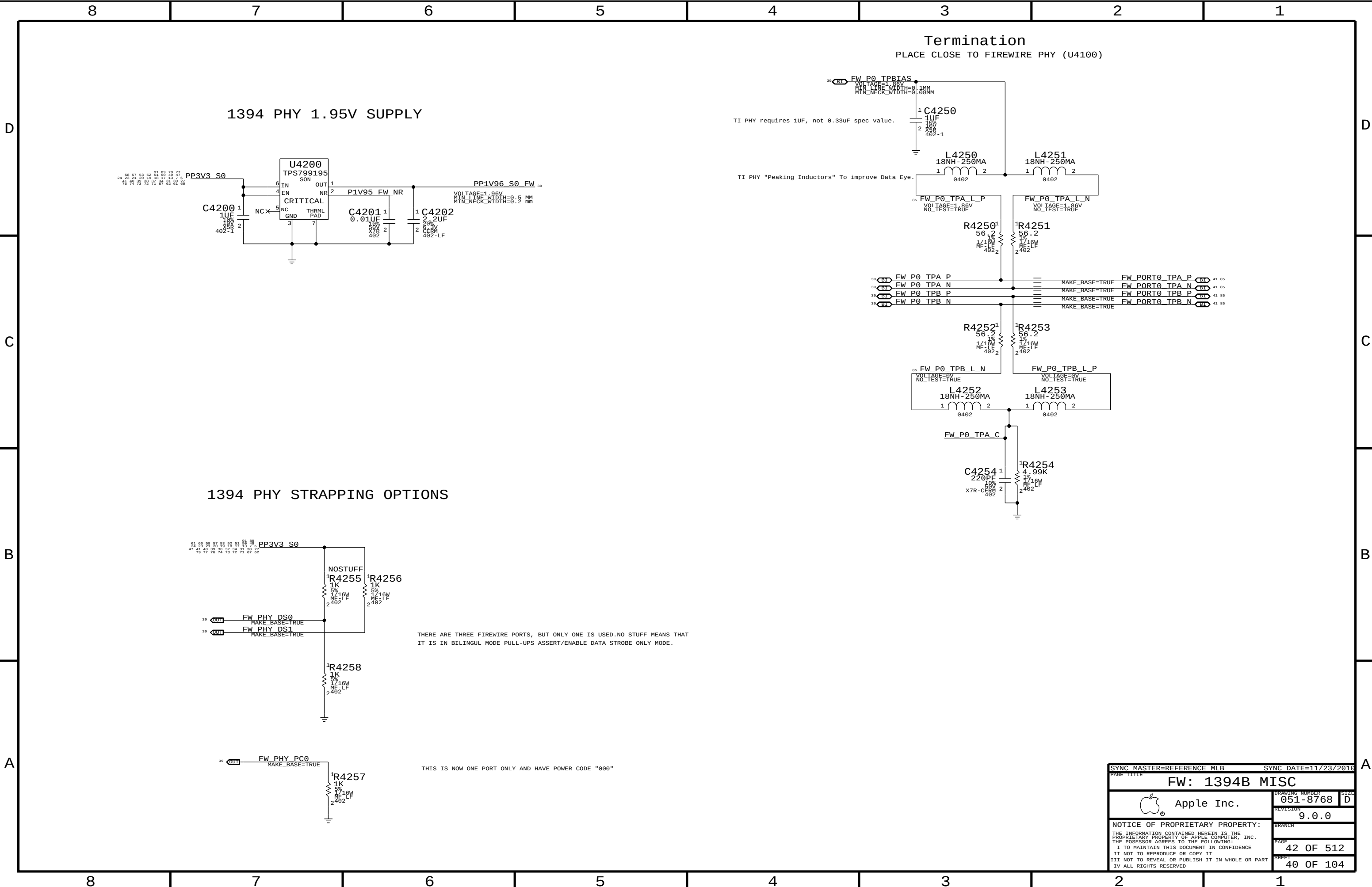




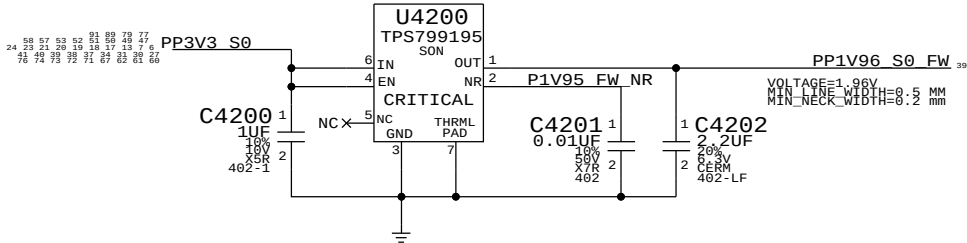


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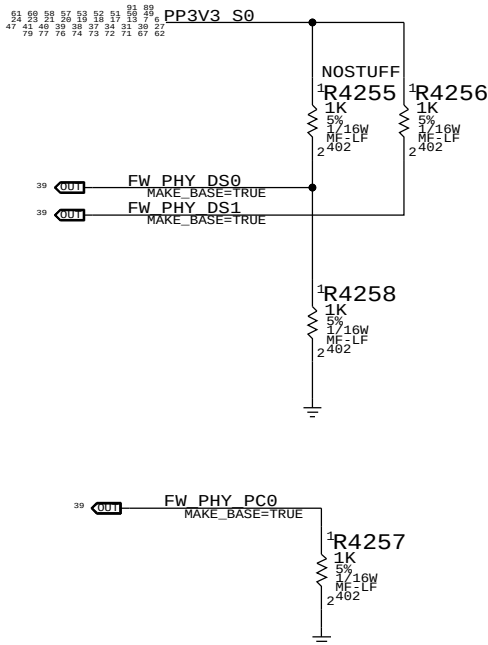




1394 PHY 1.95V SUPPLY



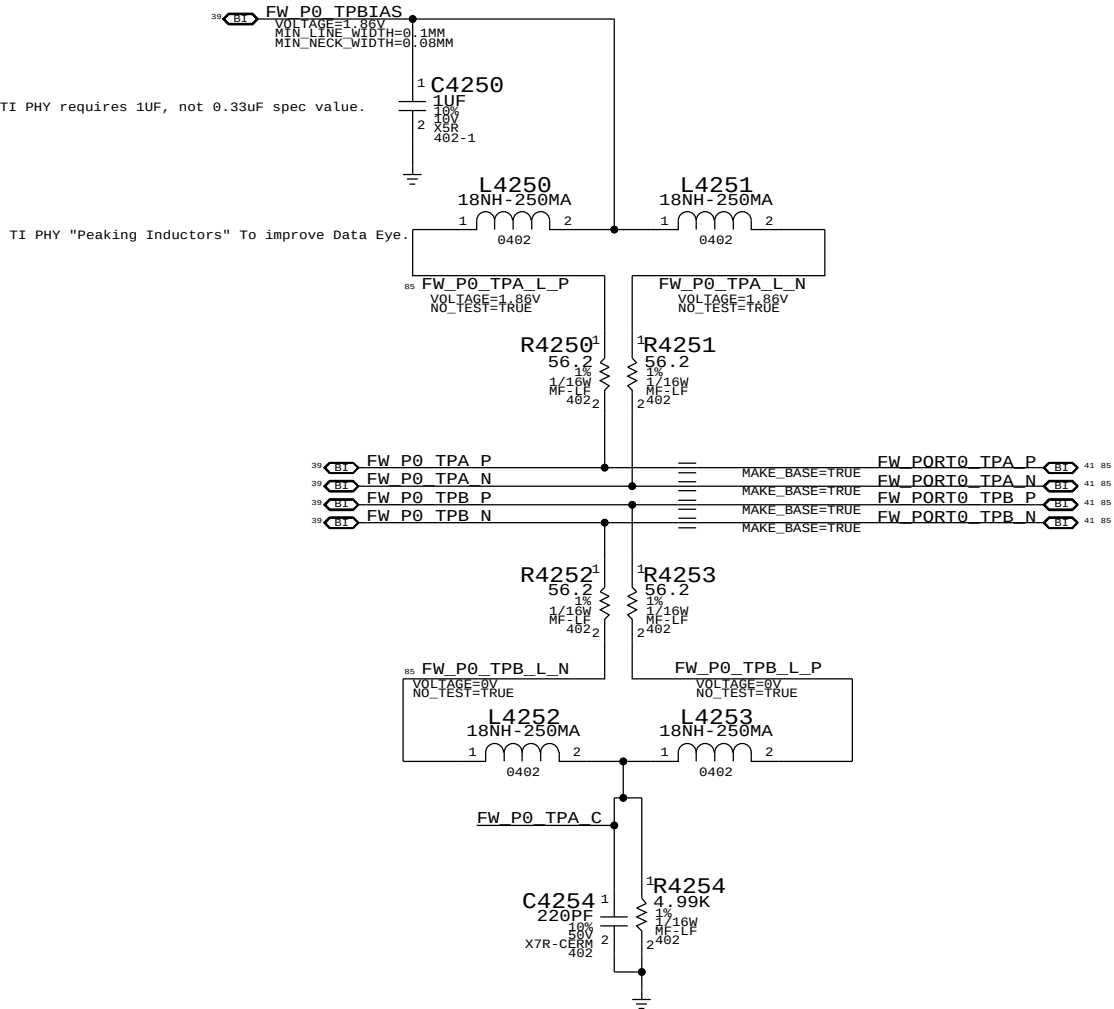
1394 PHY STRAPPING OPTIONS



THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

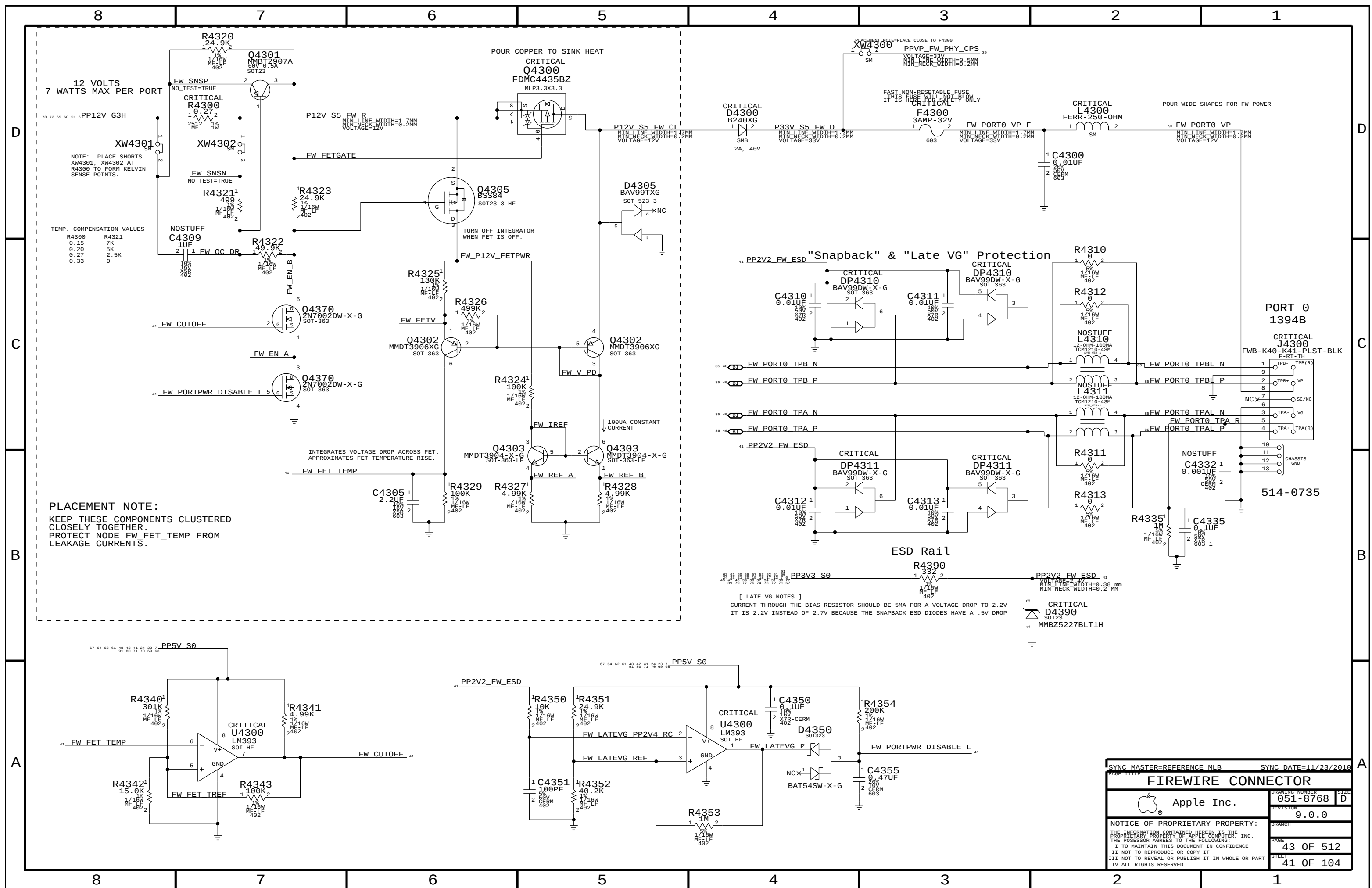
THIS IS NOW ONE PORT ONLY AND HAVE POWER CODE "000"

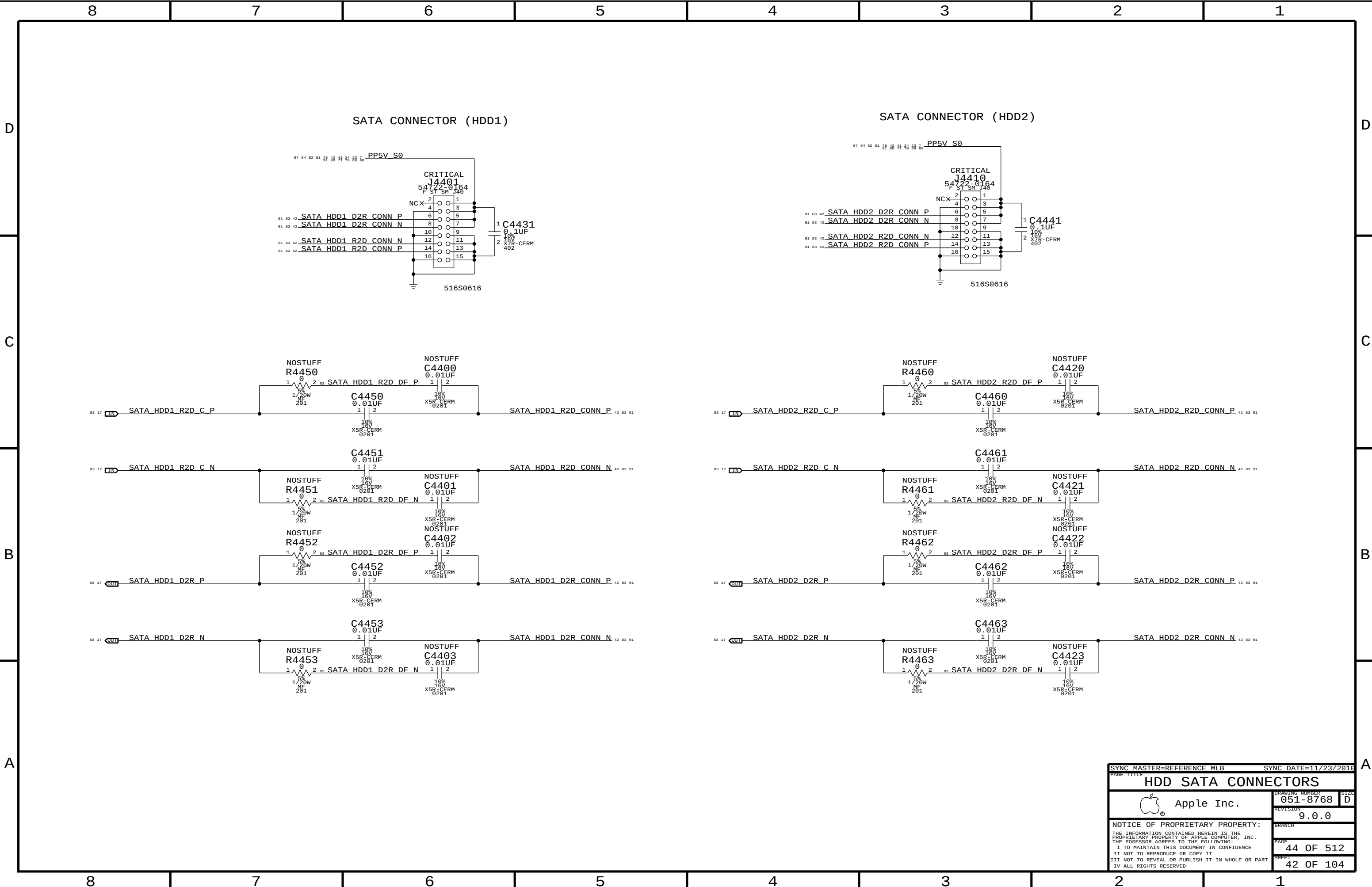
Termination
PLACE CLOSE TO FIREWIRE PHY (U4100)

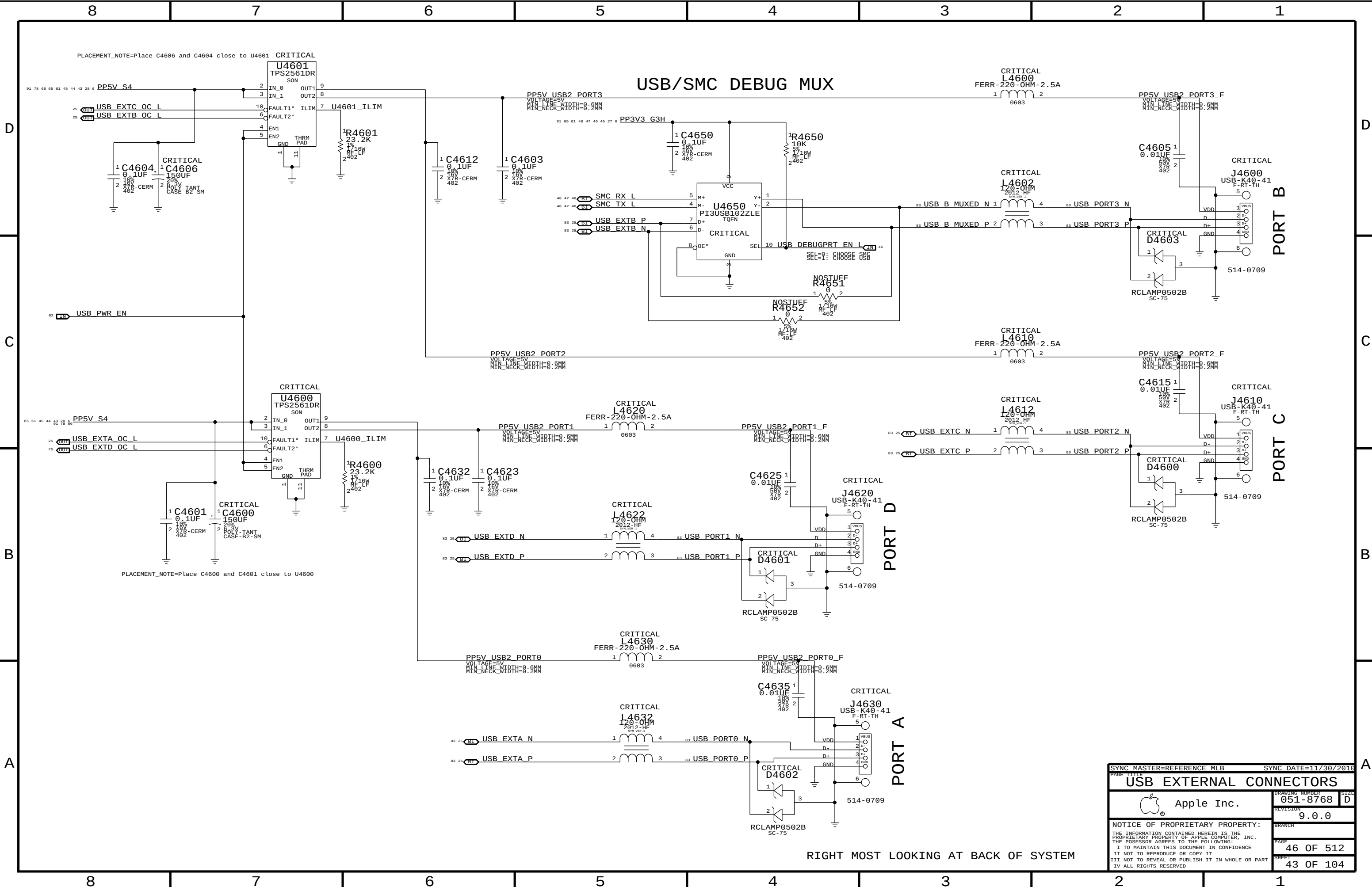


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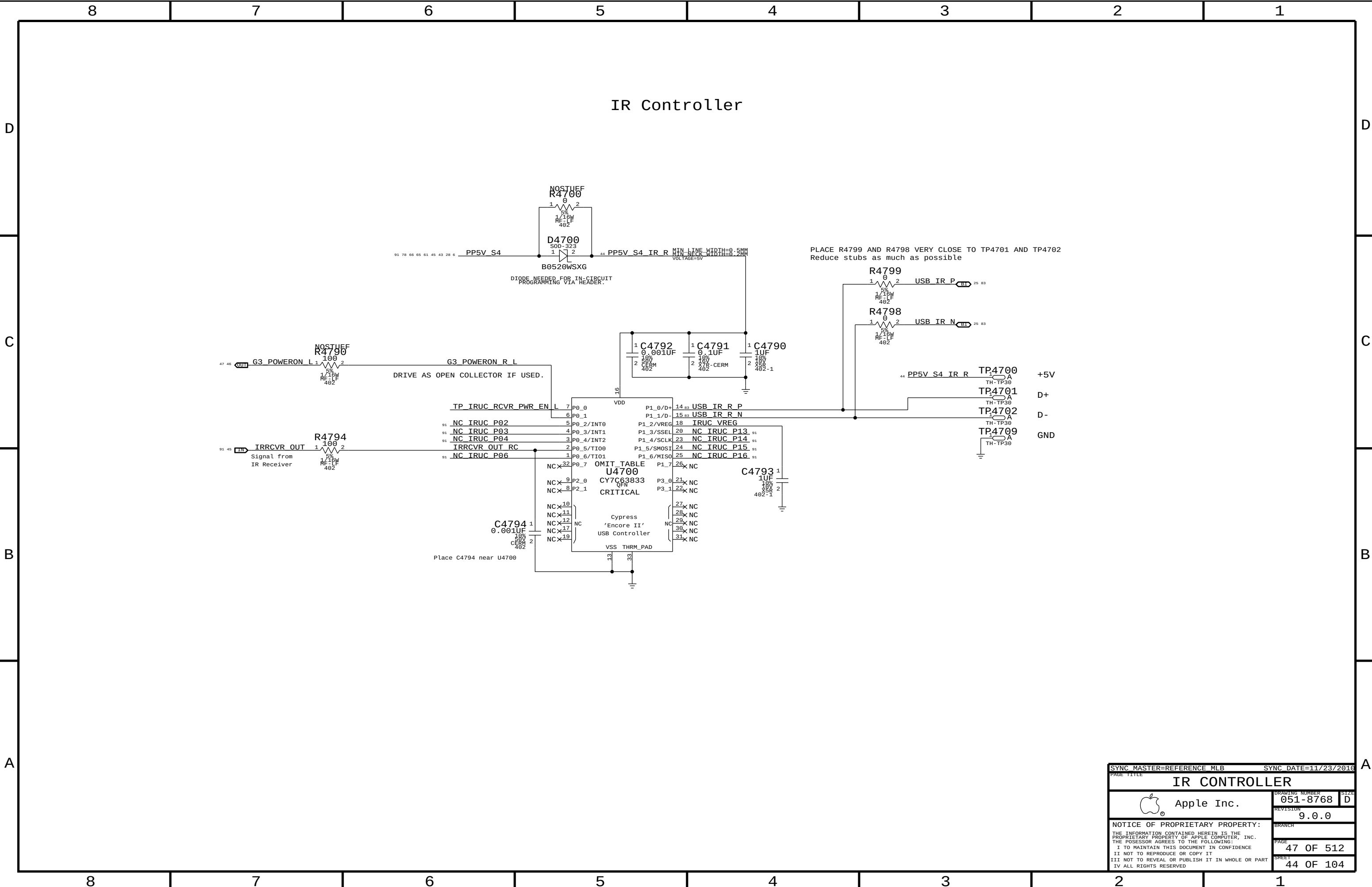


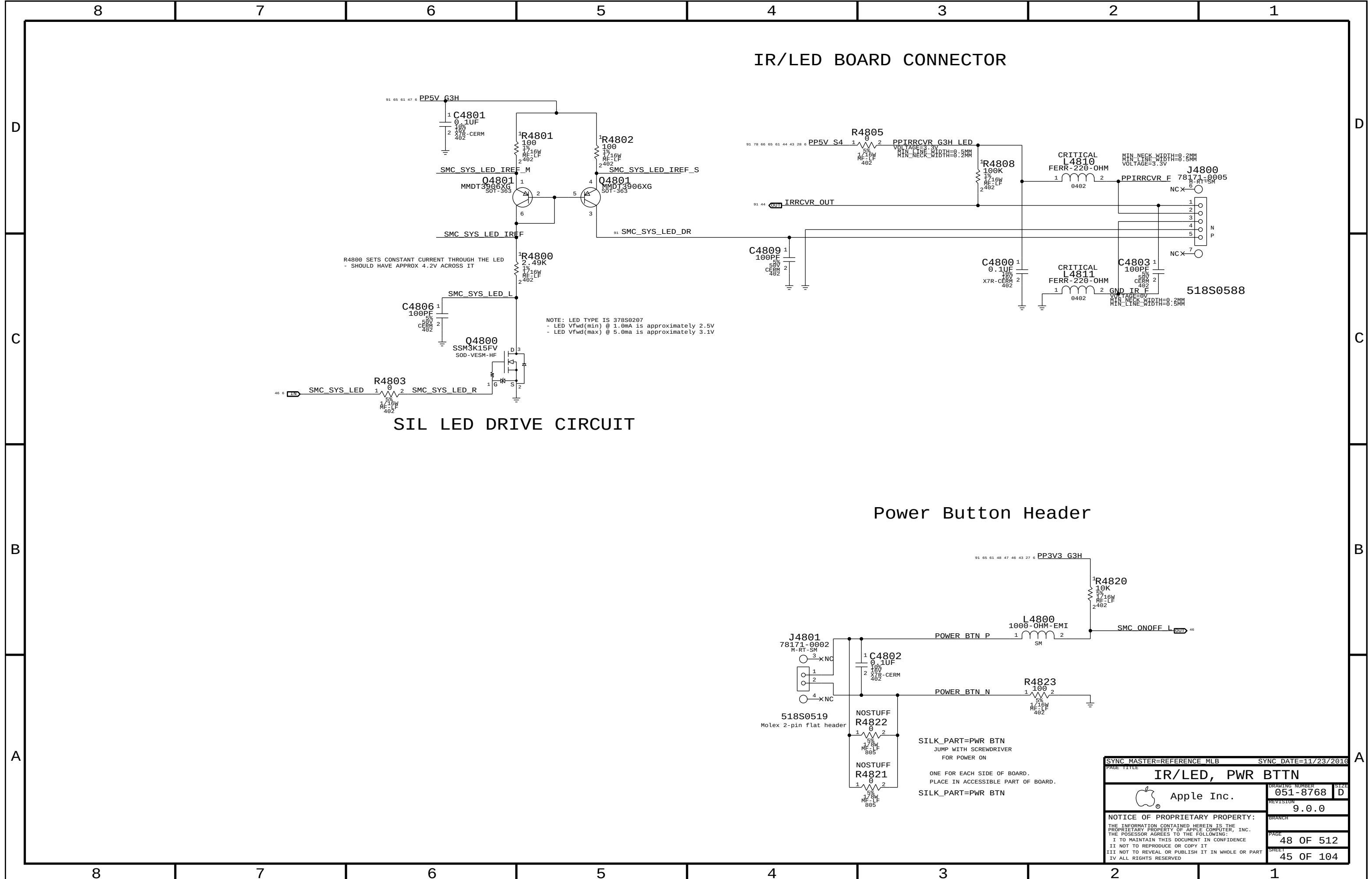




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USB EXTERNAL CONNECTORS		DRAWING NUMBER	
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RIGHT MOST LOOKING AT BACK OF SYSTEM



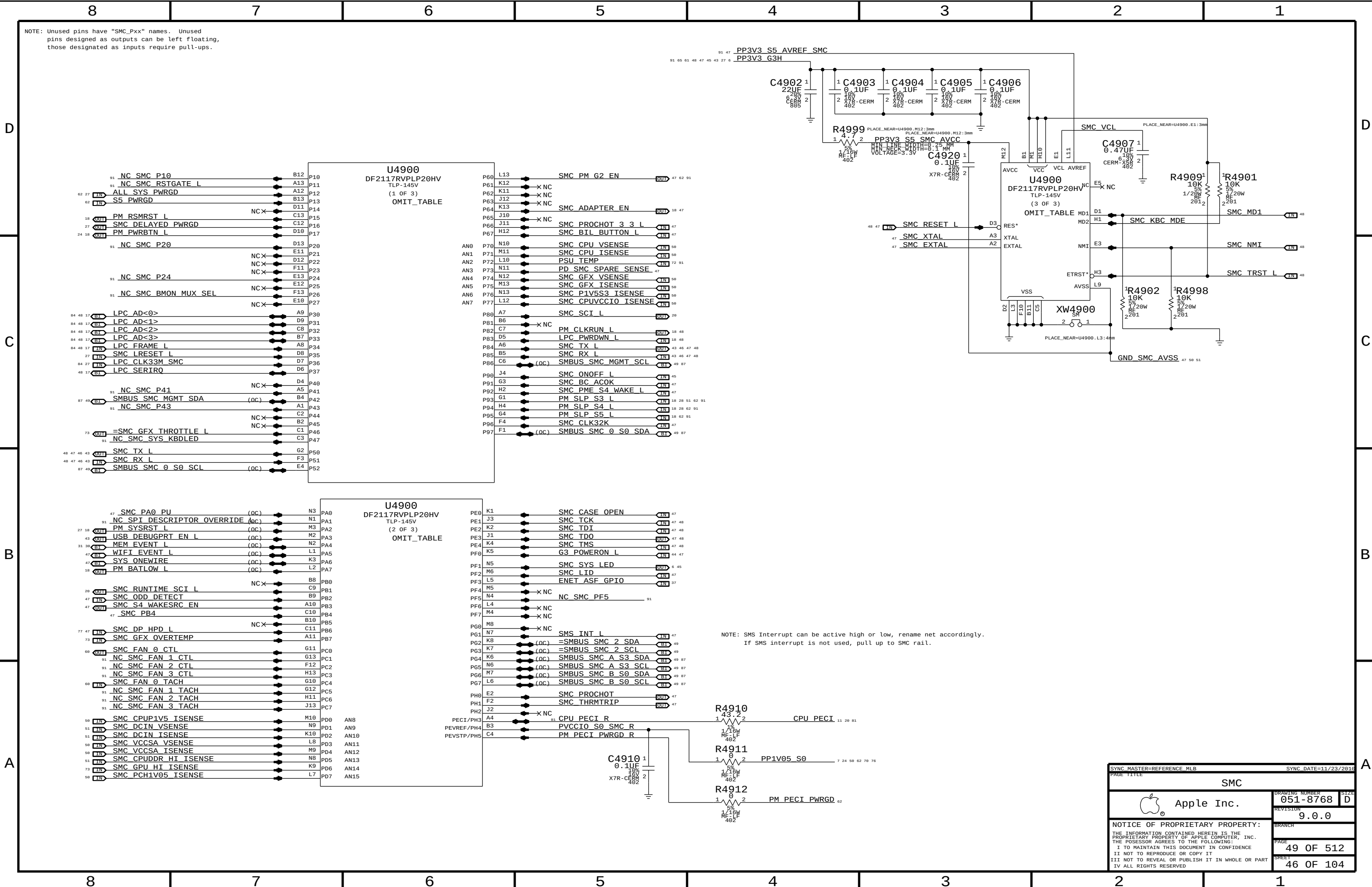


IR/LED BOARD CONNECTOR

SIL LED DRIVE CIRCUIT

Power Button Header

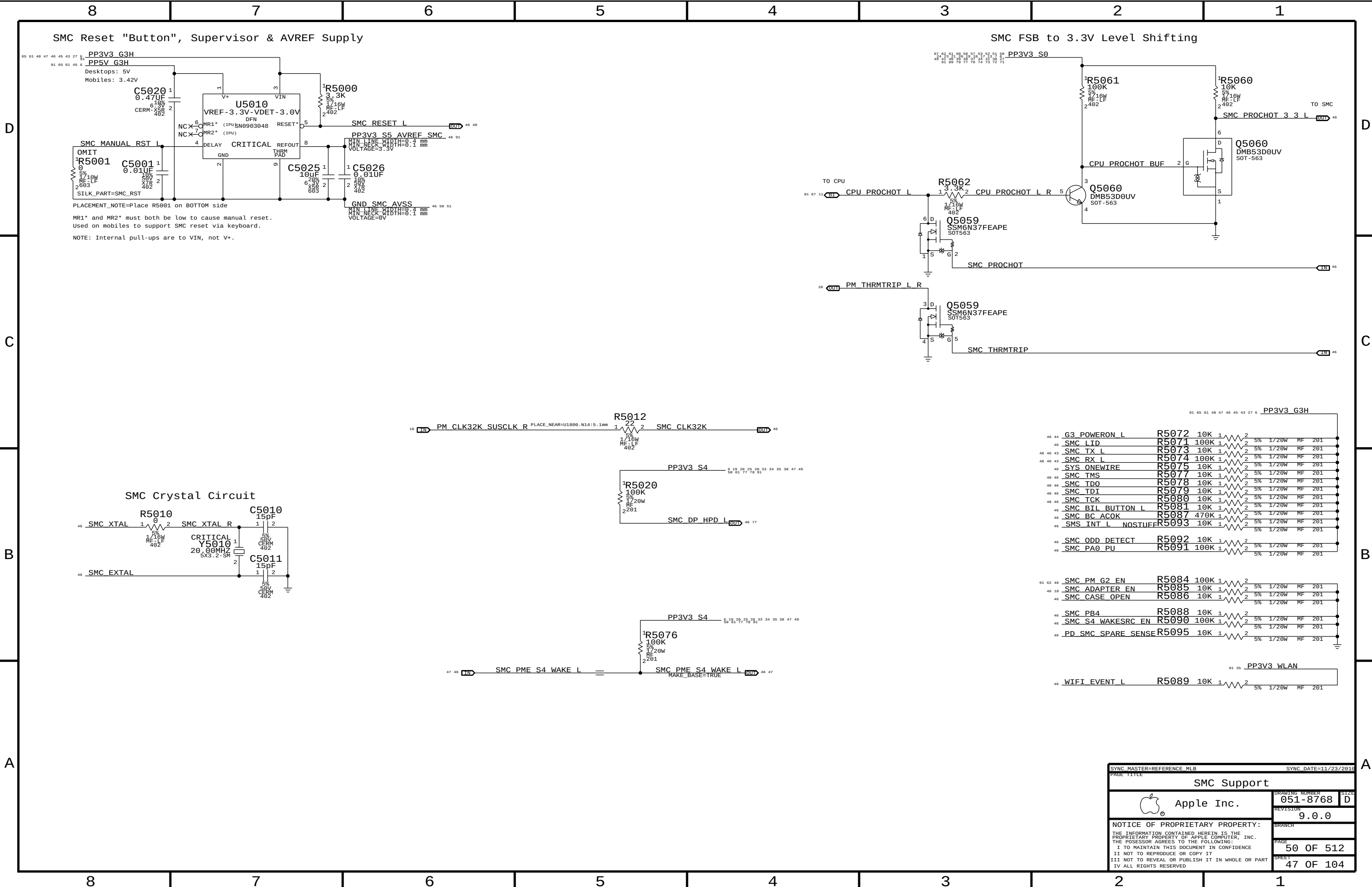
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PAGE TITLE		IR/LED, PWR BTTN	
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NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

NOTE: SMS interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

SYNC MASTER=REFERENCE_MLB		SYNC DATE=11/23/2010	
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SMC			
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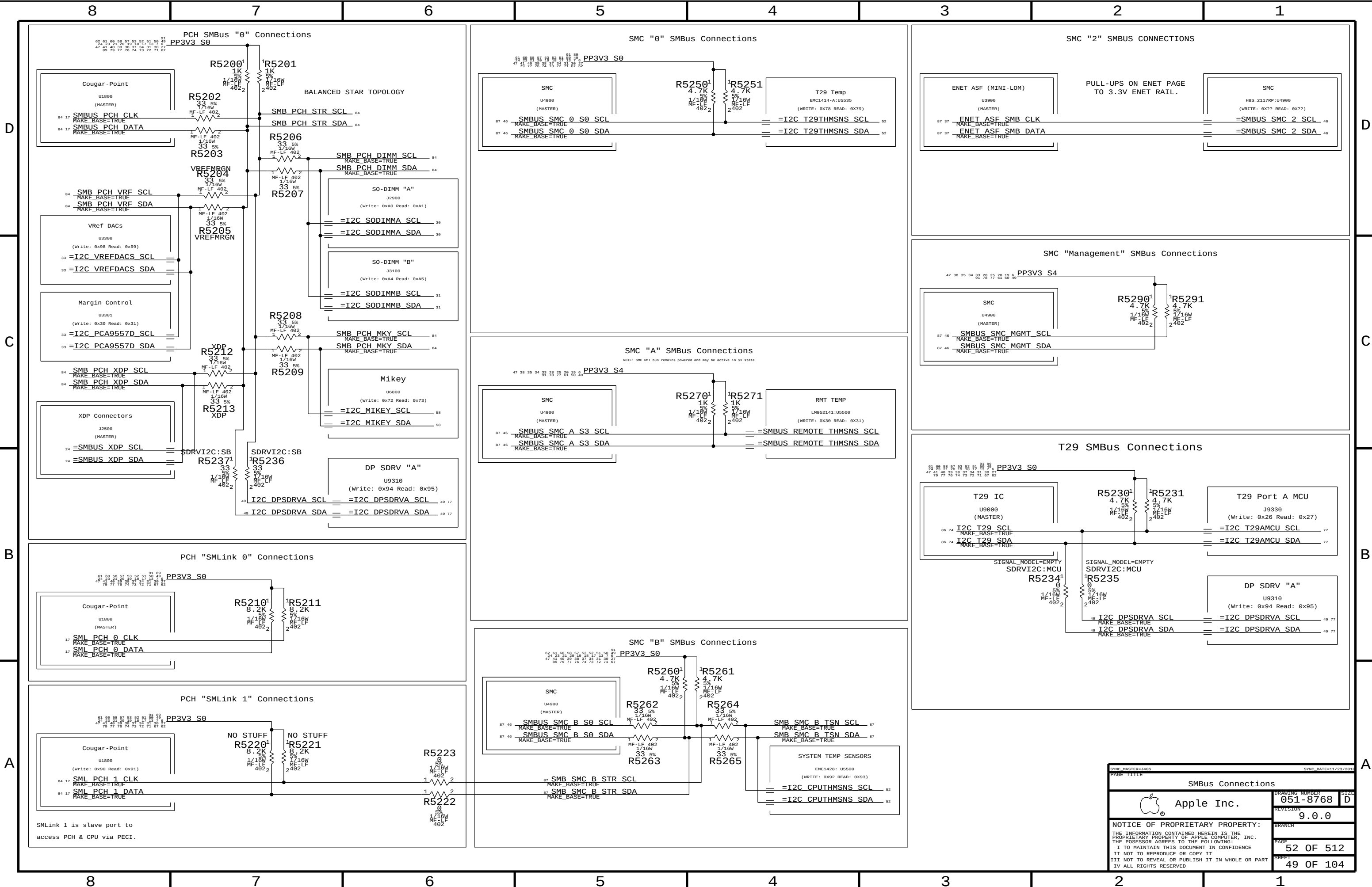
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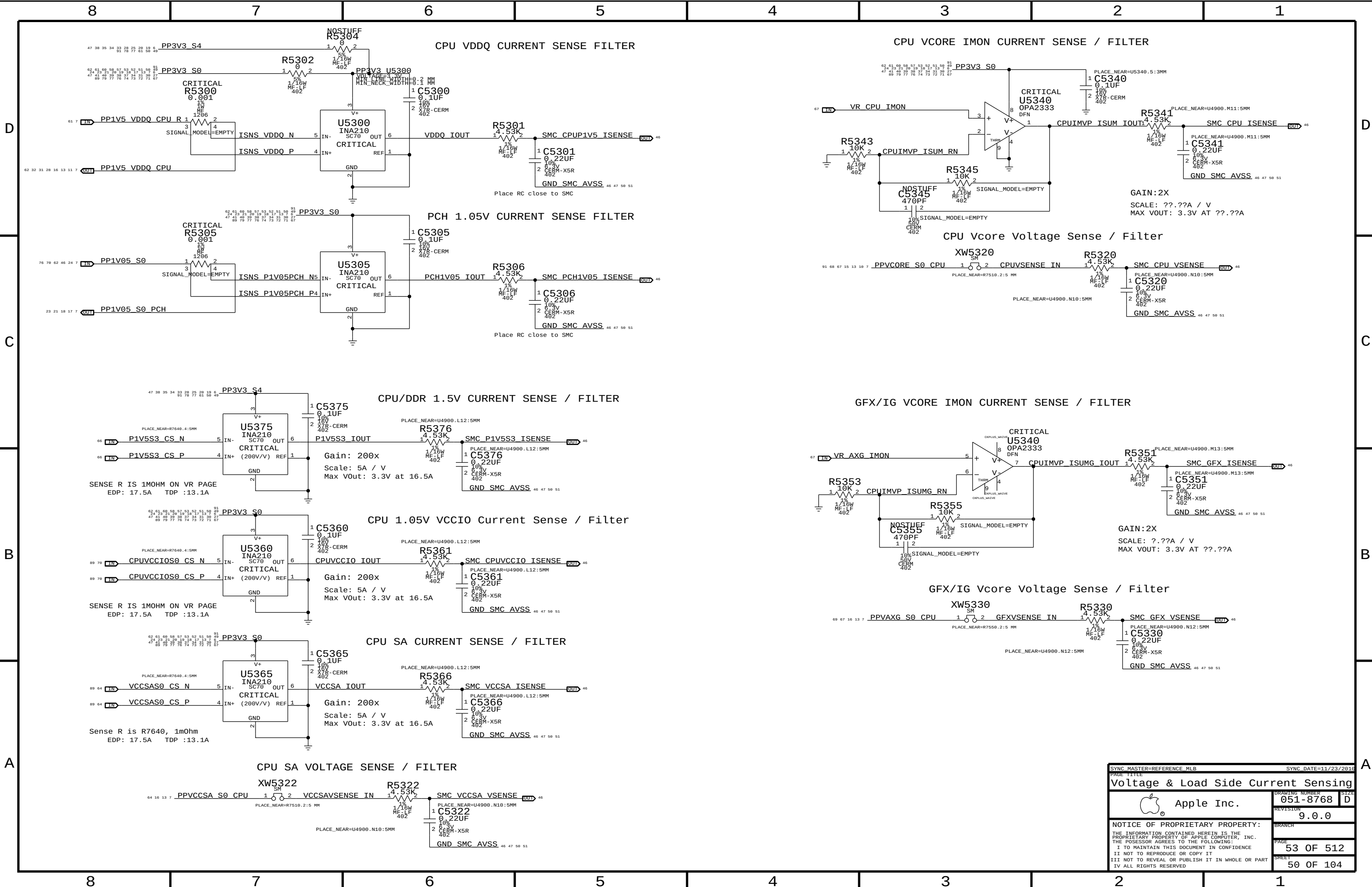
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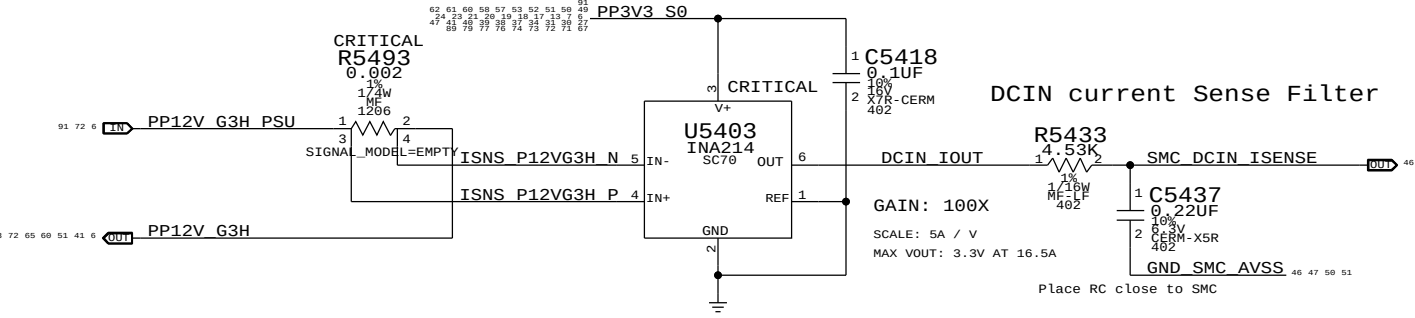




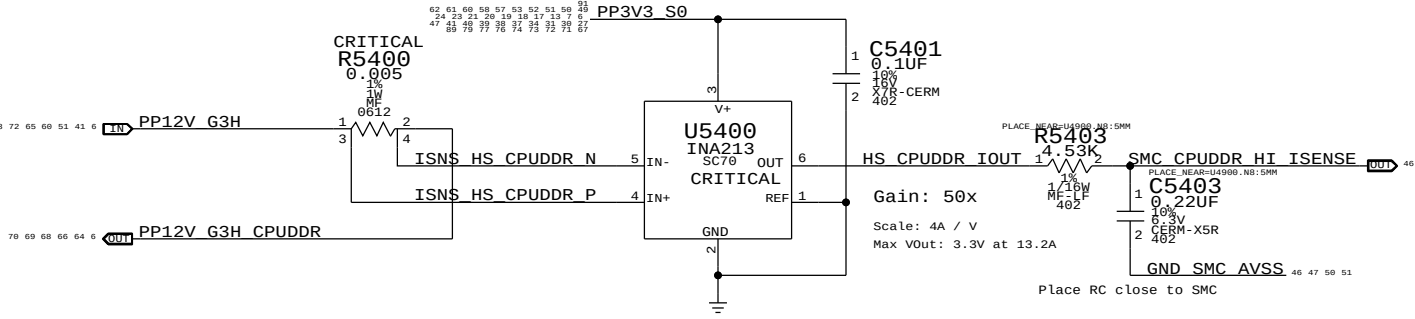
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PAGE TITLE			
Voltage & Load Side Current Sensing			
 Apple Inc.	DRAWING NUMBER	051-8768	SIZE D
	REVISION	9.0.0	
	BRANCH		
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PAGE		53 OF 512	
SHEET		50 OF 104	

J40I, J40S, AND J40

PSU CURRENT SENSE

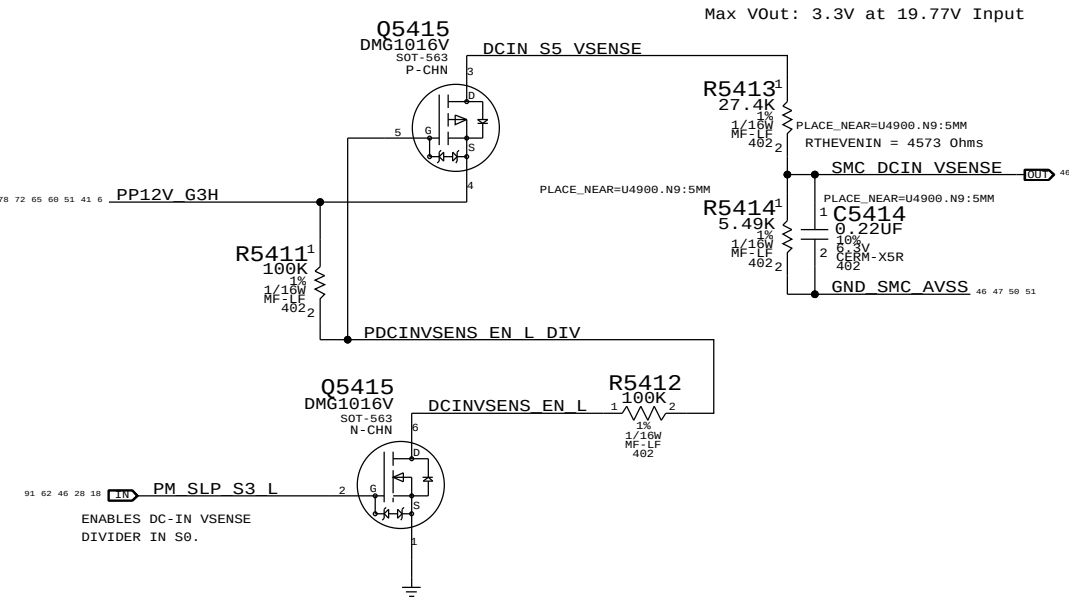


CPU / DDR HIGH SIDE CURRENT SENSE / FILTER

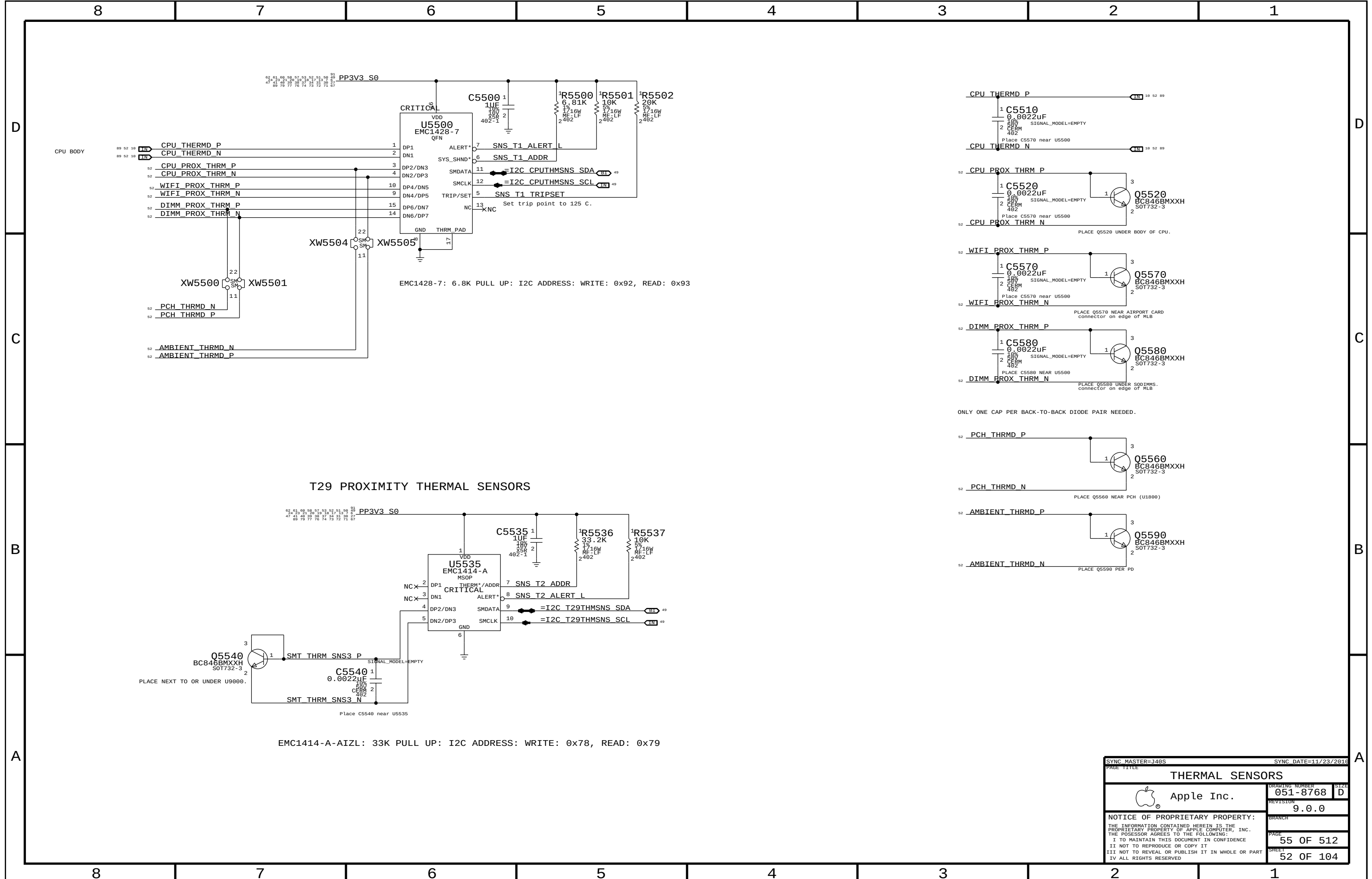


DC-In Voltage Sense Enable & Filter

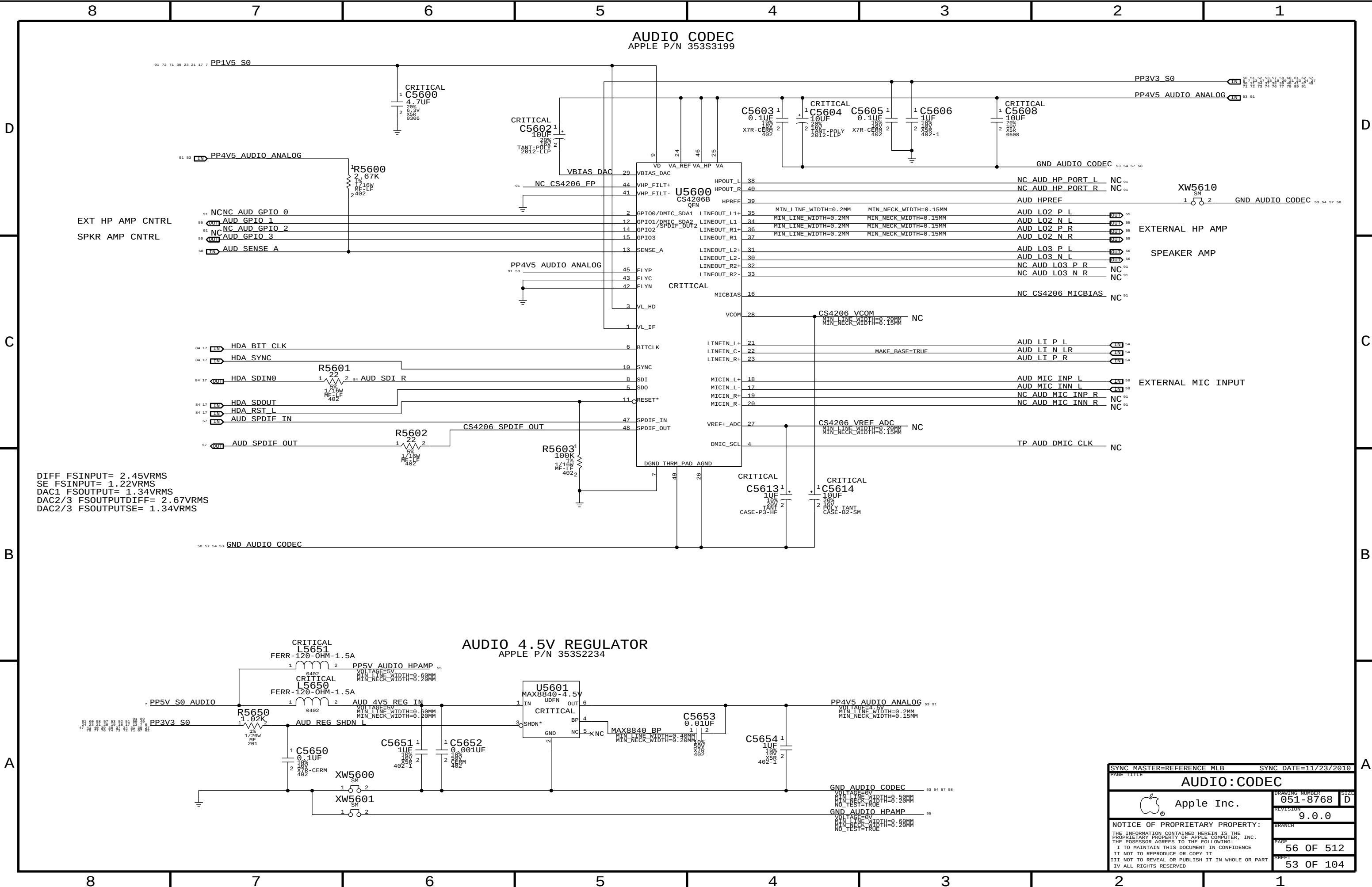
TURN ON DIVIDER ONLY IN S0.



PAGE TITLE		PAGE TITLE	
High Side Current Sensing		High Side Current Sensing	
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		PAGE	55 OF 512
		SHEET	52 OF 104



DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

PAGE TITLE		SYNC DATE=11/23/2010	
AUDIO:CODEC		DRAWING NUMBER	
Apple Inc.		051-8768	
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		9.0.0	
		BRANCH	
		PAGE	
		56 OF 512	
		SHEET	
		53 OF 104	

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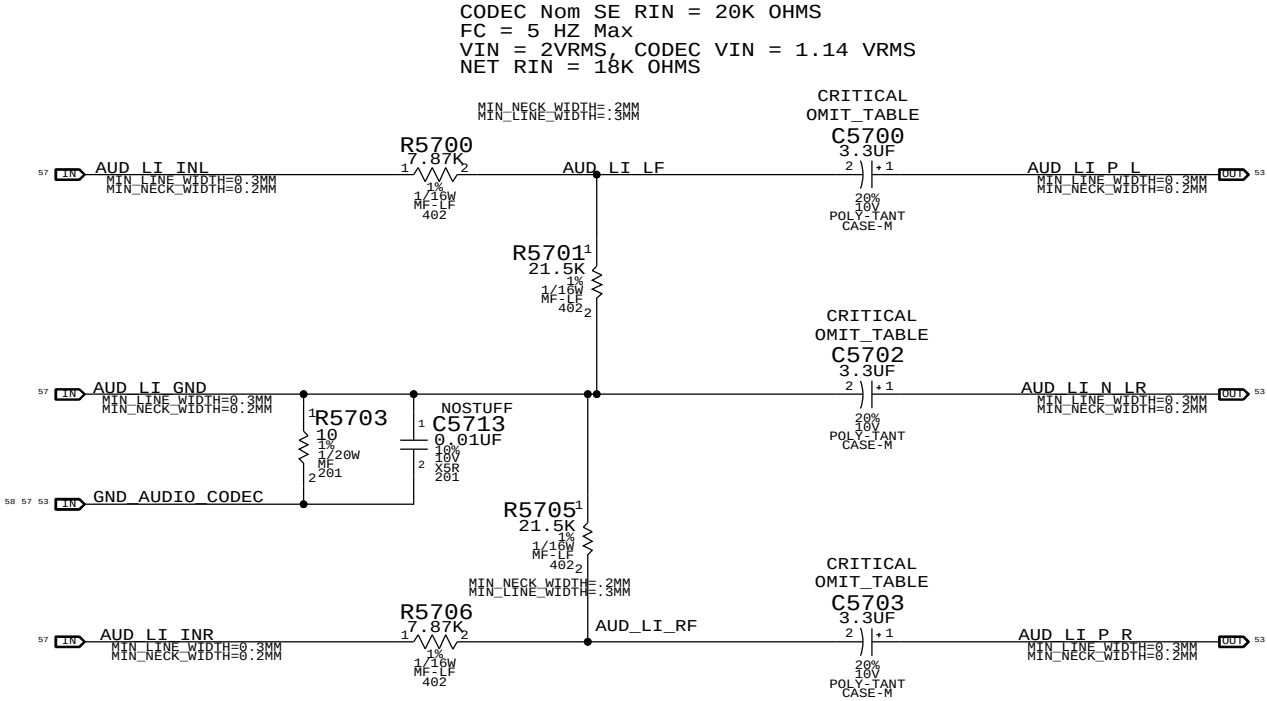
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A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
128S9399	3	CAP, TANT, POLY, 4.7UF, 20%, 16V, SMD	C5700, C5702, C5703	CRITICAL	?

SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

PAGE TITLE

AUDIO: LINE - IN

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DRAWING NUMBER
051-8768

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8

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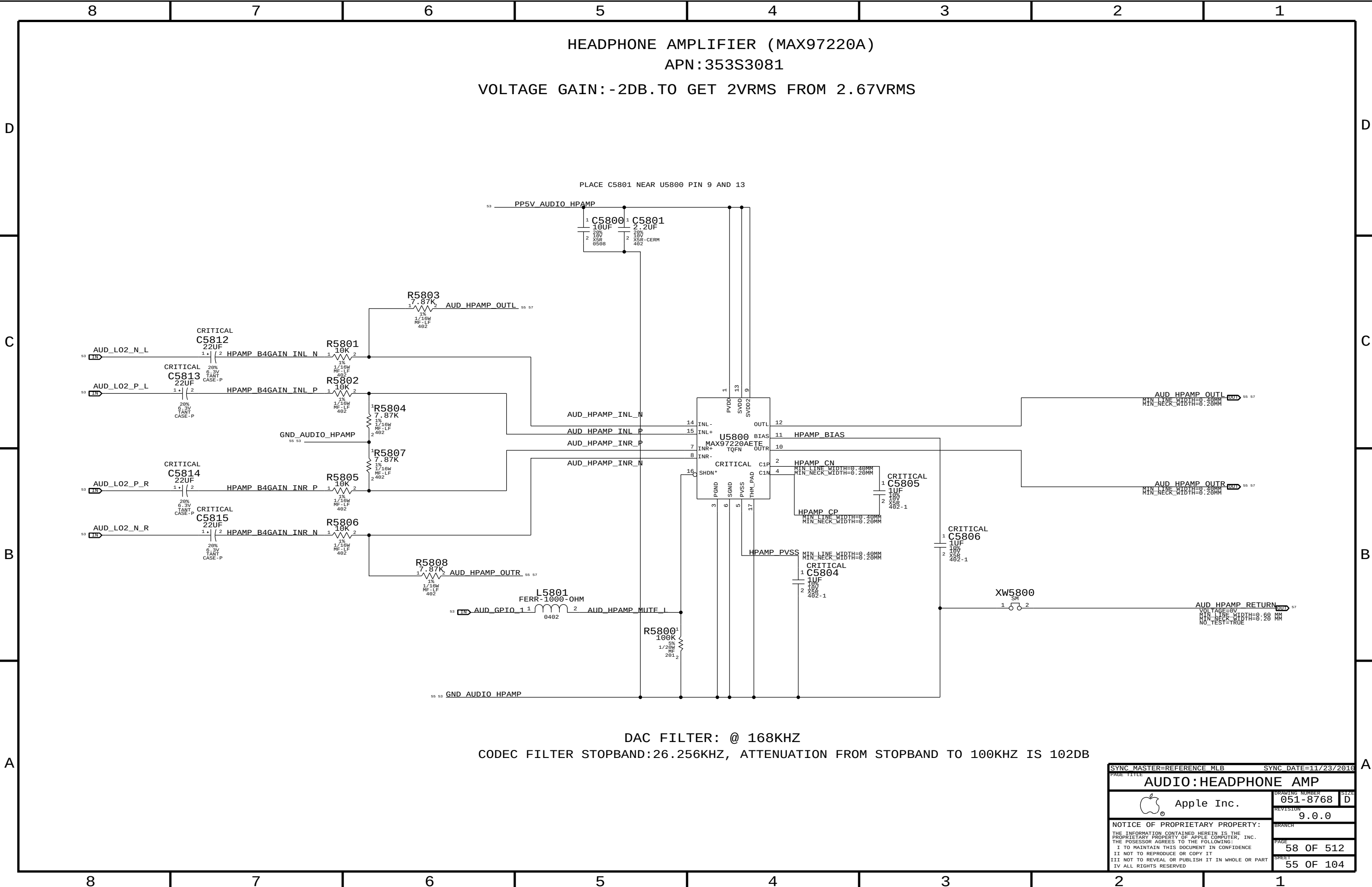
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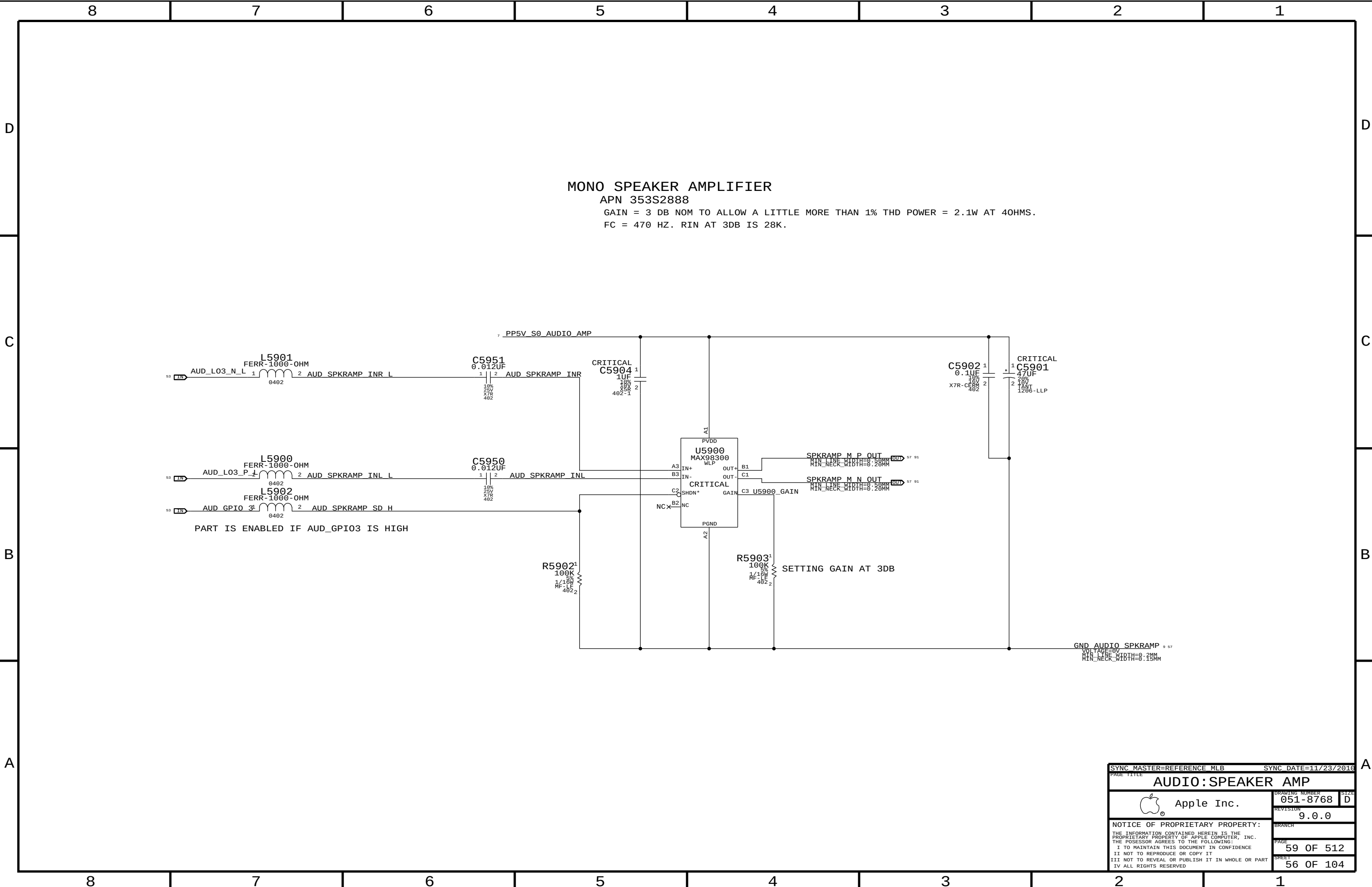
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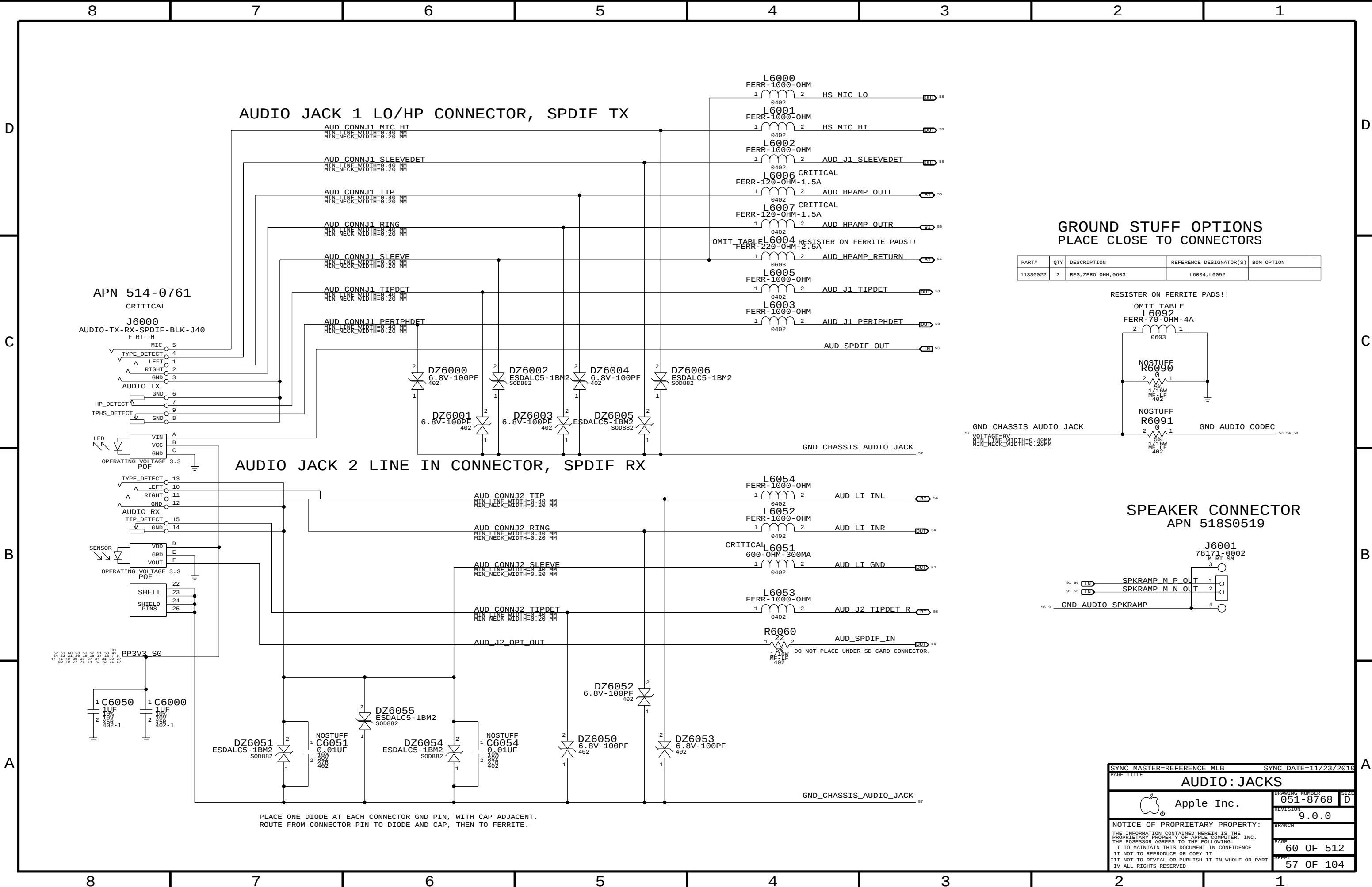
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1



PAGE TITLE		SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2016	
AUDIO:HEADPHONE AMP		DRAWING NUMBER		SIZE	
Apple Inc.		051-8768		D	
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		9.0.0			
		PAGE		58 OF 512	
		SHEET		55 OF 104	





GROUND STUFF OPTIONS
PLACE CLOSE TO CONNECTORS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
113S0022	2	RES,ZERO OHM,0603	L6004,L6092	

RESISTER ON FERRITE PADS!!

OMIT TABLE

L6092

FERR-70-OHM-4A

0603

NOSTUFF

R6090

0

1/10W

MF-LF

402

NOSTUFF

R6091

0

1/10W

MF-LF

402

SPEAKER CONNECTOR
APN 518S0519

J6001

78171-0002

M-RT-SM

3

SPKRAMP M P OUT

SPKRAMP M N OUT

GND AUDIO SPKRAMP

SYNC MASTER=REFERENCE MLB SYNC DATE=11/23/2016

PAGE TITLE

AUDIO: JACKS



Apple Inc.

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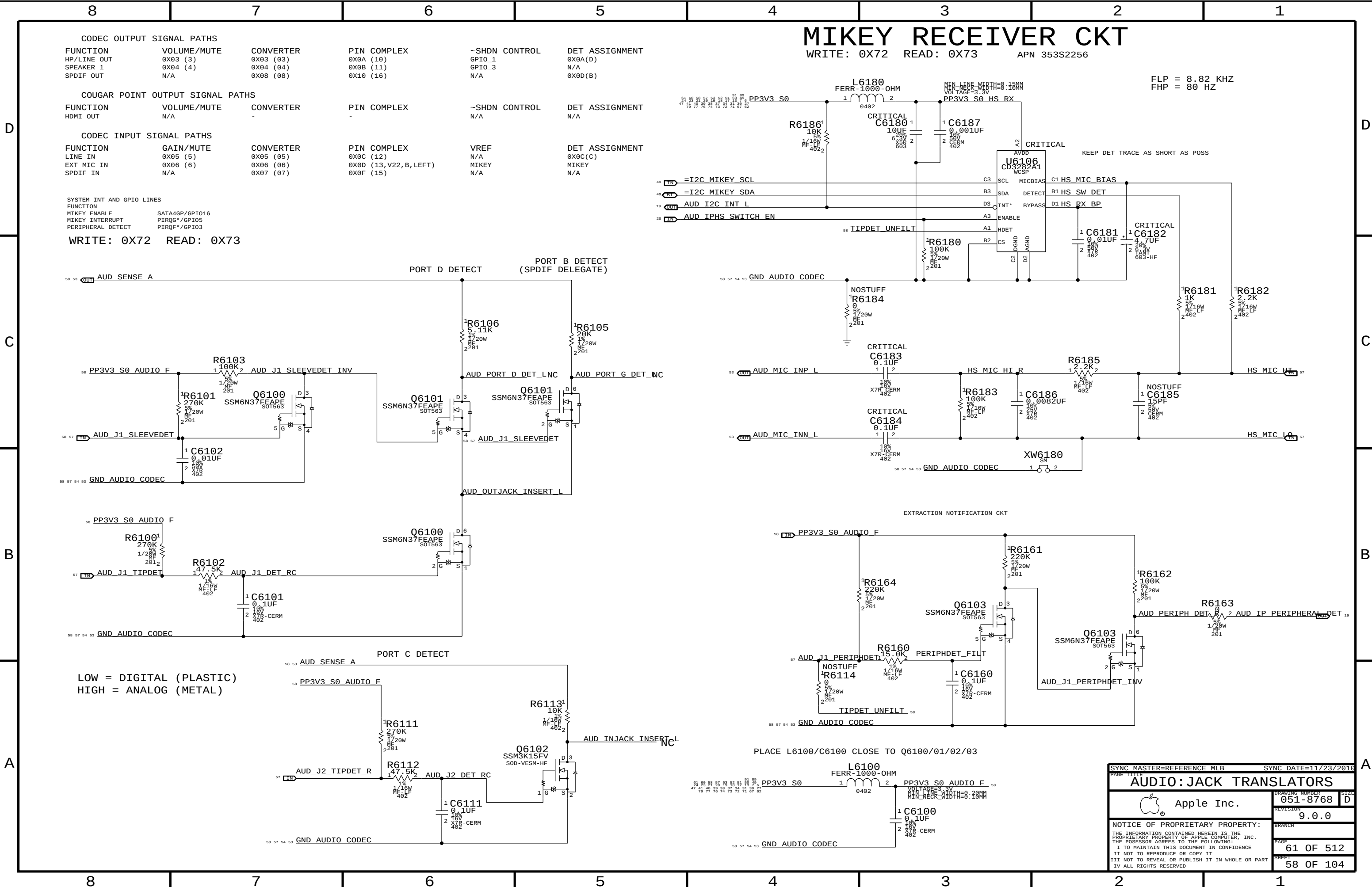
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256

FLP = 8.82 KHZ
FHP = 80 HZ

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_1	0X0A(D)
SPEAKER 1	0X04 (4)	0X04 (04)	0X0B (11)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (08)	0X10 (16)	N/A	0X0D(B)

COUGAR POINT OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	~SHDN CONTROL	DET ASSIGNMENT
HDMI OUT	N/A	-	-	N/A	N/A

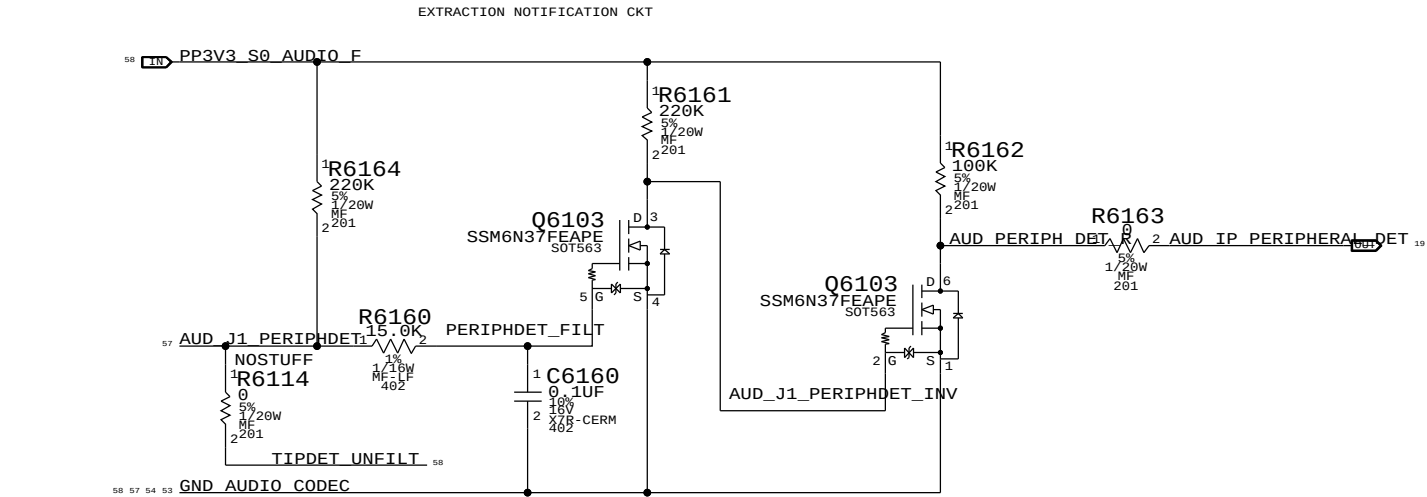
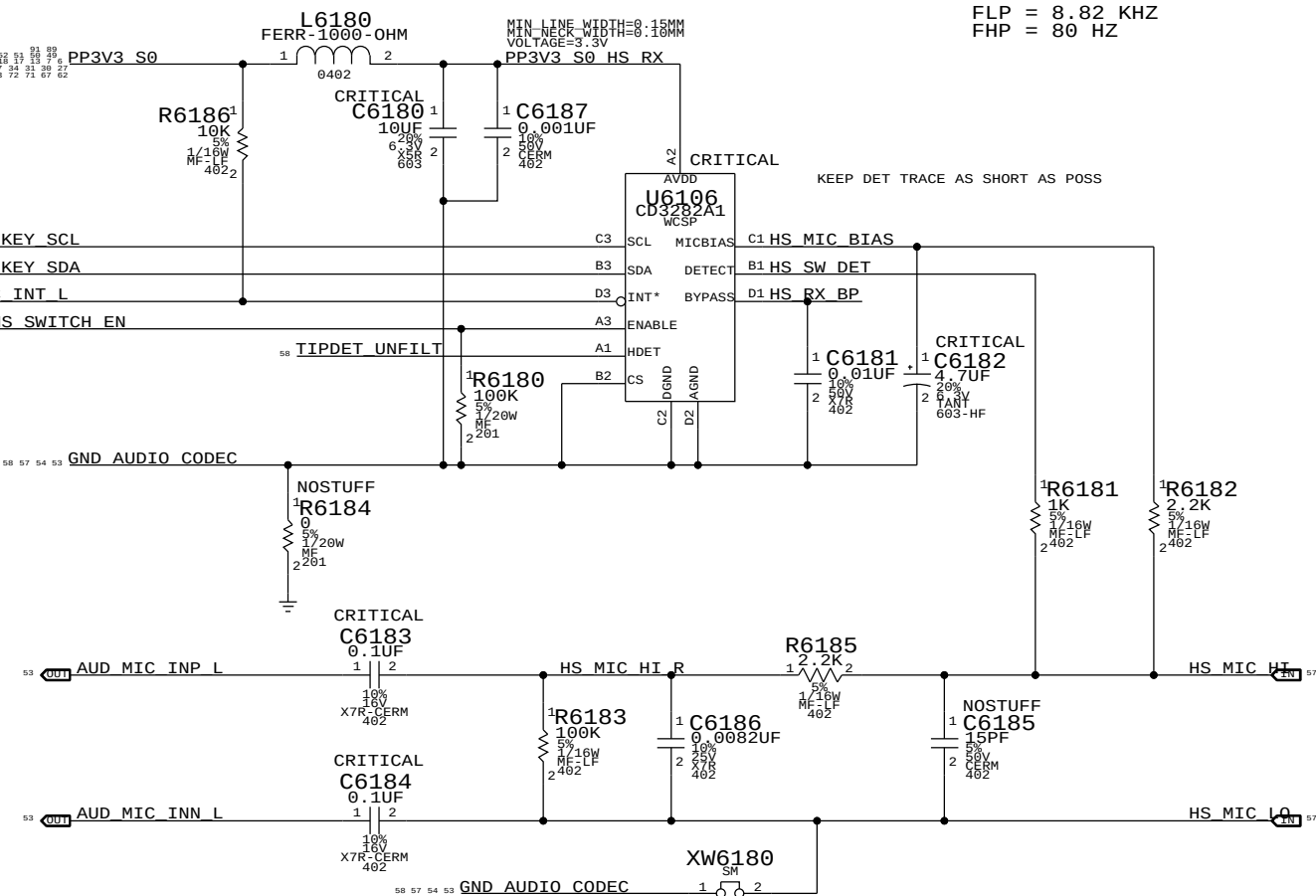
CODEC INPUT SIGNAL PATHS

FUNCTION	GAIN/MUTE	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X05 (05)	0X0C (12)	N/A	0X0C(C)
EXT MIC IN	0X06 (6)	0X06 (06)	0X0D (13, V22, B, LEFT)	MIKEY	MIKEY
SPDIF IN	N/A	0X07 (07)	0X0F (15)	N/A	N/A

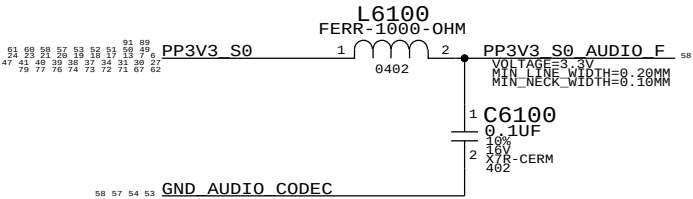
SYSTEM INT AND GPIO LINES

FUNCTION	
MIKEY ENABLE	SATA4GP/GPIO16
MIKEY INTERRUPT	PIRQG*/GPIO5
PERIPHERAL DETECT	PIRQF*/GPIO3

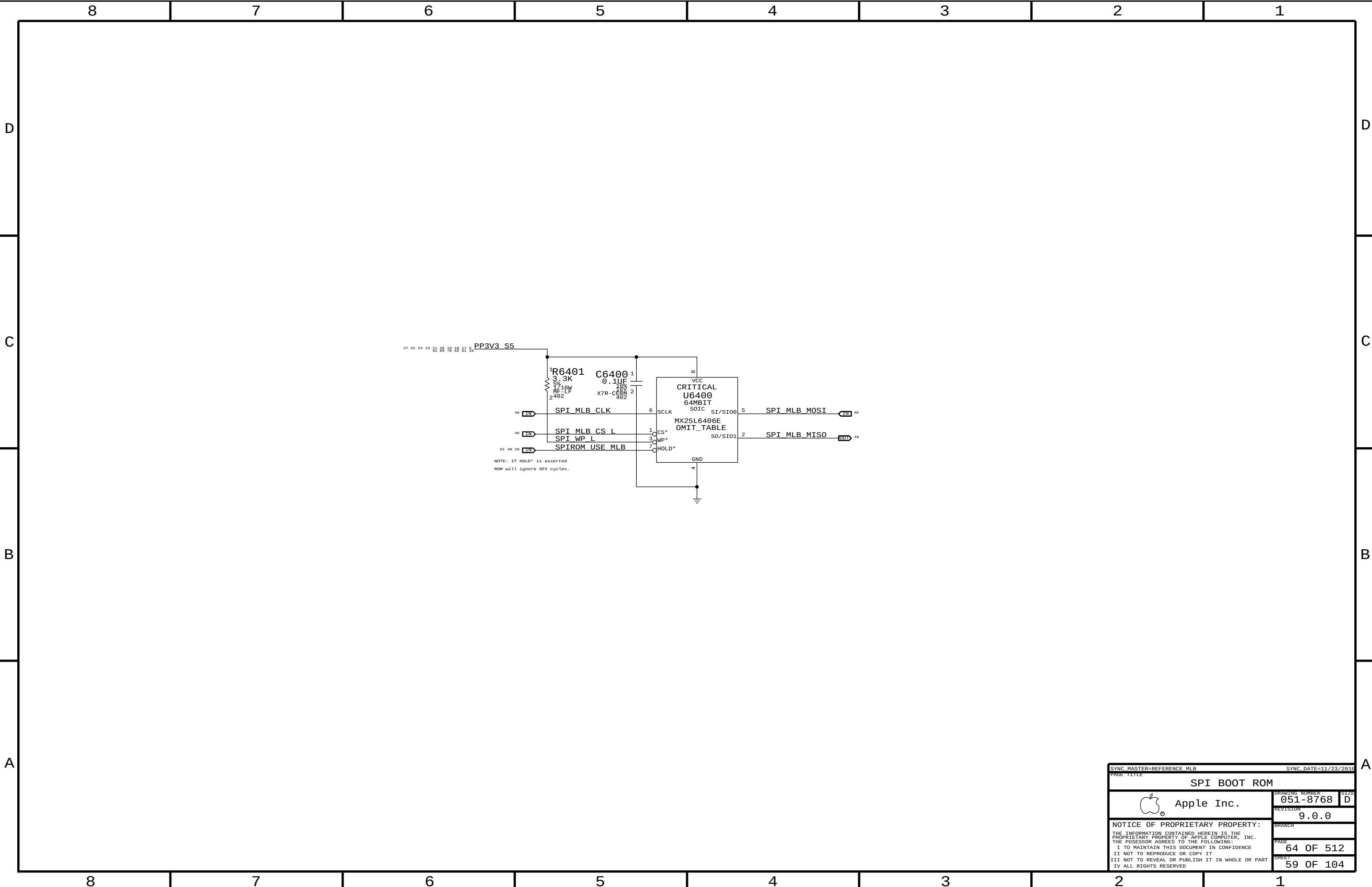
WRITE: 0X72 READ: 0X73



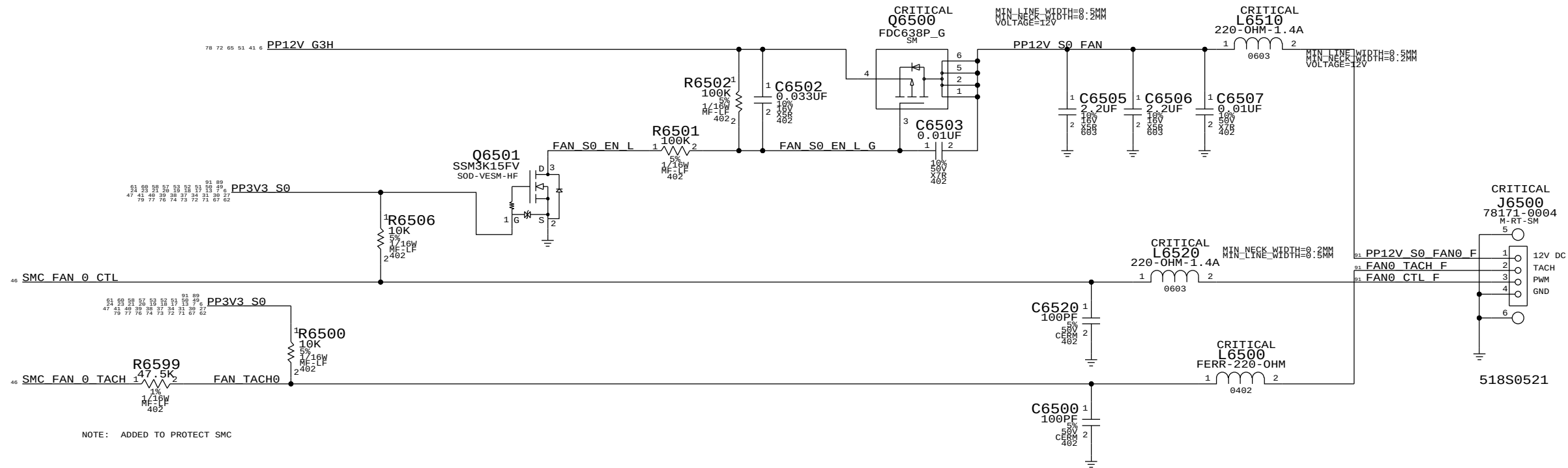
PLACE L6100/C6100 CLOSE TO Q6100/01/02/03

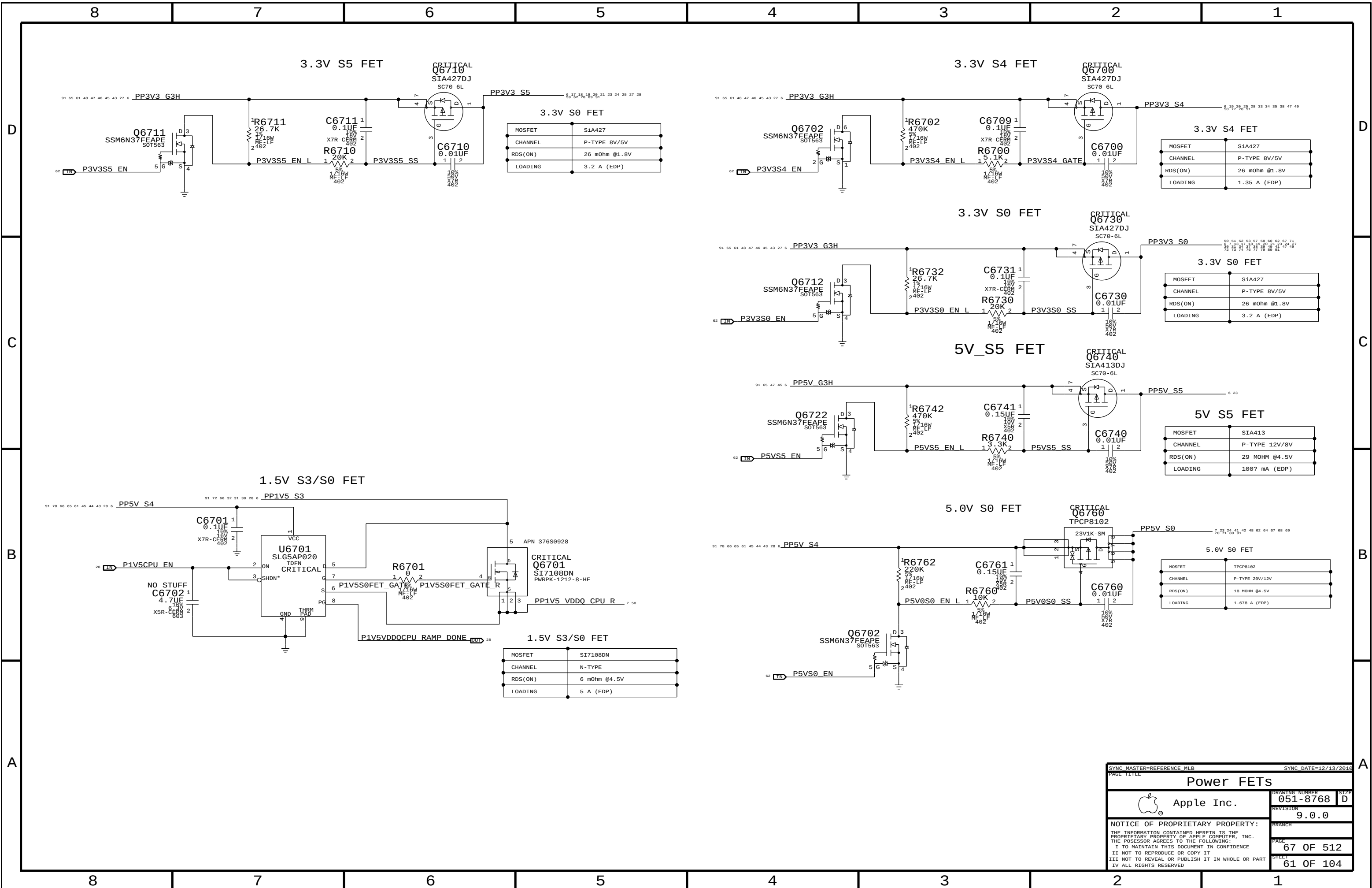


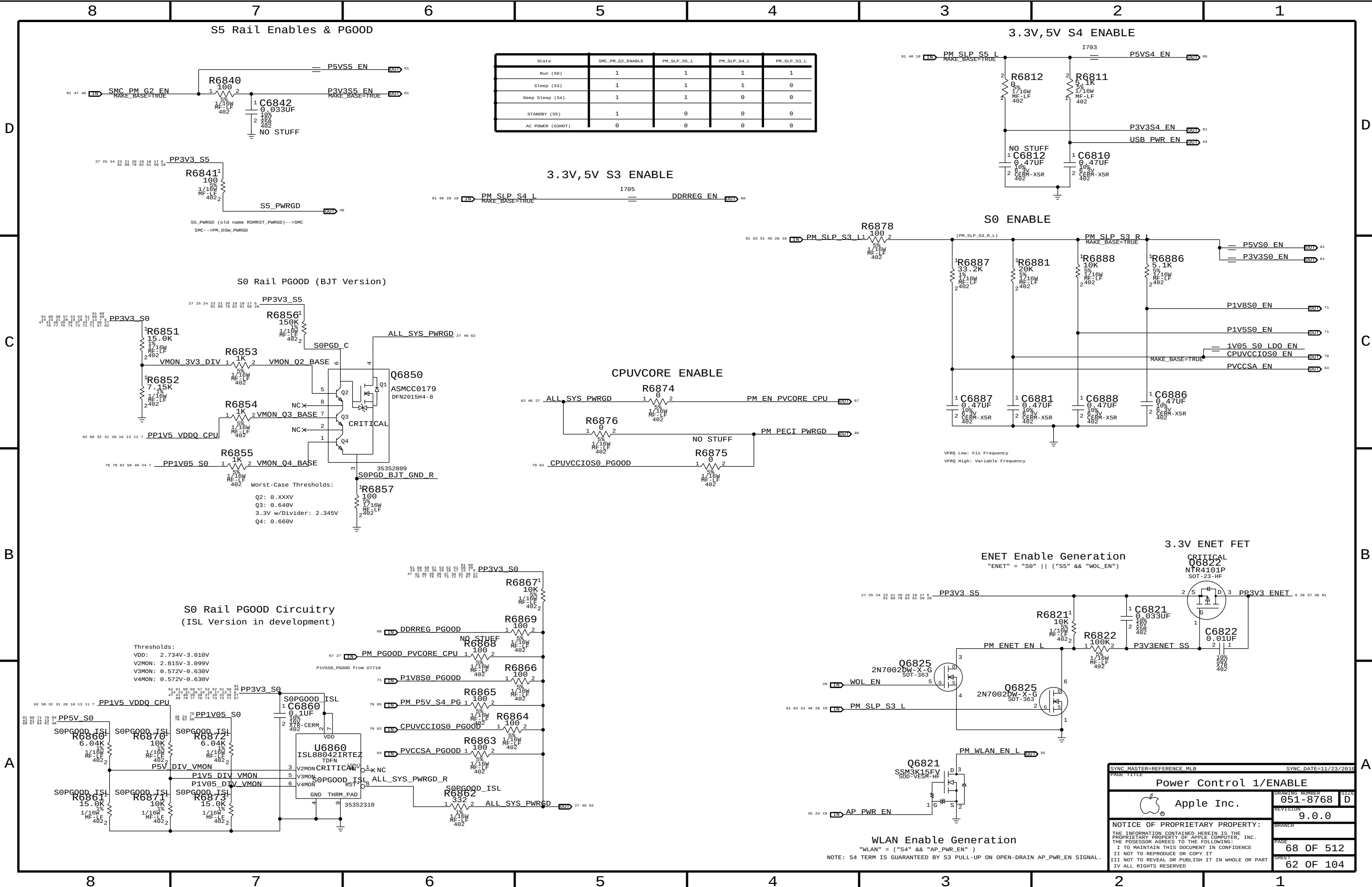
SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2016	
PAGE TITLE			
AUDIO: JACK TRANSLATORS			
 Apple Inc.		DRAWING NUMBER	051-8768
		SIZE	D
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		SHEET	58 OF 104



FAN CONTROL CIRCUIT







State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
STANDBY (S5)	1	0	0	0
AC POWER (G3MGT)	0	0	0	0

SYNC MASTER=REFERENCE_MLB

SYNC DATE=11/23/2010

Power Control 1/ENABLE

Apple Inc.

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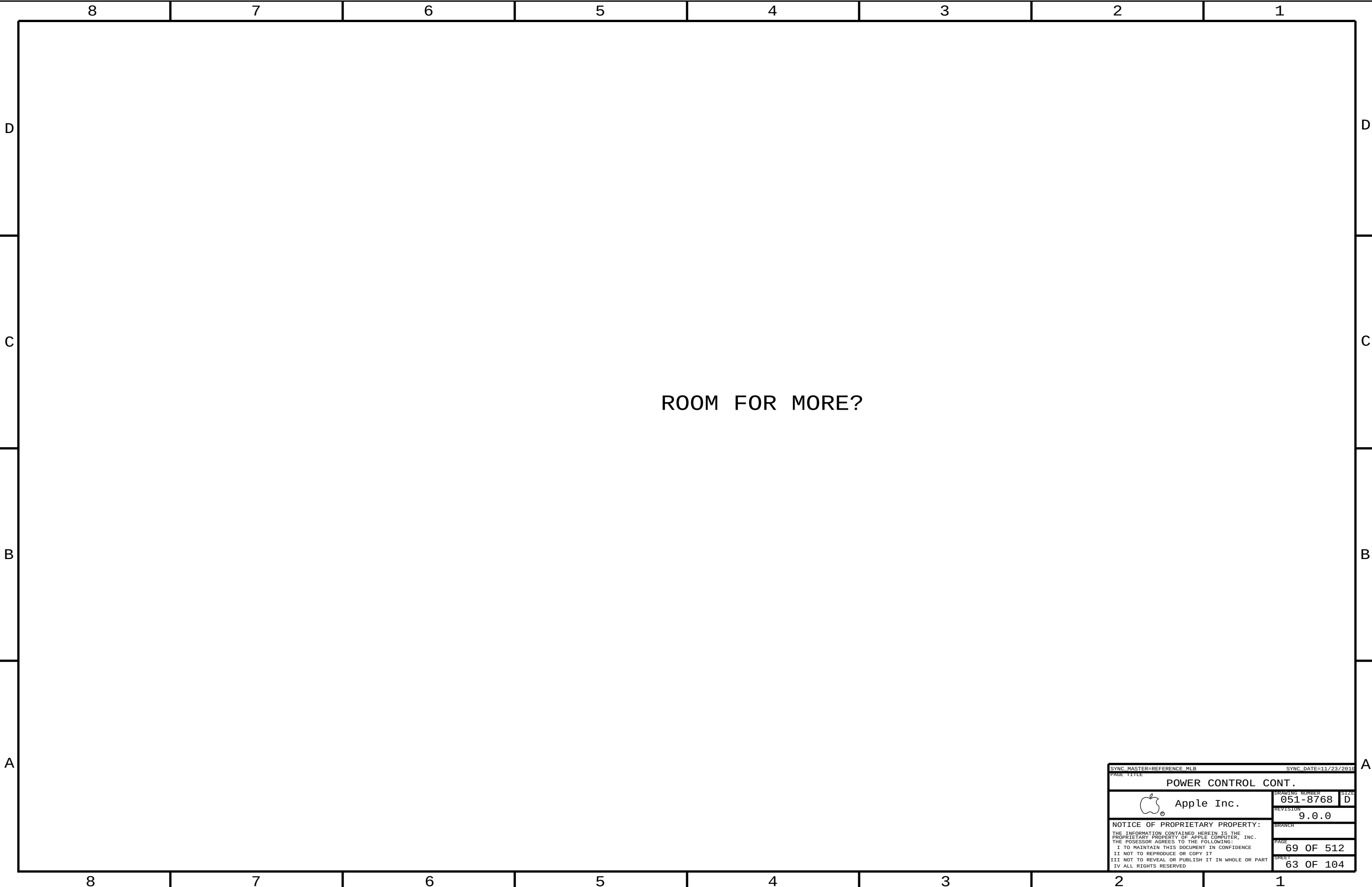
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PAGE

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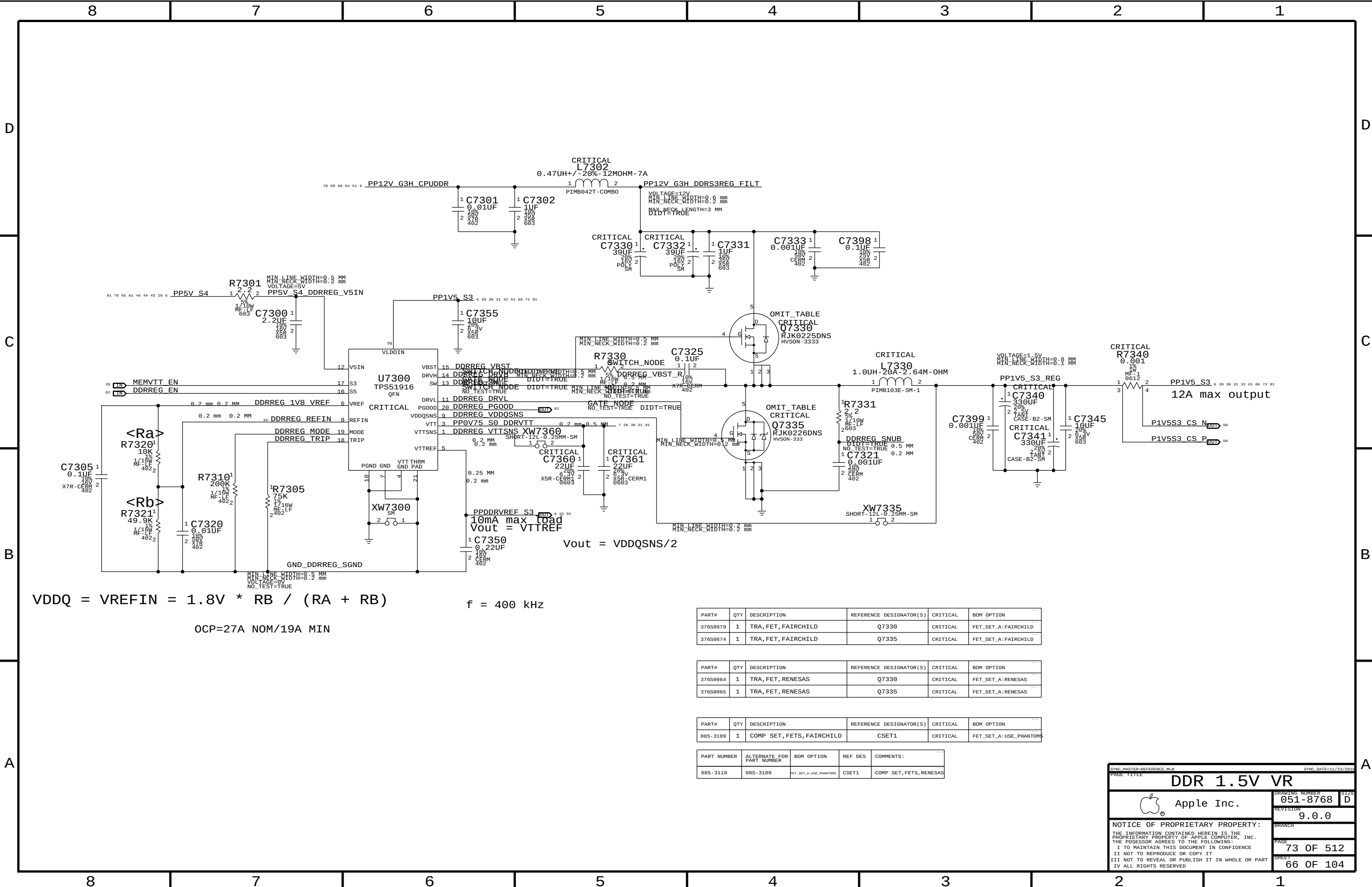
SHEET

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ROOM FOR MORE?

SYNC MASTER=REFERENCE_MLB		SYNC DATE=11/23/2010	
PAGE TITLE			
POWER CONTROL CONT.			
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		PAGE	69 OF 512
		SHEET	63 OF 104



$$VDDQ = VREFIN = 1.8V * RB / (RA + RB)$$

$$OCP=27A \text{ NOM}/19A \text{ MIN}$$

$$f = 400 \text{ kHz}$$

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S9979	1	TRA, FET, FAIRCHILD	Q7330	CRITICAL	FET_SET_A:FAIRCHILD
376S9874	1	TRA, FET, FAIRCHILD	Q7335	CRITICAL	FET_SET_A:FAIRCHILD

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
376S9964	1	TRA, FET, RENESAS	Q7330	CRITICAL	FET_SET_A:RENESAS
376S9965	1	TRA, FET, RENESAS	Q7335	CRITICAL	FET_SET_A:RENESAS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-3109	1	COMP SET, FETS, FAIRCHILD	CSET1	CRITICAL	FET_SET_A:USE_PHANTOMS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-3110	085-3109	FET_SET_A:USE_PHANTOMS	CSET1	COMP SET, FETS, RENESAS

SYNC MASTER-REFERENCE: HLB

SYNC DATE=11/23/2011

DDR 1.5V VR

Apple Inc.

051-8768

D

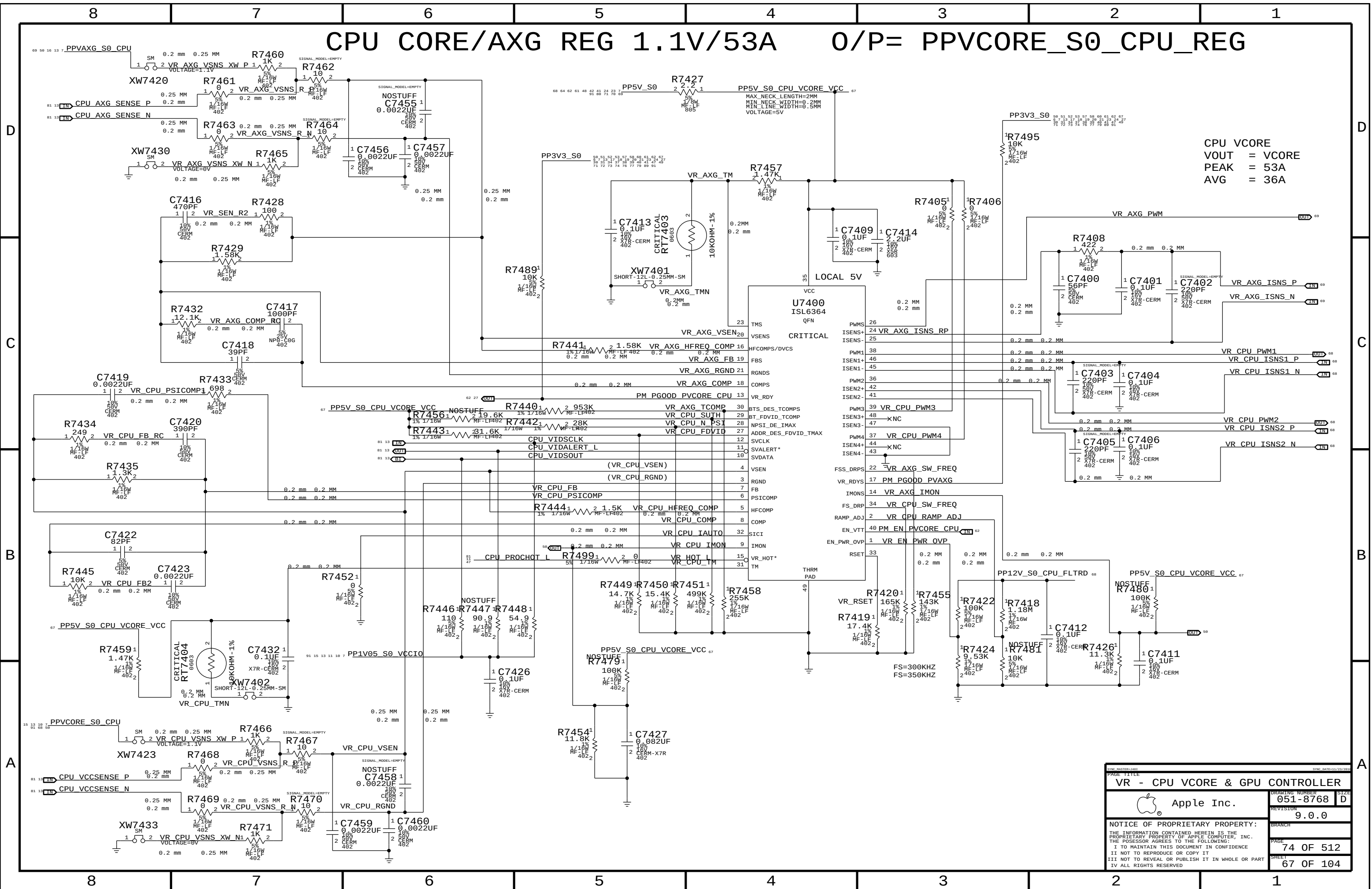
REVISION 9.0.0

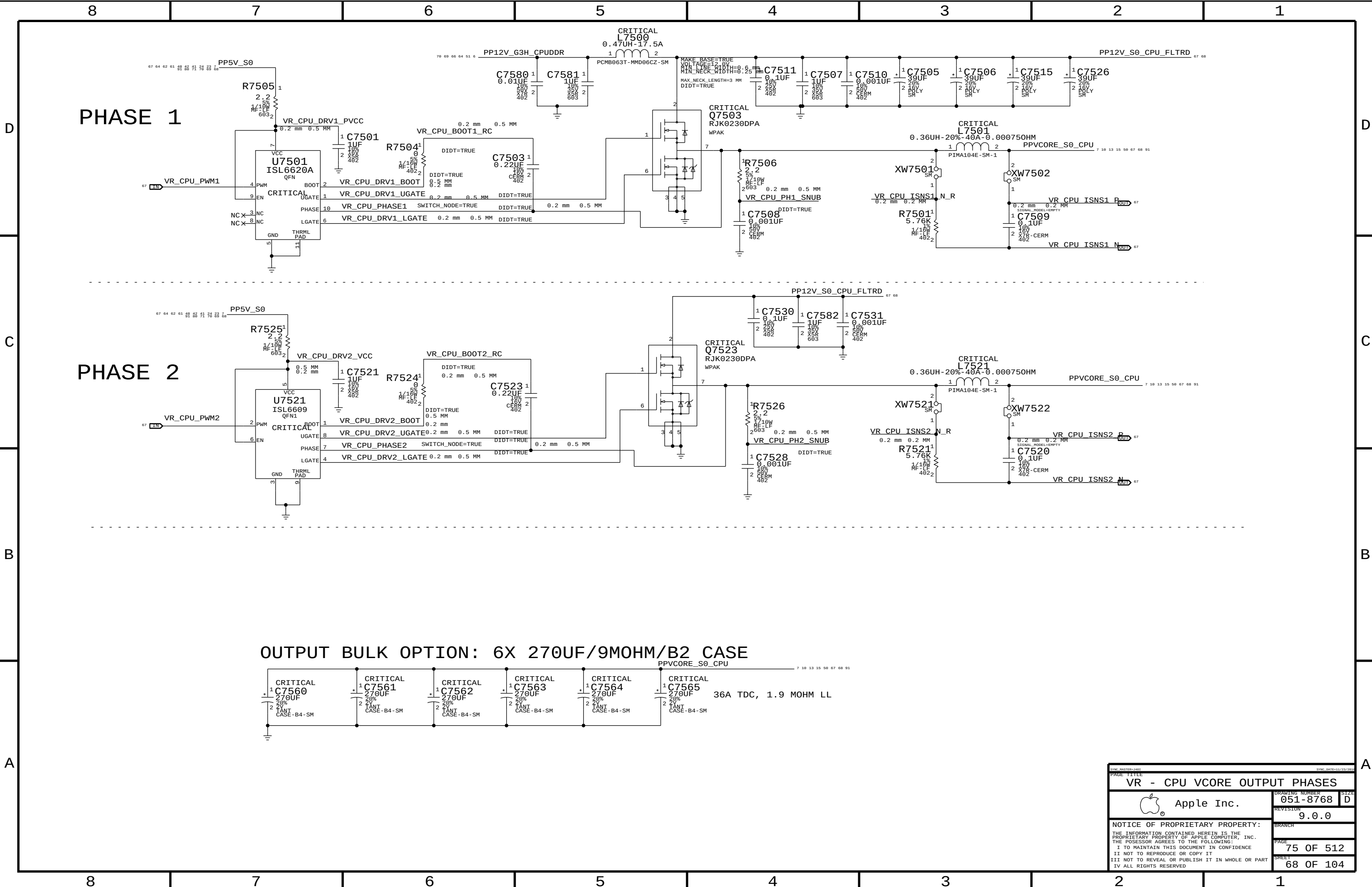
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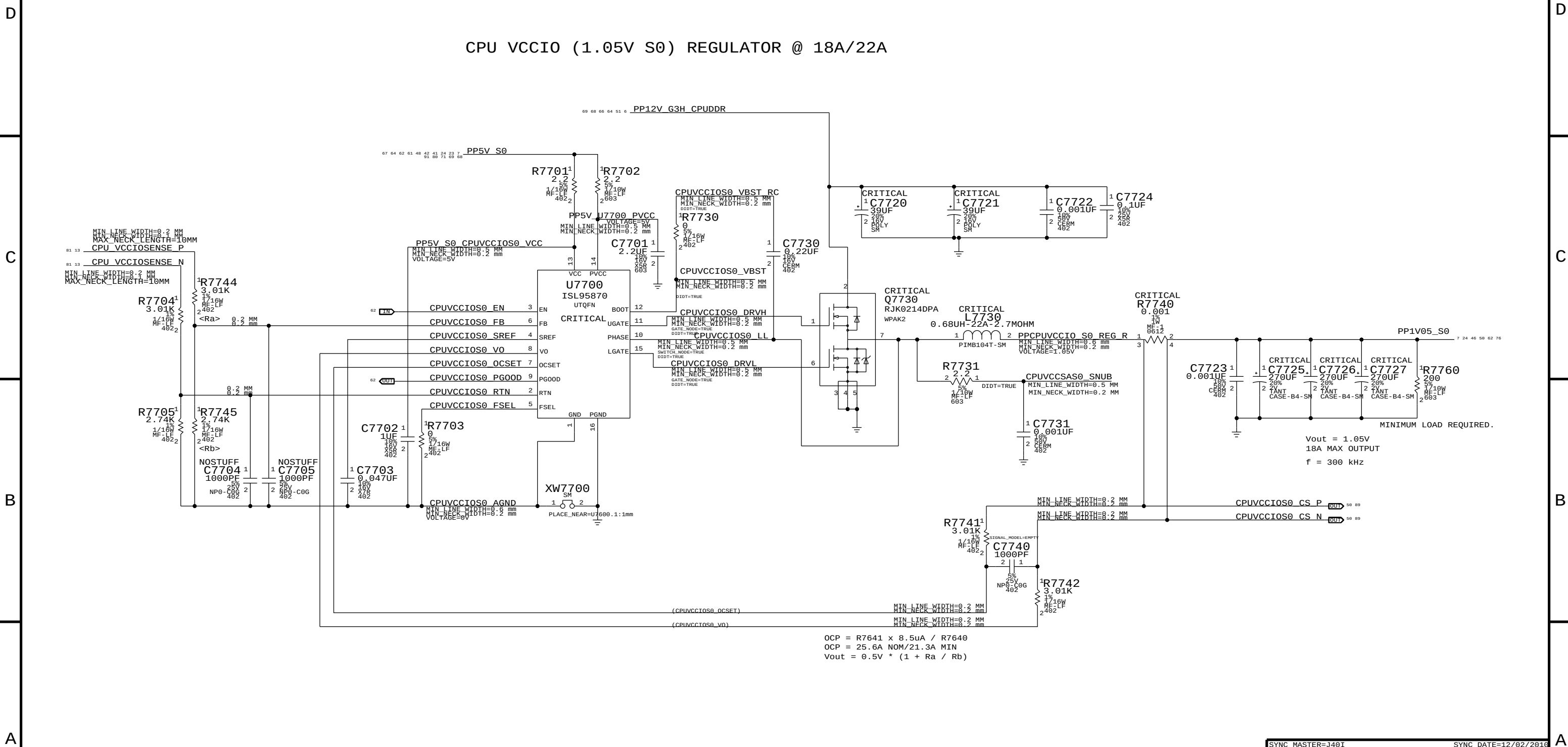


VR - CPU VCore Output Phases	
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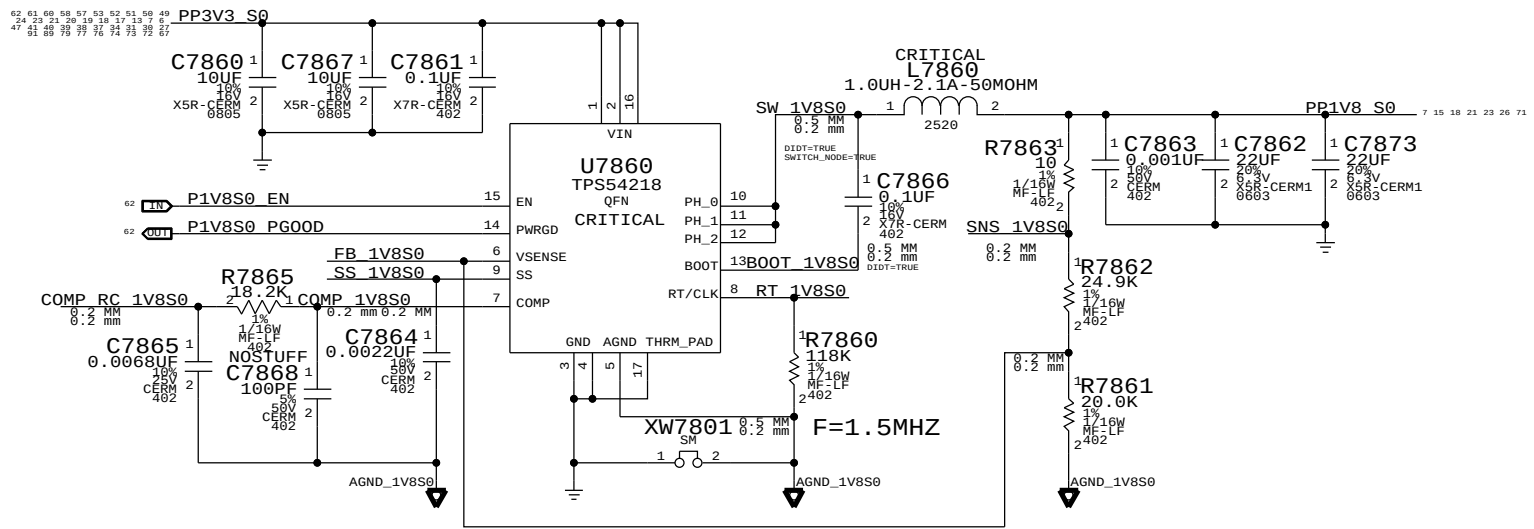


8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

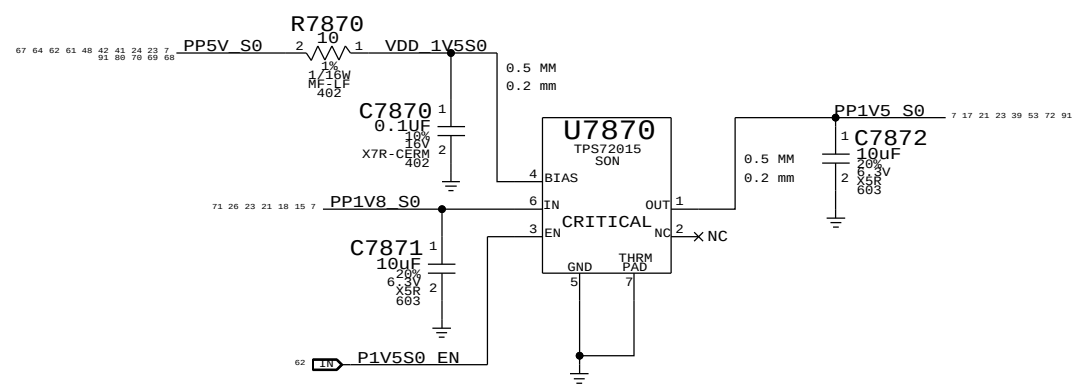


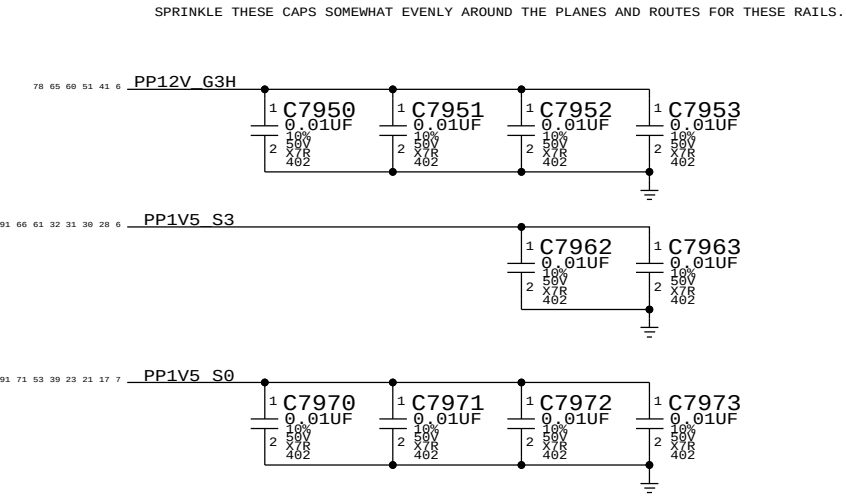
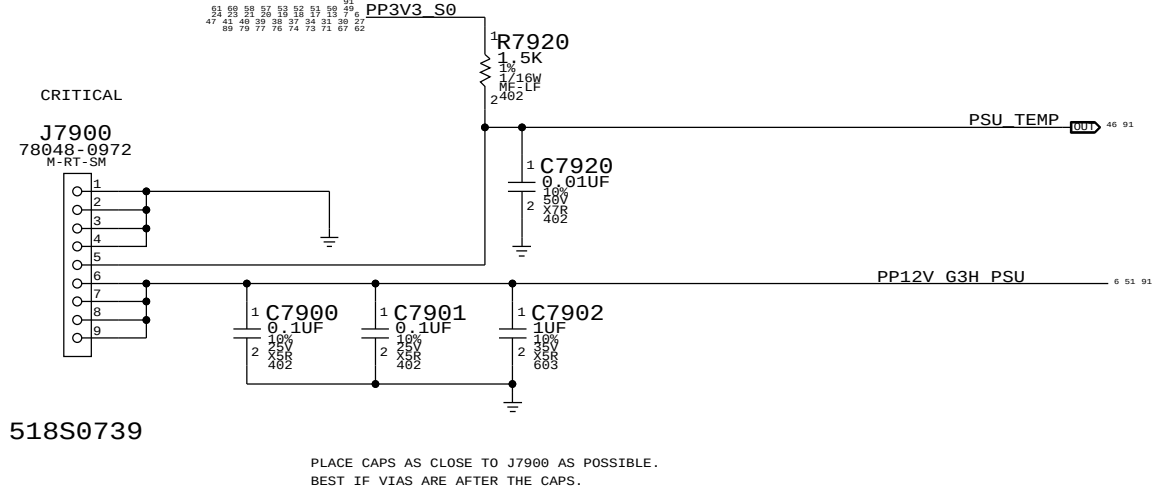
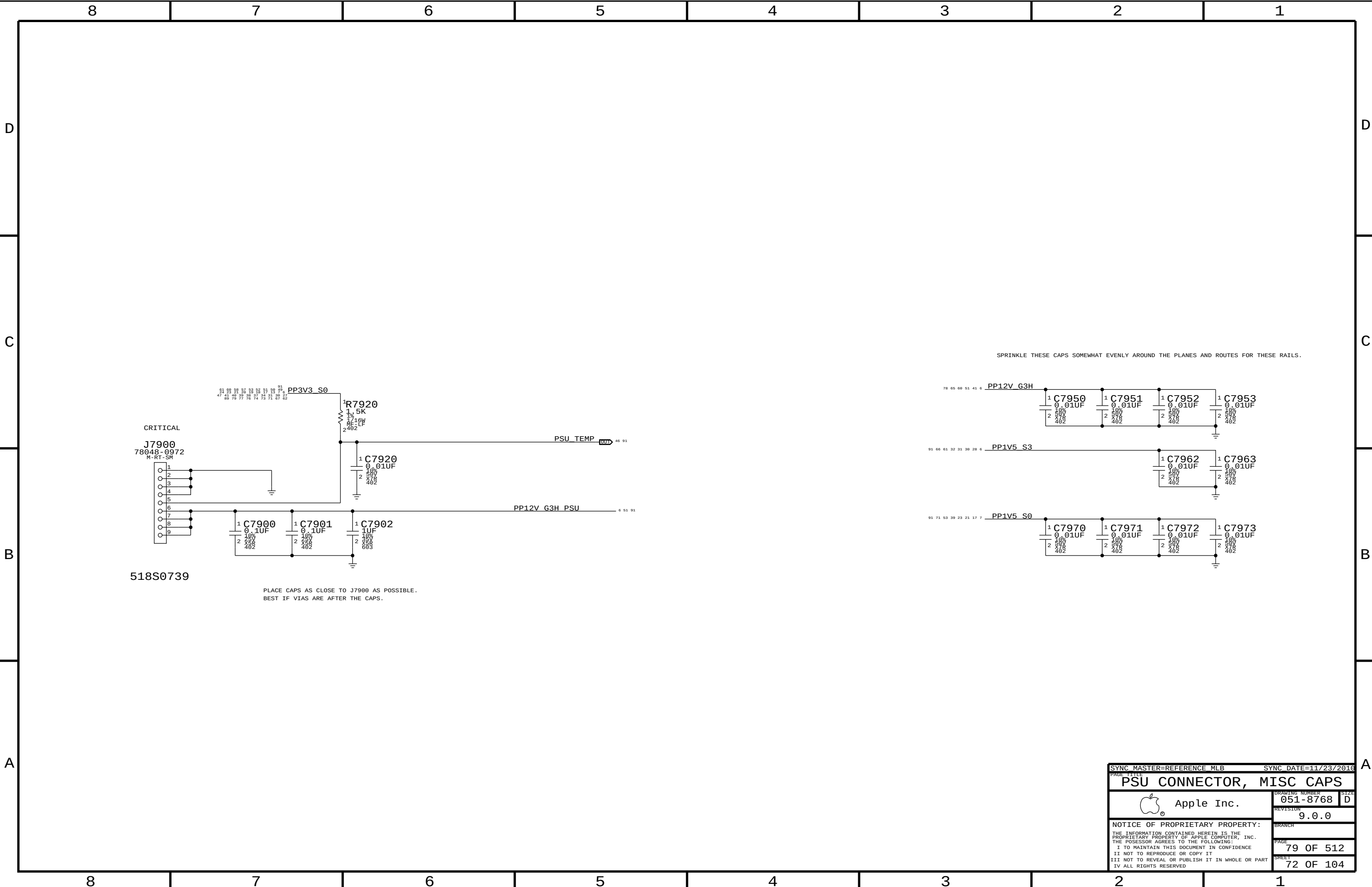
								PAGE TITLE VR - 1.05V S0				
								 Apple Inc.		DRAWING NUMBER 051-8768		SIZE D
										REVISION 9.0.0		
								NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH		
										PAGE 77 OF 512		
										SHEET 70 OF 104		
8	7	6	5	4	3	2	1					

1.8V S0 SWITCHER @ 1.7A/2A

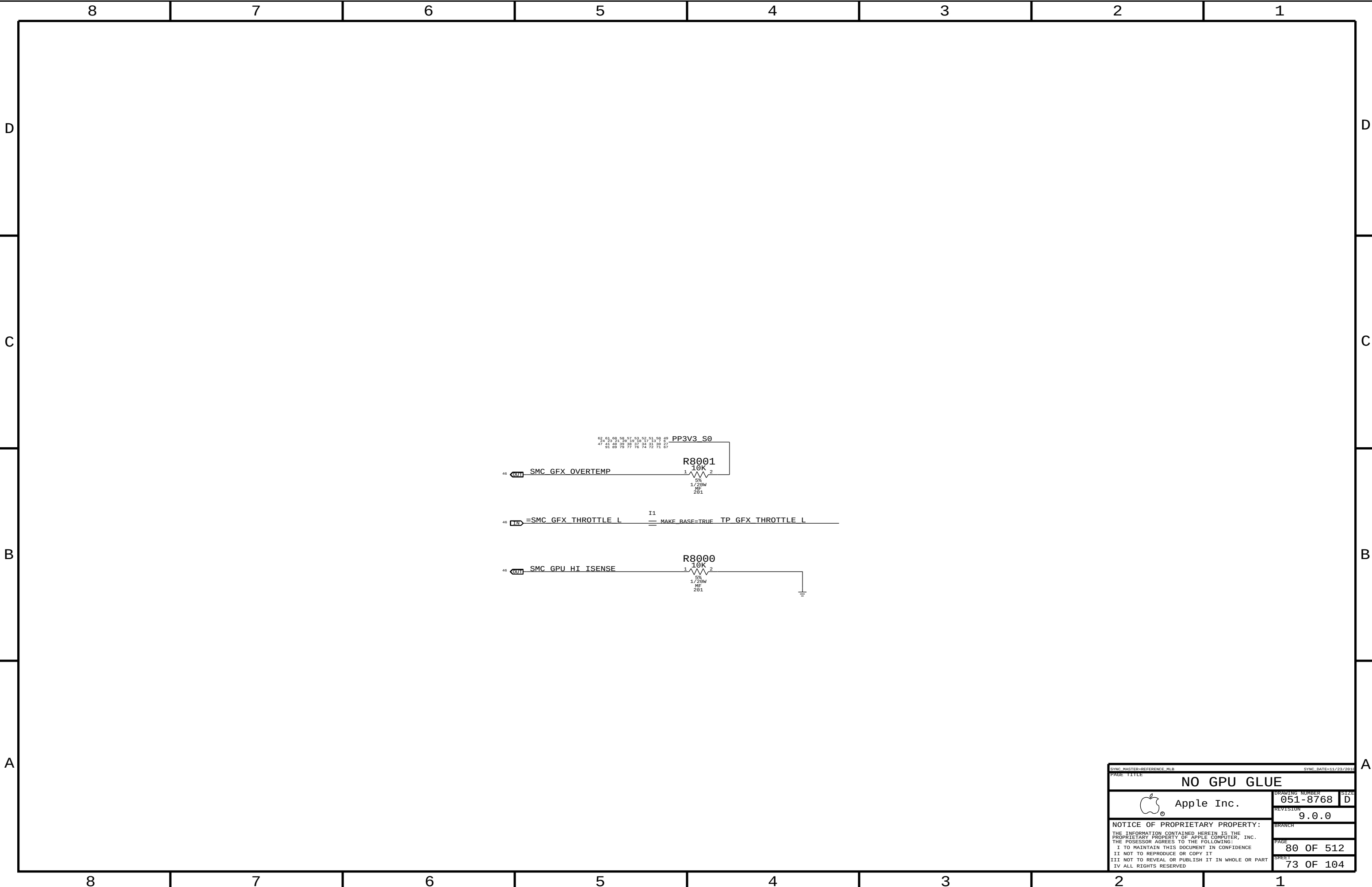


1.5V S0 @ 0.3A/0.5A

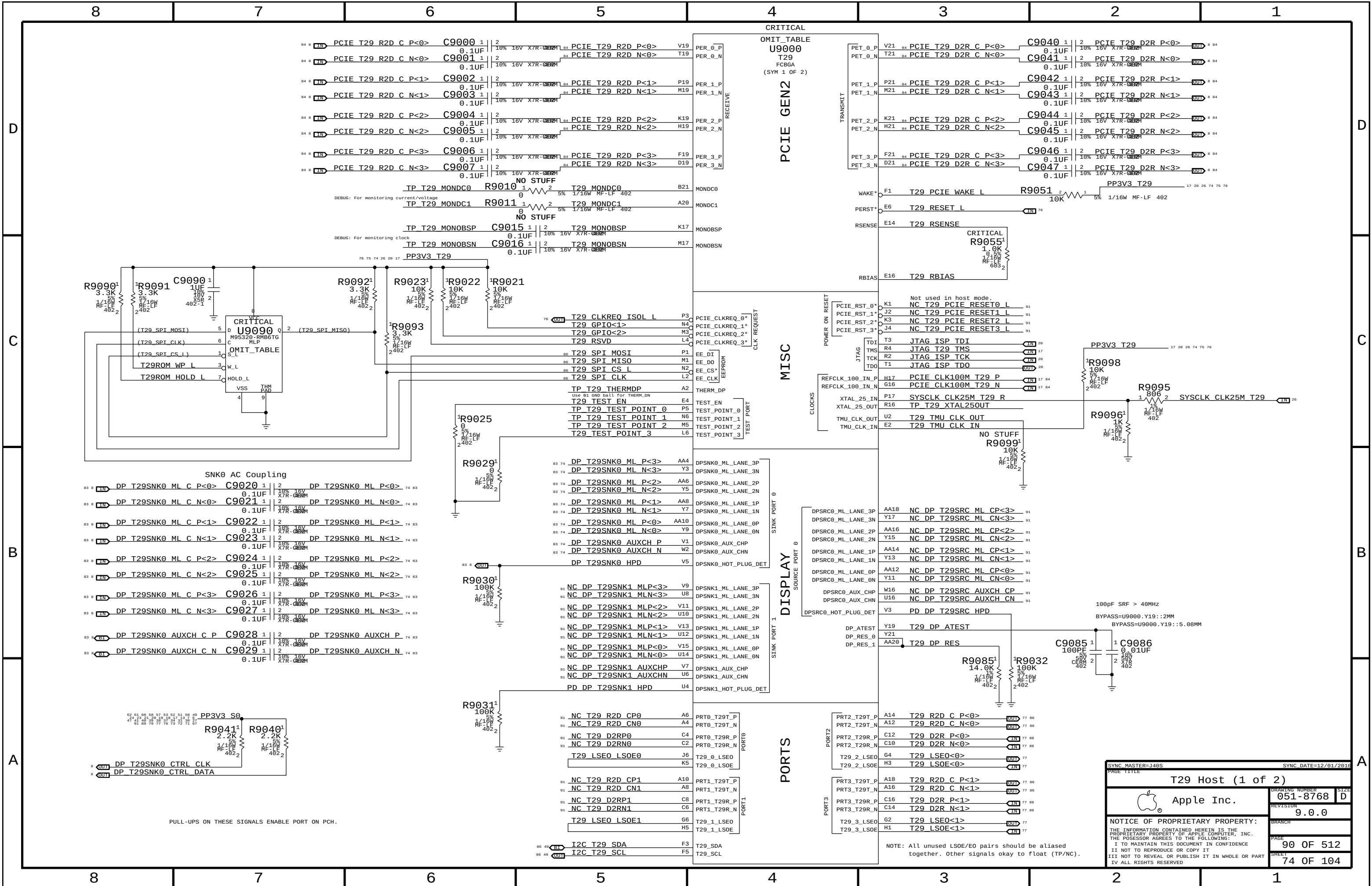




SYNC MASTER=REFERENCE MLB		SYNC DATE=11/23/2016	
PAGE TITLE PSU CONNECTOR, MISC CAPS			
 Apple Inc.		DRAWING NUMBER 051-8768	SIZE D
		REVISION 9.0.0	
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PAGE TITLE		PAGE TITLE	
NO GPU GLUE		NO GPU GLUE	
DRAWING NUMBER		DRAWING NUMBER	
051-8768		051-8768	
REVISION		REVISION	
9.0.0		9.0.0	
BRANCH		BRANCH	
PAGE		PAGE	
80 OF 512		80 OF 512	
SHEET		SHEET	
73 OF 104		73 OF 104	



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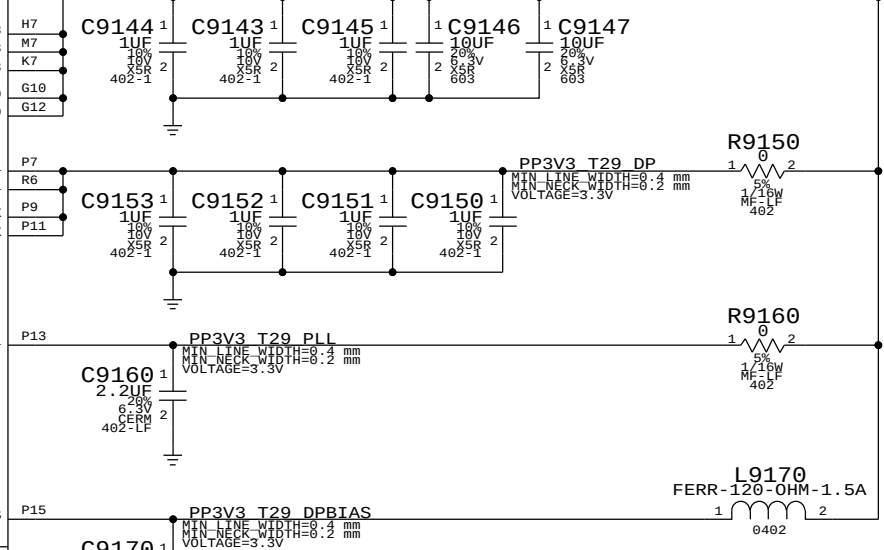
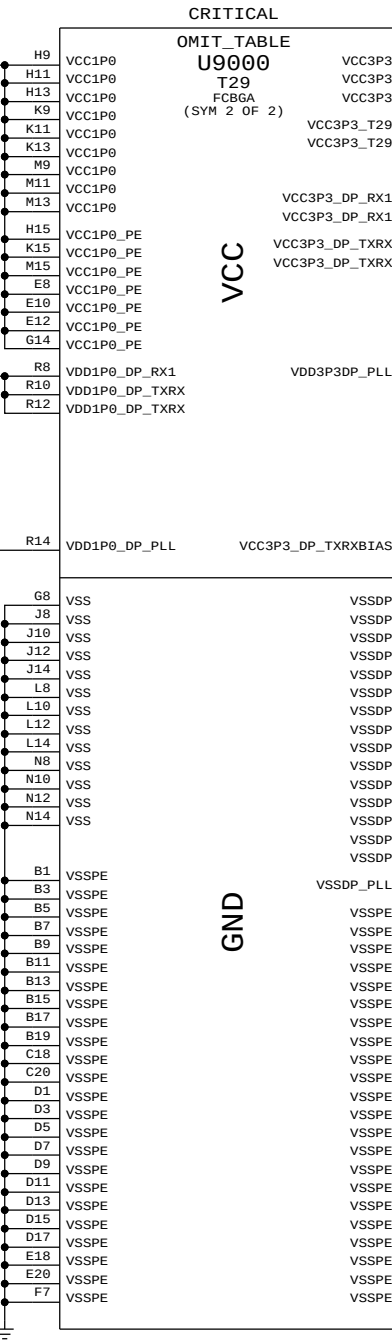
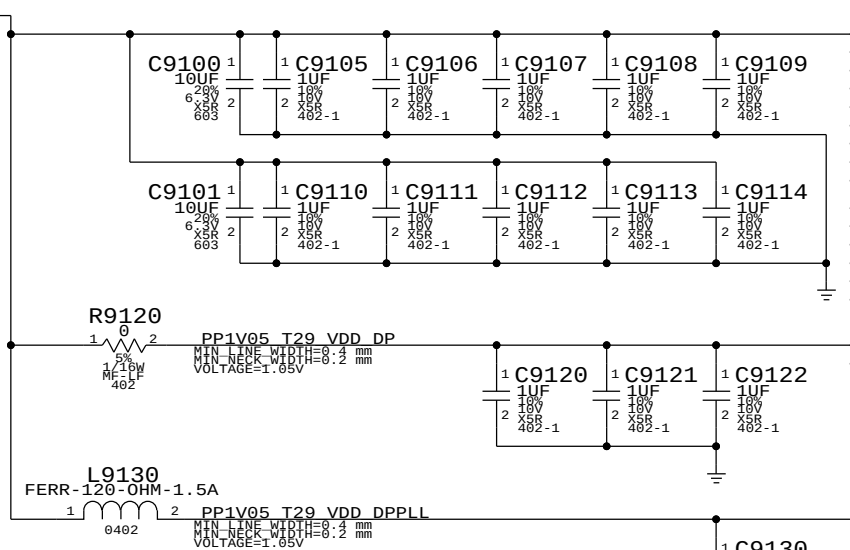
C

B

A

8 7 6 5 4 3 2 1

76 PP1V05 T29
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



PP3V3 T29
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.

8 7 6 5 4 3 2 1

SYNC MASTER=J46S		SYNC DATE=11/23/2010	
PAGE TITLE			
T29 Host (2 of 2)			
 Apple Inc.		DRAWING NUMBER	051-8768
		REVISION	9.0.0
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Page Notes

Power aliases required by this page:

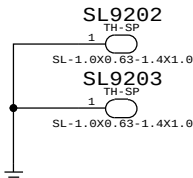
- PPVIN_SW_T29BST (8-13V Boost Input)
- PP18V_T29_REG (18V Boost Output)
- PP3V3_T29_P3V3T29FET (3.3V FET Input)
- PP3V3_T29_FET (3.3V FET Output)
- PP3V3_S0_T29PWRCTL
- PP1V05_T29_P1V05T29FET (1.05V FET Input)
- PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:

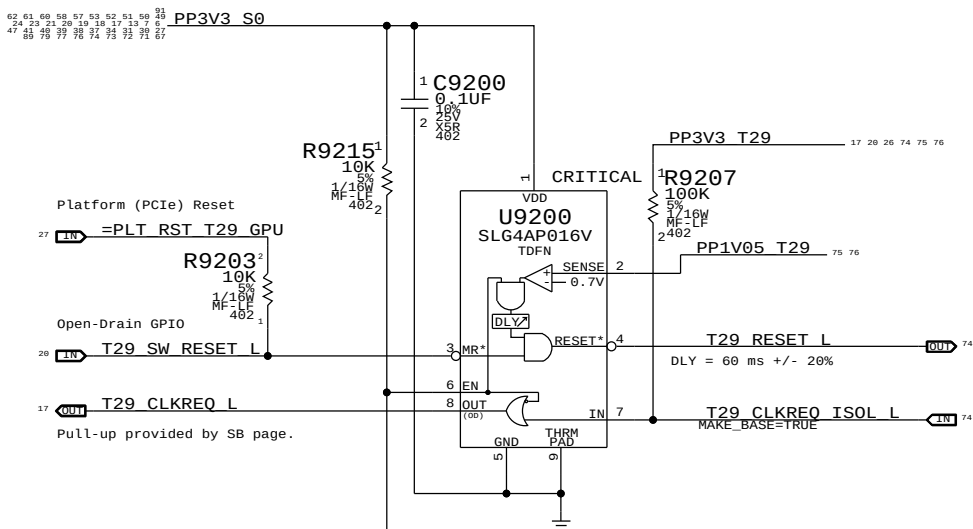
- T29_CLKREQ_L
- T29_RESET_L

BOM options provided by this page:

T29BST:Y - Stuffs 18V boost circuitry.

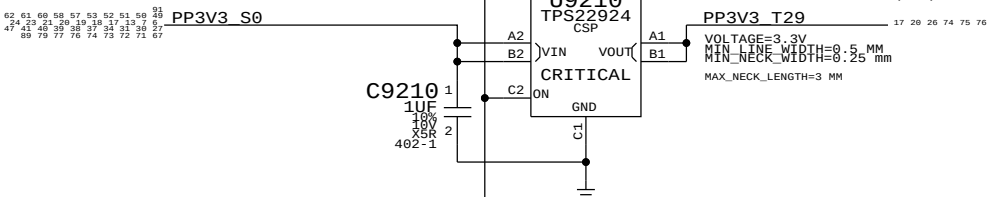


Supervisor & CLKREQ# Isolation



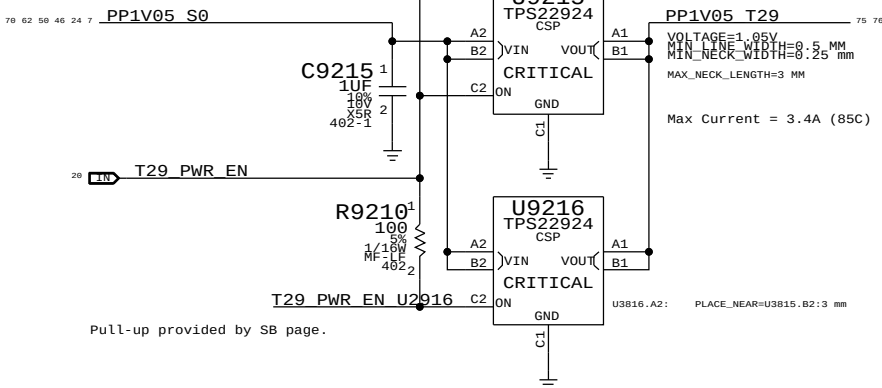
3.3V T29 Switch

Max Current = 1.7A (85C)



1.05V T29 Switch

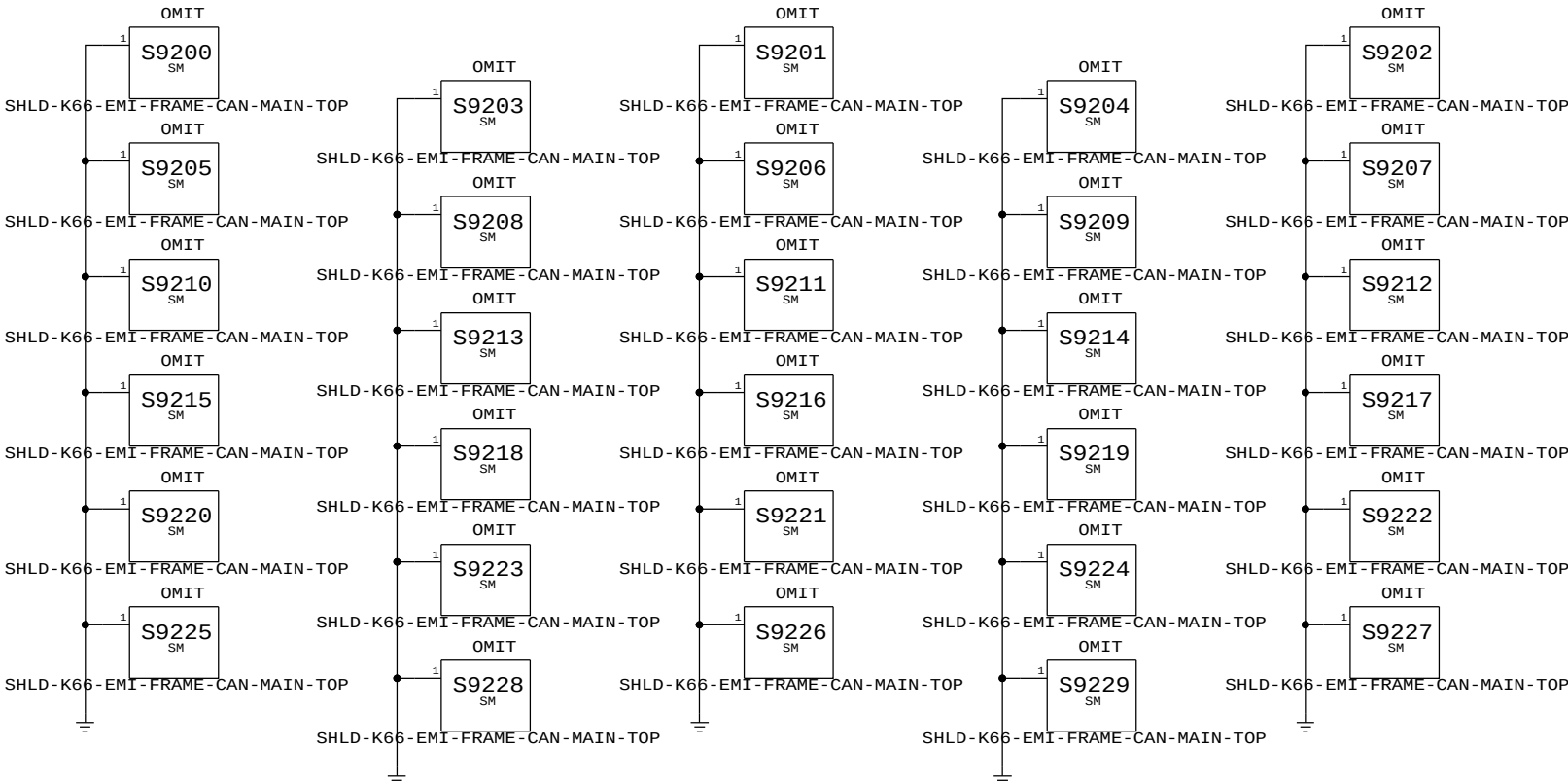
Max Current = 3.4A (85C)



U9210 & U9215/U9216

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

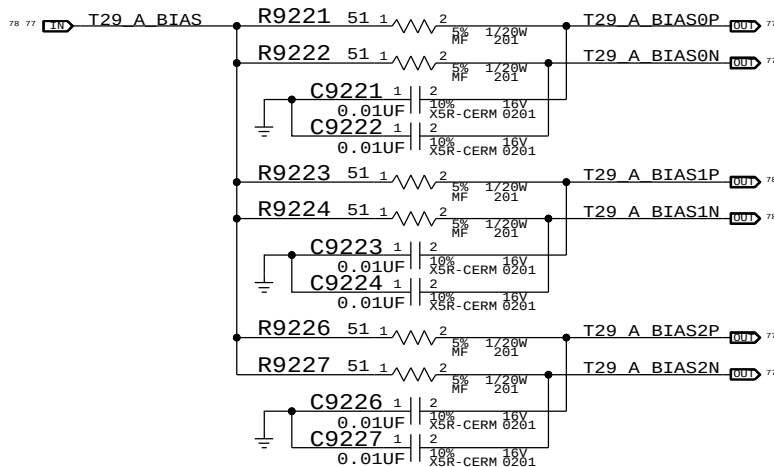
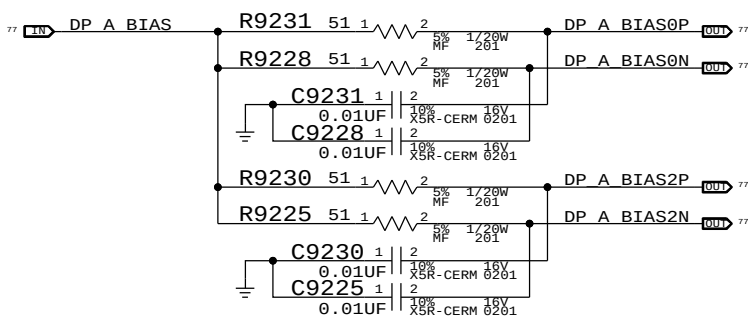
Max Output: 2A per IC



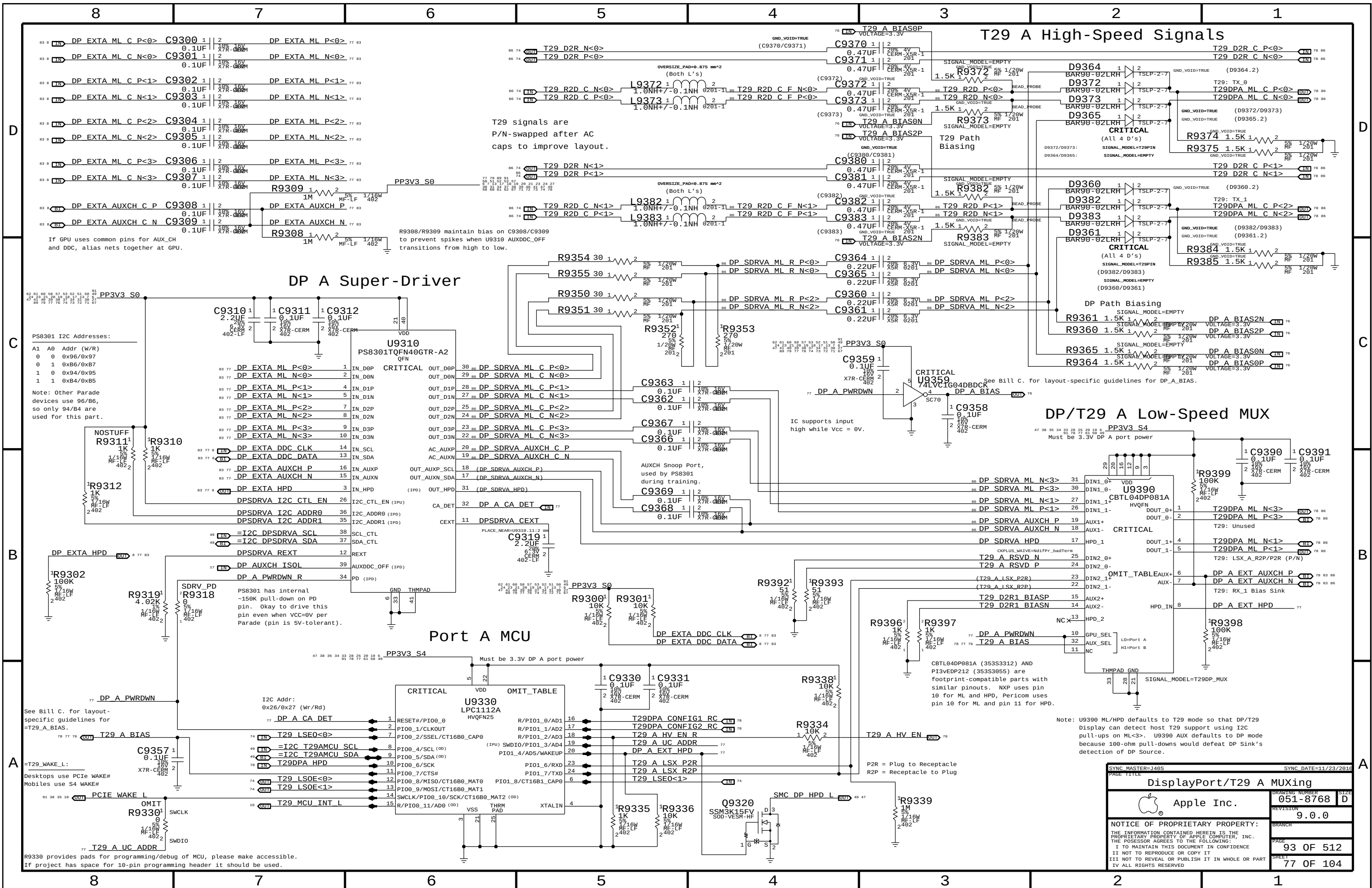
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
806-2254	1	FRAME,EMI SHIELD,TB,340	SH9200	CRITICAL	?

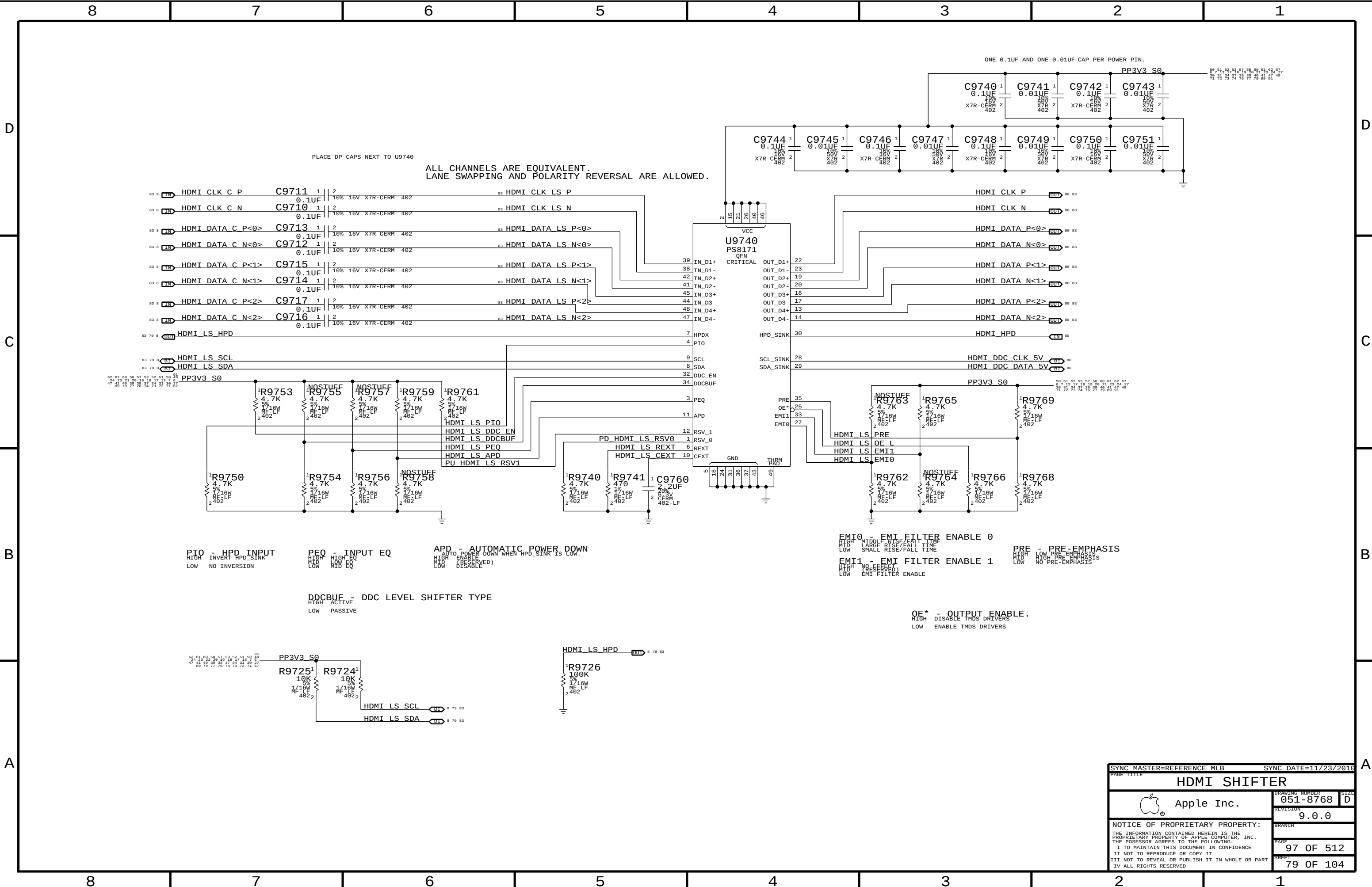
DP/T29 Bias Filters

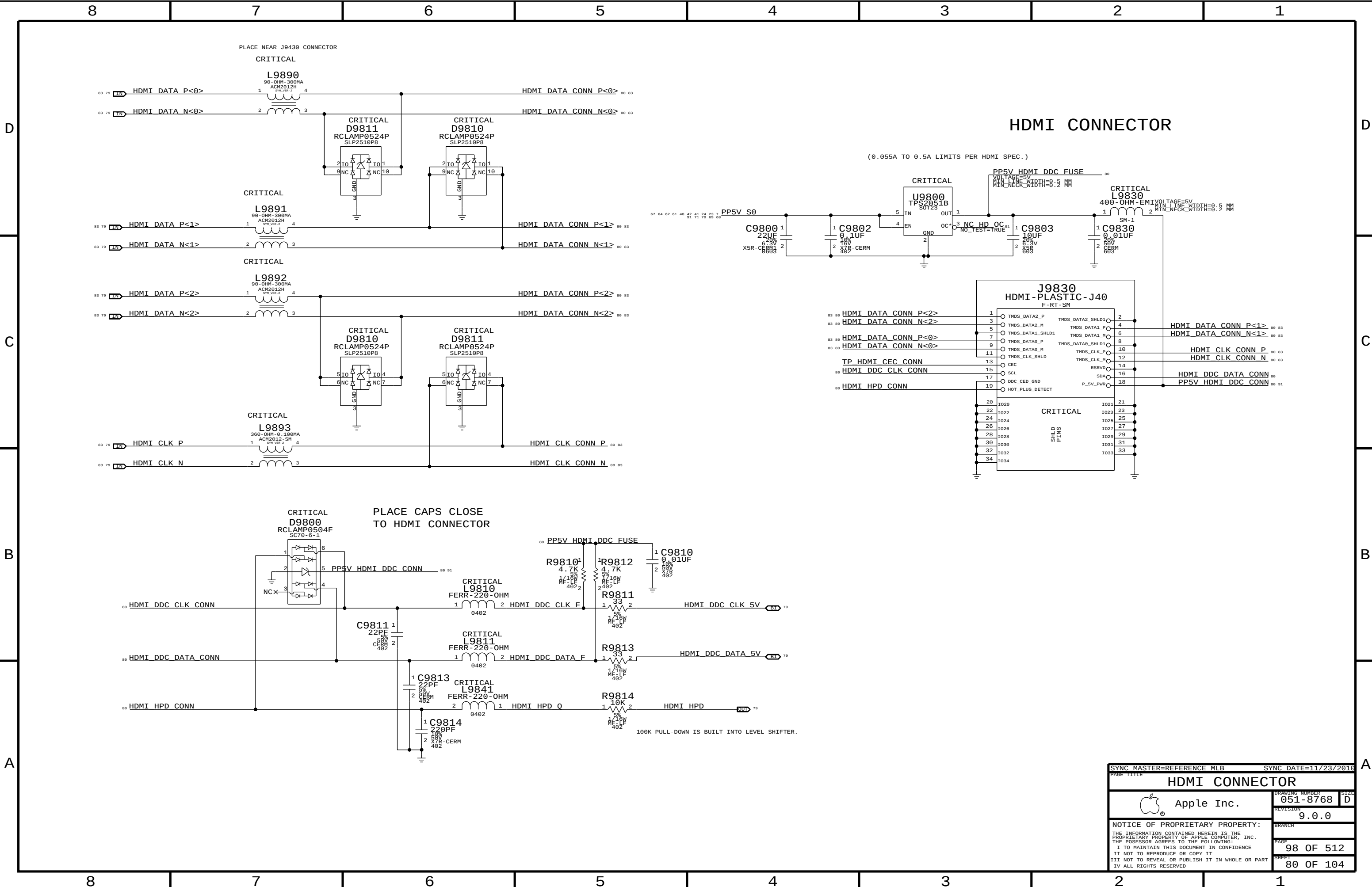
Prevents high-frequency coupling between 1.5K bias resistors.



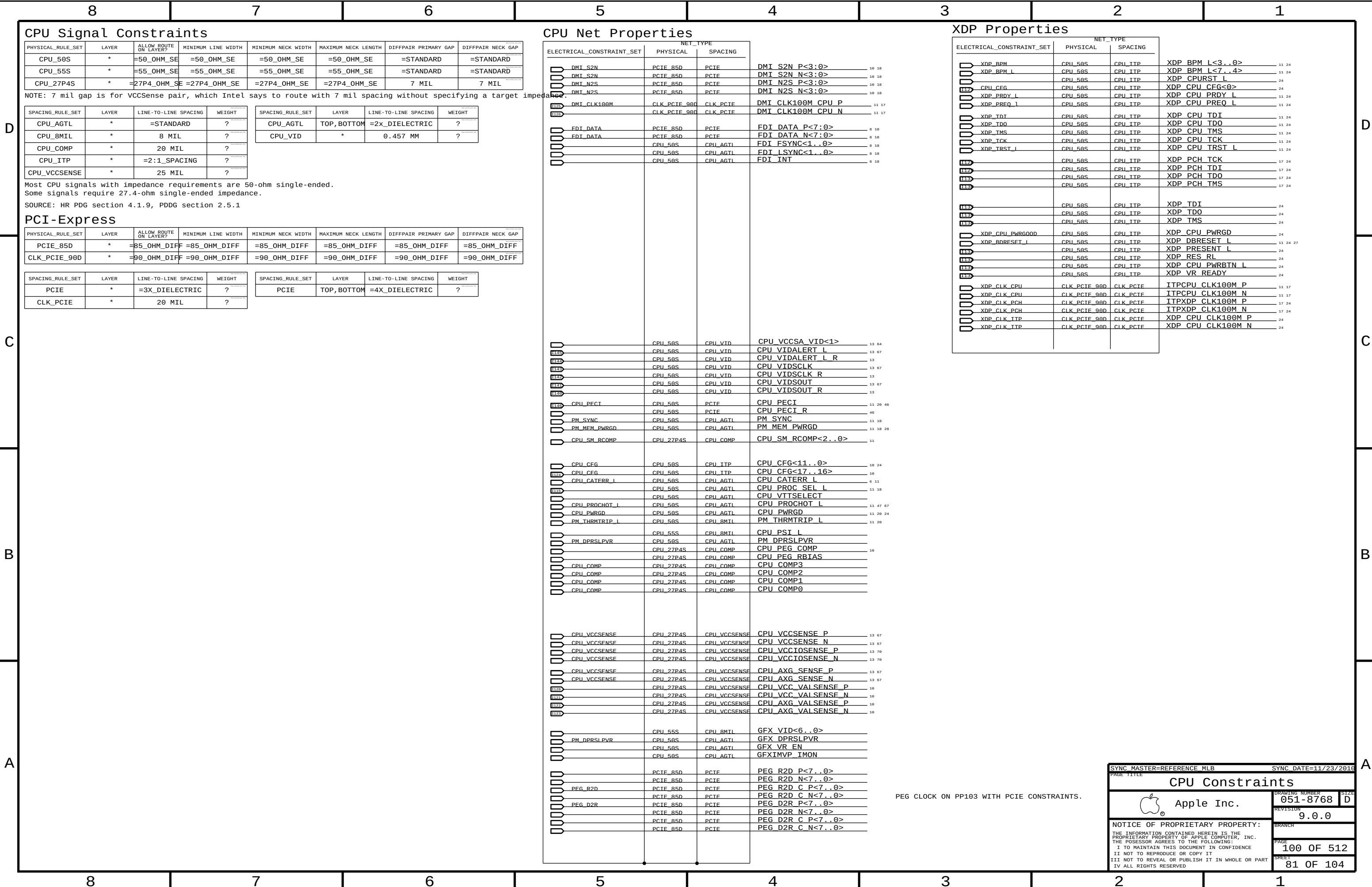
SYNC MASTER=J40S		SYNC DATE=11/23/2010	
PAGE TITLE			
T29 Power Support			
 Apple Inc.		DRAWING NUMBER	051-8768
		SIZE	D
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		PAGE	92 OF 512
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PAGE TITLE		SYNC DATE=11/23/2016	
HDMI CONNECTOR		DRAWING NUMBER	
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPT_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		PHYSICAL	SPACING	
	LPC_AD	LPC_50S	LPC		LPC_AD<3...0>	17 46 48
		LPC_50S	LPC		LPC_R_AD<3...0>	17
	LPC_FRAME_L	LPC_50S	LPC		LPC_FRAME_L	17 46 48
		LPC_50S	LPC		LPC_FRAME_R_L	17
	LPC_RESET_L	LPC_50S	LPC		LPCPLUS_RESET_L	27 48
	PCH_LPC_CLK0	CLK_LPC_50S	CLK_LPC		LPC_CLK33M_SMC_R	19 27
		CLK_LPC_50S	CLK_LPC		LPC_CLK33M_SMC	27 46
		CLK_LPC_50S	CLK_LPC		LPC_CLK33M_LPCPLUS	27 48
		CLK_LPC_50S	CLK_LPC		LPC_CLK33M_LPCPLUS_R	19 27
		SMB_50S	SMB		SMB_PCH_STR_SCL	49
		SMB_50S	SMB		SMB_PCH_STR_SDA	49
		SMB_50S	SMB		SMB_PCH_DIMM_SCL	49
		SMB_50S	SMB		SMB_PCH_DIMM_SDA	49
		SMB_50S	SMB		SMB_PCH_MKY_SCL	49
		SMB_50S	SMB		SMB_PCH_MKY_SDA	49
		SMB_50S	SMB		SMB_PCH_VRF_SCL	49
		SMB_50S	SMB		SMB_PCH_VRF_SDA	49
	SMBUS_PCH_CLK	SMB_50S	SMB		SMBUS_PCH_CLK	17 49
		SMB_50S	SMB		SMBUS_PCH_DATA	17 49
	SMBUS_PCH_0_CLK	SMB_50S	SMB		SMB_PCH_XDP_SCL	49
		SMB_50S	SMB		SMB_PCH_XDP_SDA	49
		SMB_50S	SMB		SML_PCH_1_CLK	17 49
		SMB_50S	SMB		SML_PCH_1_DATA	17 49
	SPT_CLK	SPT_55S	SPT		SPI_CLK_R	17 48
		SPT_55S	SPT		SPI_CLK	48 91
	SPT_MOST	SPT_55S	SPT		SPI_MOSI_R	17 48
		SPT_55S	SPT		SPI_MOSI	48 91
	SPT_MISO	SPT_55S	SPT		SPI_MISO	17 48 91
	SPT_CS0	SPT_55S	SPT		SPI_CS0_R_L	17 48
		SPT_55S	SPT		SPI_CS0_L	48 91
	HDA_BIT_CLK	HDA_50S	HDA		HDA_BIT_CLK	17 53
		HDA_50S	HDA		HDA_BIT_CLK_R	17
	HDA_SYNC	HDA_50S	HDA		HDA_SYNC	17 53
		HDA_50S	HDA		HDA_SYNC_R	17
	HDA_RST_L	HDA_50S	HDA		HDA_RST_R_L	17
		HDA_50S	HDA		HDA_RST_L	17 53
	HDA_SDIN0	HDA_50S	HDA		HDA_SDIN0	17 53
		HDA_50S	HDA		AUD_SDI_R	53
	HDA_SDOUT	HDA_50S	HDA		HDA_SDOUT	17 53
		HDA_50S	HDA		HDA_SDOUT_R	17

9250	CLK_PCTF_90D	CLK_PCTF	PCH_CLK96M_DOT_P	17
9250	CLK_PCTF_90D	CLK_PCTF	PCH_CLK96M_DOT_N	17
9250	CLK_PCTF_90D	CLK_PCTF	PCH_CLK100M_SATA_P	17
9250	CLK_PCTF_90D	CLK_PCTF	PCH_CLK100M_SATA_N	17
9250	CPU1_50S	CLK_PCTF	PCH_CLK14P3M_REFCLK	17
9250	CPU1_50S	CLK_PCTF	PCH_CLK33M_PCIIN	17 27

	PCIE_85D	PCIE	PCIE_ENET_R2D_P	37
	PCIE_85D	PCIE	PCIE_ENET_R2D_N	37
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P	17 37
	PCIE_85D	PCIE	PCIE_ENET_R2D_C_N	17 37
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_P	17 37
	PCIE_85D	PCIE	PCIE_ENET_D2R_N	17 37
	PCIE_85D	PCIE	PCIE_ENET_D2R_C_P	37
	PCIE_85D	PCIE	PCIE_ENET_D2R_C_N	37
	PCIE_85D	PCIE	PCIE_AP_R2D_P	35 91
	PCIE_85D	PCIE	PCIE_AP_R2D_N	35 91
PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_C_P	17 35
	PCIE_85D	PCIE	PCIE_AP_R2D_C_N	17 35
PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_D2R_P	17 35 91
	PCIE_85D	PCIE	PCIE_AP_D2R_N	17 35 91
	PCIE_85D	PCIE	PCIE_FW_R2D_P	39
	PCIE_85D	PCIE	PCIE_FW_R2D_N	39
PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	17 39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_N	17 39
PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_P	17 39
	PCIE_85D	PCIE	PCIE_FW_D2R_N	17 39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_N	39

1F039	CLK_PCTF_90D	CLK_PCTF	PCIE_CLK100M_PCH_P	17	
1F03A	CLK_PCTF_90D	CLK_PCTF	PCIE_CLK100M_PCH_N	17	
1F03B	PCTF_CLK100M_T29	CLK_PCTF_90D	CLK_PCTF	PCIE_CLK100M_T29_P	17 74
1F03C	CLK_PCTF_90D	CLK_PCTF	PCIE_CLK100M_T29_N	17 74	

	PCIE_CLK100M	CLK_PCIE_96D	CLK_PCIE	PEG CLK100M P	8 37
		CLK_PCIE_96D	CLK_PCIE	PEG CLK100M N	8 37
	PCIE_CLK100M_ENET	CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_ENET P	17 37
		CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_ENET N	17 37
100	PCIE_CLK100M_AP	CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_AP_CONN P	30 39
100		CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_AP_CONN N	30 39
	PCIE_CLK100M_AP	CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_AP P	17 35
		CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_AP N	17 35
	PCIE_CLK100M_FW	CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_FW P	17 39
		CLK_PCIE_96D	CLK_PCIE	PCIE_CLK100M_FW N	17 39

PCIE T29_R2D	PCIE_85D	PCIE	PCIE T29_R2D C P<3...0>	8 74
PCIE T29_R2D	PCIE_85D	PCIE	PCIE T29_R2D C N<3...0>	8 74
PCIE T29_R2D	PCIE_85D	PCIE	PCIE T29_R2D P<3...0>	74
PCIE T29_R2D	PCIE_85D	PCIE	PCIE T29_R2D N<3...0>	74
PCIE T29_D2R	PCIE_85D	PCIE	PCIE T29_D2R P<3...0>	8 74
PCIE T29_D2R	PCIE_85D	PCIE	PCIE T29_D2R N<3...0>	8 74
PCIE T29_D2R	PCIE_85D	PCIE	PCIE T29_D2R C P<3...0>	74
PCIE T29_D2R	PCIE_85D	PCIE	PCIE T29_D2R C N<3...0>	74

ETHERNET CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	*	0.6 MM	?	ENET_HV	*	*	2KV_SPACING
2KV_SPACING	*	0.50 MM	100	ENET_HV	ENET_HV	*	ENET_MDI
2KV_SPACING	TOP,BOTTOM	1.27 MM	100				

NOTE: 2.2kV isolation needed on all primary-side ethernet signals. Min 1.27mm separation from all other copper on the board!

SD CARD INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

FireWire Interface Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

Ethernet Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		ENET_50S	ENET_3X	BCM5764 CLK25M XTALI 26 37
		ENET_50S	ENET_3X	BCM5764 CLK25M XTALO 26 37
		ENET_50S	ENET_3X	ENET RESET L 27 34
	ENET_MDI	ENET_100D	ENET_MDI	ENET MDI P<3..0> 36 37
		ENET_100D	ENET_MDI	ENET MDI N<3..0> 36 37
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				

High-voltage isolated ethernet signals.

SD Interface Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
SD55S		SD 55S	SD INTERFACE	SDCONN CLK 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN CLK FF 34
SD55S		SD 55S	SD INTERFACE	SDCONN CMD 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN DATA<7..0> 34 37 91
SD55S		SD 55S	SD INTERFACE	SDCONN CLK R 34
SD55S		SD 55S	SD INTERFACE	SDCONN CMD R 34
SD55S		SD 55S	SD INTERFACE	SDCONN DATA R<7..0> 34
SD55S		SD 55S	SD INTERFACE	ENET CR CLK 37
SD55S		SD 55S	SD INTERFACE	ENET CR CMD 37
SD55S		SD 55S	SD INTERFACE	ENET CR DATA<7..0> 37
SD55S		SD 55S	SD INTERFACE	SDCONN F CLK 34
SD55S		SD 55S	SD INTERFACE	SDCONN F CLK FF 34

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
LEADS	FW_P1_TPA	FW_118D	FW_TP	FW_P0_TPA_L_N 46
LEADS		FW_118D	FW_TP	FW_P0_TPA_L_P 46
LEADS	FW_P1_TPB	FW_118D	FW_TP	FW_P0_TPB_L_N 46
LEADS		FW_118D	FW_TP	FW_P0_TPB_L_P 46
PORTS		FW_118D	FW_TP	FW_PORT0_TPAL_N 41
PORTS		FW_118D	FW_TP	FW_PORT0_TPAL_P 41
LEADS		FW_118D	FW_TP	FW_PORT0_TPA_N 46 41
PORTS		FW_118D	FW_TP	FW_PORT0_TPA_P 46 41
LEADS		FW_118D	FW_TP	FW_PORT0_TPBL_N 41
LEADS		FW_118D	FW_TP	FW_PORT0_TPBL_P 41
PORTS		FW_118D	FW_TP	FW_PORT0_TPB_N 46 41
PORTS		FW_118D	FW_TP	FW_PORT0_TPB_P 46 41
LEADS		50_OHM_SE	CLK_PCIE	CLK98M_FW_XI 39
LEADS		50_OHM_SE	CLK_PCIE	CLK98M_FW_XI_R 39

Port 0 and 2 Not Used

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

T29/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29DP	*	=5x_DIELECTRIC	?	T29DP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

T29 IC Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
		T29_I2C_55S	T29_I2C
		T29_I2C_55S	T29_I2C
			I2C T29_SCL
			I2C T29_SDA
	T29_SPI_CLK	T29_SPI_55S	T29_SPI
	T29_SPI_MOSI	T29_SPI_55S	T29_SPI
	T29_SPI_MISO	T29_SPI_55S	T29_SPI
	T29_SPI_CS_L	T29_SPI_55S	T29_SPI
			T29_SPI_CLK
			T29_SPI_MOSI
			T29_SPI_MISO
			T29_SPI_CS_L
		T29DP_90D	T29DP
		T29DP_90D	T29DP
			T29_R2D_C P<3..0>
			T29_R2D_C N<3..0>
			T29_D2R P<3..0>
			T29_D2R N<3..0>

T29/DP Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
T29_R2D0	T29DP_90D	T29DP	T29_R2D P<0>	77
T29_R2D0	T29DP_90D	T29DP	T29_R2D N<0>	77
T29_R2D1	T29DP_90D	T29DP	T29_R2D P<1>	77
T29_R2D1	T29DP_90D	T29DP	T29_R2D N<1>	77
T29_D2R0	T29DP_90D	T29DP	T29_D2R C P<0>	77 78
T29_D2R0	T29DP_90D	T29DP	T29_D2R C N<0>	77
T29_D2R1	T29DP_90D	T29DP	T29_D2R C P<1>	77 78
T29_D2R1	T29DP_90D	T29DP	T29_D2R C N<1>	77 78
	T29DP_90D	T29DP	T29_R2D C F P<1..0>	77
	T29DP_90D	T29DP	T29_R2D C F N<1..0>	77
	T29DP_90D	T29DP	T29DPA_D2R1_AUXCH P	78
	T29DP_90D	T29DP	T29DPA_D2R1_AUXCH N	78
	T29DP_90D	T29DP	DP_SDRVA_ML C P<3..0>	77
	T29DP_90D	T29DP	DP_SDRVA_ML C N<3..0>	77
	T29DP_90D	T29DP	DP_SDRVA_ML R P<3..0>	77
	T29DP_90D	T29DP	DP_SDRVA_ML R N<3..0>	77
	T29DP_90D	T29DP	DP_SDRVA_ML P<2>	77
	T29DP_90D	T29DP	DP_SDRVA_ML N<2>	77
DP_SDRVA_ML	T29DP_90D	T29DP	DP_SDRVA_ML P<3>	77
DP_SDRVA_ML	T29DP_90D	T29DP	DP_SDRVA_ML N<3>	77
	T29DP_90D	T29DP	DP_SDRVA_ML P<0>	77
	T29DP_90D	T29DP	DP_SDRVA_ML N<0>	77
DP_SDRVA_ML	T29DP_90D	T29DP	DP_SDRVA_ML P<1>	77
DP_SDRVA_ML	T29DP_90D	T29DP	DP_SDRVA_ML N<1>	77
DP_SDRVA_AUXCH	T29DP_90D	T29DP	DP_SDRVA_AUXCH P	77
DP_SDRVA_AUXCH	T29DP_90D	T29DP	DP_SDRVA_AUXCH N	77
	T29DP_90D	T29DP	DP_SDRVA_AUXCH C P	77
	T29DP_90D	T29DP	DP_SDRVA_AUXCH C N	77
	T29DP_90D	T29DP	T29DPA_ML P<3..0>	77 78
	T29DP_90D	T29DP	T29DPA_ML N<3..0>	77
	T29DP_90D	T29DP	T29DPA_ML C P<3..0>	77 78
	T29DP_90D	T29DP	T29DPA_ML C N<3..0>	77 78
	T29DP_90D	T29DP	DP_A_EXT_AUXCH P	77 78
	T29DP_90D	T29DP	DP_A_EXT_AUXCH N	77 78
	T29DP_90D	T29DP	T29DPA_D2R1_AUXCH_CC P	77
	T29DP_90D	T29DP	T29DPA_D2R1_AUXCH_CC N	77
	T29DP_90D	T29DP	T29_D2R_CC P<0>	77
	T29DP_90D	T29DP	T29_D2R_CC N<0>	77



PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
 SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_A_S3_SCL 46 49
 SMBUS_SMC_A_S3_SDA	SMB_50S	SMB	SMBUS_SMC_A_S3_SDA 46 49
 SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 46 49
 SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 46 49
 SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL 46 49
 SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA 46 49
 SMBUS_SMC_BSA_SCL	SMB_50S	SMB	ENET_ASF_SMB_CLK 37 49
 SMBUS_SMC_BSA_SDA	SMB_50S	SMB	ENET_ASF_SMB_DATA 37 49
 SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT_SCL 46 49
 SMBUS_SMC_MGMT_SDA	SMB_50S	SMB	SMBUS_SMC_MGMT_SDA 46 49
 SMB_SMC_B_STR_SCL	SMB_50S	SMB	SMB_SMC_B_STR_SCL 49
 SMB_SMC_B_STR_SDA	SMB_50S	SMB	SMB_SMC_B_STR_SDA 49
 SMB_SMC_B_TSN_SCL	SMB_50S	SMB	SMB_SMC_B_TSN_SCL 49
 SMB_SMC_B_TSN_SDA	SMB_50S	SMB	SMB_SMC_B_TSN_SDA 49

SYNC_MASTER=REFERENCE_MLB

SYNC_DATE=11/23/2016

PAGE_TITLE

SMC Constraints

 Apple Inc.

DRAWING_NUMBER

051-8768

SIZE

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REVISION

9.0.0

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GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

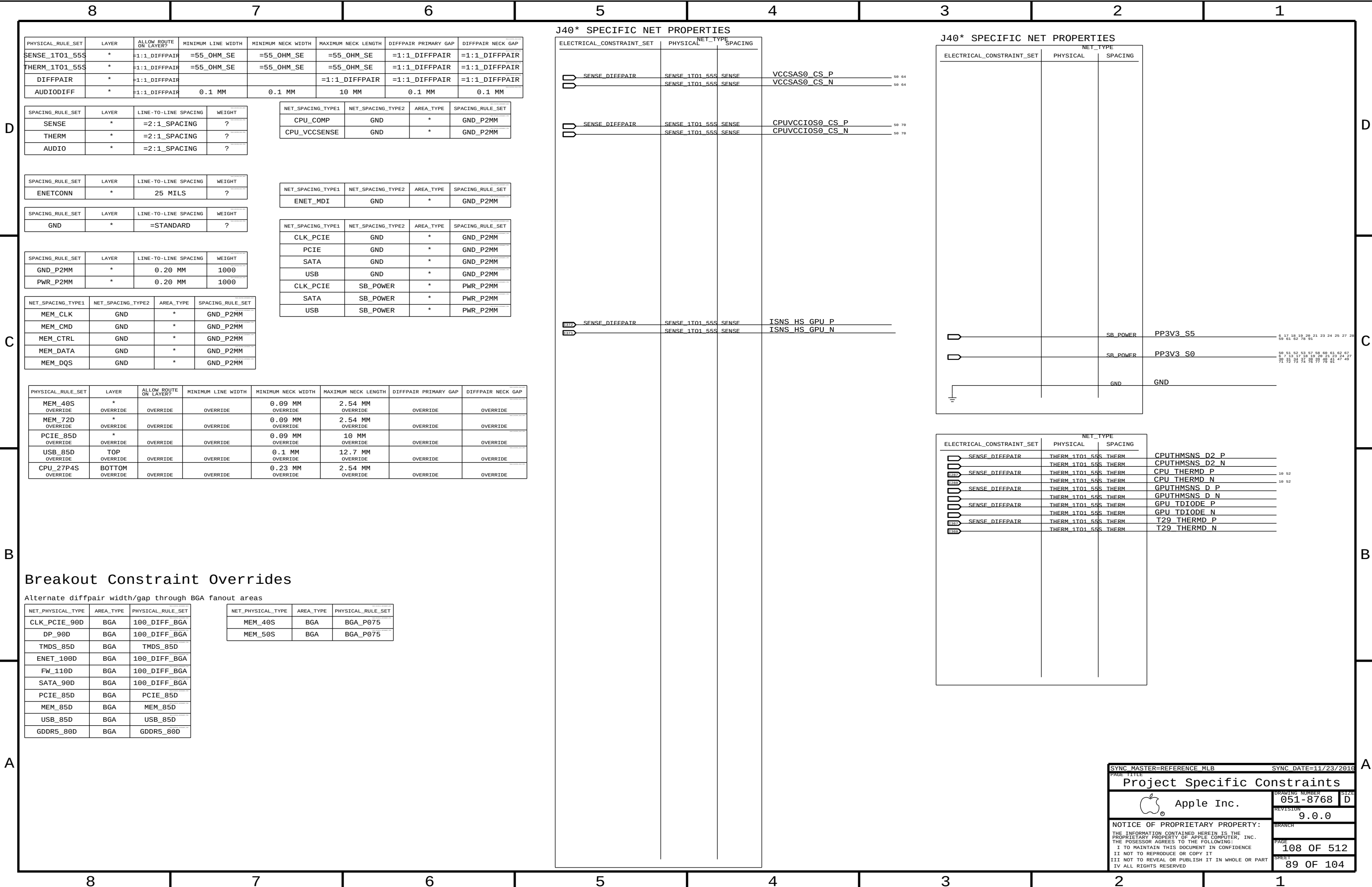
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=2x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

GDDR5 FB B Net Properties

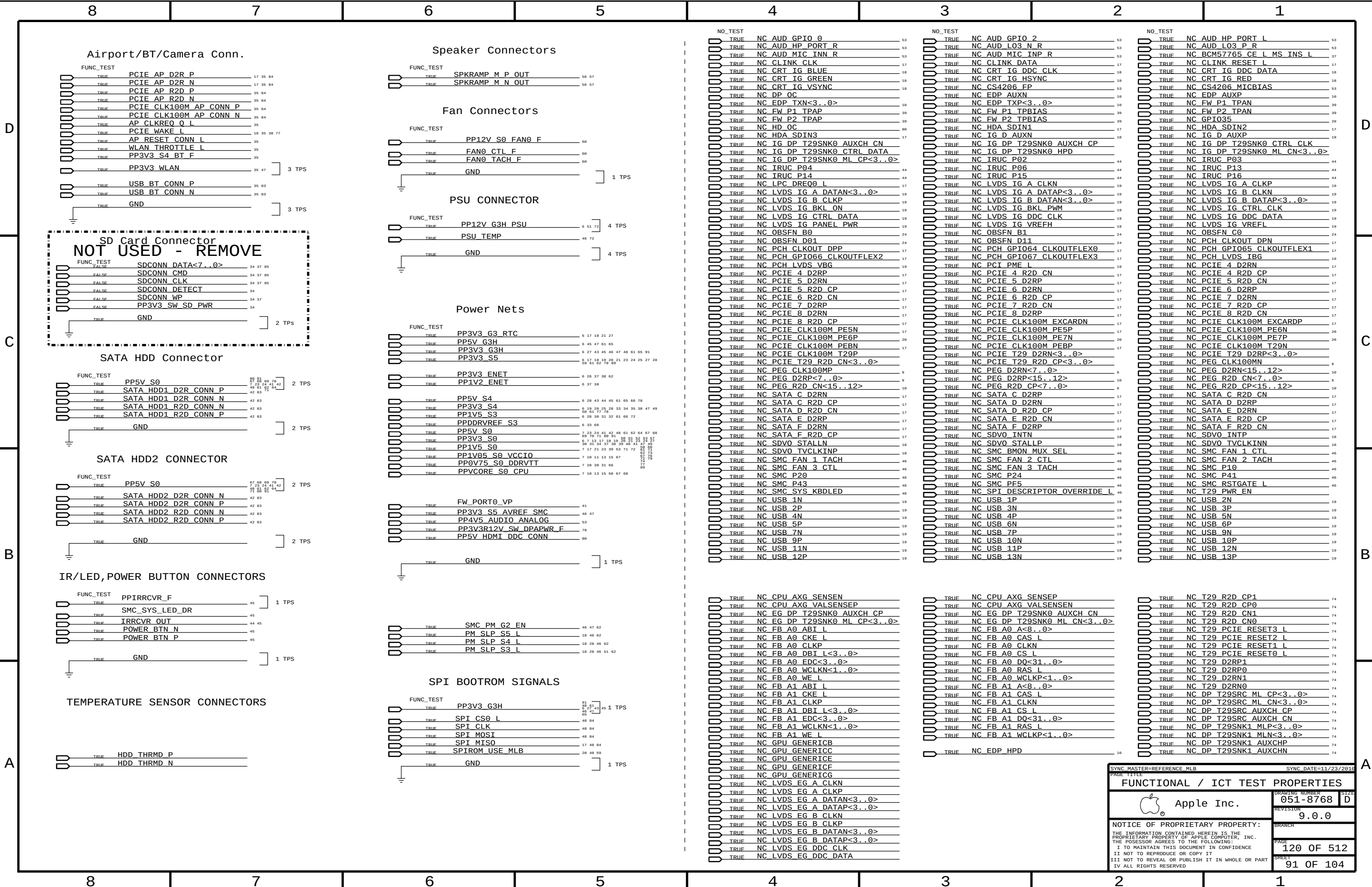
ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK P
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK N
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK P
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK N
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 A<8..0>
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 A<8..0>
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 ABI L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 ABI L
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 RAS L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 RAS L
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CAS L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CAS L
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 WE L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 WE L
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CKE L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CKE L
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CS L
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CS L
FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<0>
FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<1>
FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<2>
FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<3>
FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<0>
FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<1>
FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<2>
FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<3>
FB_B0_DBI L0	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<0>
FB_B0_DBI L1	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<1>
FB_B0_DBI L2	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<2>
FB_B0_DBI L3	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<3>
FB_B1_DBI L0	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<0>
FB_B1_DBI L1	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<1>
FB_B1_DBI L2	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<2>
FB_B1_DBI L3	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<3>
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<0>
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<0>
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<1>
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<1>
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<0>
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<0>
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<1>
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<1>
FB_B0_DQ BYTE0	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<7..0>
FB_B0_DQ BYTE1	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<15..8>
FB_B0_DQ BYTE2	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<23..16>
FB_B0_DQ BYTE3	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<31..24>
FB_B1_DQ BYTE0	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<7..0>
FB_B1_DQ BYTE1	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<15..8>
FB_B1_DQ BYTE2	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<23..16>
FB_B1_DQ BYTE3	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<31..24>
FB_AB_RESET	GDDR5_45R50SE	GDDR5_CMD	FB RESET L

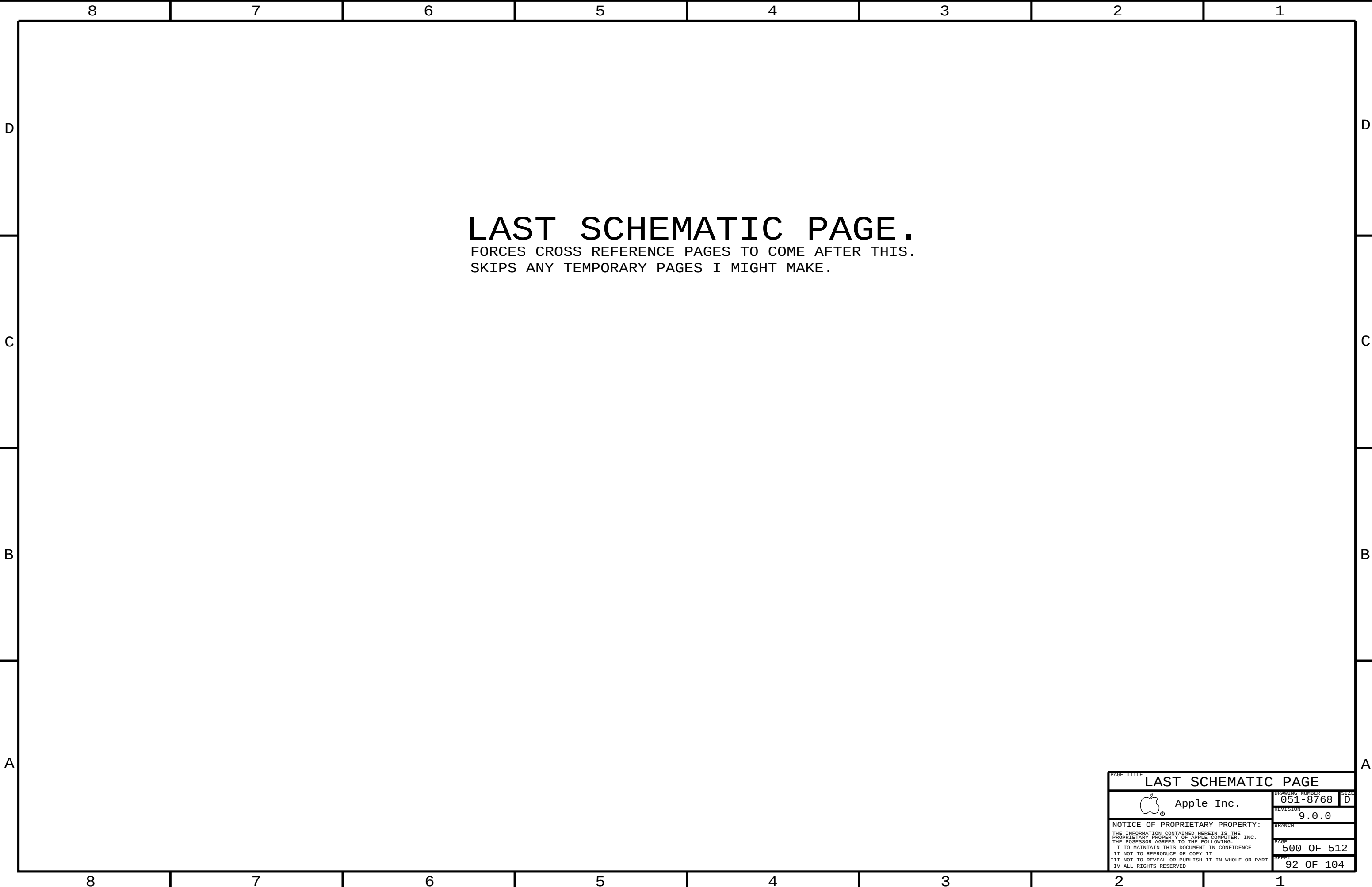
Whistler Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M
GPU_CLK100M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK100M
B31P	100_OHM_DIEF	3X_DIELECTRIC	GPU_CLK_TEST_N
B32P	100_OHM_DIEF	3X_DIELECTRIC	GPU_CLK_TEST_P



	8	7	6	5	4	3	2	1
J40* 12 LAYER BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS								
BOARD LAYERS				BOARD AREAS		BOARD UNITS (ALL OF MM)	ALLEGRO VERSION	
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA		MM	15.5.1	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM	
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.095 MM				
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
37_OHM_SE	TOP, BOTTOM	Y	0.185 MM	0.095 MM				
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
40_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.095 MM				
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
45_OHM_SE	TOP, BOTTOM	Y	0.13 MM	0.13 MM				
45_OHM_SE	*	Y	0.099 MM	0.099 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.095 MM				
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM				
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
72_OHM_DIFF	*	Y	0.154 MM	0.154 MM	=STANDARD	0.200 MM	0.200 MM	
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
80_OHM_DIFF	*	Y	0.105 MM	0.105 MM	=STANDARD	0.120 MM	0.120 MM	
80_OHM_DIFF	TOP, BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
85_OHM_DIFF	*	Y	0.110 MM	0.090 MM	=STANDARD	0.180 MM	0.180 MM	
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
90_OHM_DIFF	*	Y	0.102 MM	0.090 MM	=STANDARD	0.220 MM	0.220 MM	
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
100_OHM_DIFF	*	Y	0.080 MM	0.080 MM	=STANDARD	0.200 MM	0.200 MM	
100_OHM_DIFF	TOP, BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
110_OHM_DIFF	*	Y	0.065 MM	0.065 MM	=STANDARD	0.200 MM	0.200 MM	
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM	
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE_ON_LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP								
100_DIFF_BGA * =100_OHM_DIFF =100_OHM_DIFF =100_OHM_DIFF =100_OHM_DIFF =100_OHM_DIFF =100_OHM_DIFF								
100_DIFF_BGA ISL3, ISL4 Y 0.075 MM 0.075 MM 0.125 MM 0.125 MM								
100_DIFF_BGA ISL9, ISL10 Y 0.075 MM 0.075 MM 0.125 MM 0.125 MM								
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.								
PHYSICAL_RULE_SET LAYER ALLOW_ROUTE_ON_LAYER? MINIMUM LINE WIDTH MINIMUM NECK WIDTH MAXIMUM NECK LENGTH DIFFPAIR PRIMARY GAP DIFFPAIR NECK GAP								
BGA_P075 TOP Y 0.110 MM 0.080 MM 10 MM								
BGA_P075 BOTTOM Y 0.110 MM 0.073 MM 10 MM								
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT								
DEFAULT * 0.1 MM ?								
STANDARD * =DEFAULT ?								
BGA_P1MM * =DEFAULT ?								
BGA_P2MM * =DEFAULT ?								
BGA_P075MM * 0.075 MM ?								
NET_SPACING_TYPE1 NET_SPACING_TYPE2 AREA_TYPE SPACING_RULE_SET								
* * BGA BGA_P075MM								
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT								
1.5:1_SPACING * 0.15 MM ?								
2:1_SPACING * 0.2 MM ?								
2.5:1_SPACING * 0.25 MM ?								
3:1_SPACING * 0.3 MM ?								
4:1_SPACING * 0.4 MM ?								
5:1_SPACING * 0.5 MM ?								
SPACING_RULE_SET LAYER LINE-TO-LINE SPACING WEIGHT								
2X_DIELECTRIC * 0.140 MM ?								
3X_DIELECTRIC * 0.210 MM ?								
4X_DIELECTRIC * 0.280 MM ?								
5X_DIELECTRIC * 0.350 MM ?								
7X_DIELECTRIC * 0.490 MM ?								
NOTE: Based on K92 m1b stackup.								
8 7 6 5 4 3 2 1								
PCB Rule Definitions								
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PP3V3_S5_SMC_AVCC

PP3V3_SW_S0_PWR

PP3V3_T29

PP3V3_T29_DP

PP3V3_T29_DPBIA5

PP3V3_T29_PLL

PP3V3_US3900

PP3V3_WLAN

PP3V3_WLAN_FET

PP4V5_AUDIO_ANALOG

PP5V_AUDIO_HPAMP

PP5V_G3H

PP5V_HDMI_DDC_CONN

PP5V_HDMI_DDC_FUSE

PP5V_S0

PP5V_S0_AUDIO

PP5V_S0_AUDIO_AMP

PP5V_S0_CPUVCCIO50_V

CC

PP5V_S0_CPU_VCORE_VC

C

PP5V_S0_PCH_V5REF

PP5V_S0_VCCSA50_VCC

PP5V_S4

PP5V_S4_DORREG_V5IN

PP5V_S4_IR_R

PP5V_S5

PP5V_S5_PCH_V5REFSU

PP5V_S5_V5REFSU

PP5V_U7100_PVCC

PP5V_U7700_PVCC

PP5V_USB2_PORT0

PP5V_USB2_PORT0_F

PP5V_USB2_PORT1

PP5V_USB2_PORT1_F

PP5V_USB2_PORT2

PP5V_USB2_PORT2_F

PP5V_USB2_PORT3

PP12V_G3H

PP12V_G3H_CPUDDR

PP12V_G3H_DDRS3REG_F

ILT

PP12V_G3H_GPU

PP12V_G3H_PSU

PP12V_S0_AXG_FLTRD

PP12V_S0_CPU_FLTRD

PP12V_S0_FAN

PP12V_S0_FAN0_F

PPCPVUCCIO_S0_REG_R

PPDDRVRREF_S3

PP1RRRCVR_F

PP1RRRCVR_G3H_LED

PPUSB_HUB1_CRFILT

PPUSB_HUB1_P1_CRFILT

PPUSB_HUB2_CRFILT

PPUSB_HUB2_P1_CRFILT

PPVAXG_S0_CPU

PPVAXG_S0_CPU_RS

PPVBATT_G3_RTC

PPVBATT_G3_RTC_R

PPVCCSA_S0_CPU

PPVCCSA_S0_REG_R

PPVCORE_S0_CPU

PPVOUIT_G3_PCH_DCPRTC

PPVOUIT_S0_PCH_DCPSTS

PPVP_FW_PHY_CPS

PSU_TEMP

PU_FW_VP

PU_HDMI_LVDS_RS1V1

PU_SDCONN_OC_L

PU_USB1_PRT4N

PU_USB1_PRT4P

PU_USB2_PRT4N

PU_USB2_PRT4P

PVCCIO_S0_SMC_R

PVCCSA_EN

PVCCSA_P000D

RTC_RESET_L

RT_1V8S0

SGPDG_BJT_GND_R

SGPDG_C

SS_PWRG0

SATARDVRV_EN

SATA_HDD1_D2R_CONN_N

SATA_HDD1_D2R_CONN_P

SATA_HDD1_D2R_DF_N

SATA_HDD1_D2R_DF_P

SATA_HDD1_D2R_N

SATA_HDD1_D2R_P

SATA_HDD1_R2D_CONN_N

SATA_HDD1_R2D_CONN_P

SATA_HDD1_R2D_C_N

SATA_HDD1_R2D_C_P

SATA_HDD1_R2D_DF_N

SATA_HDD1_R2D_DF_P

SATA_HDD2_D2R_CONN_N

SATA_HDD2_D2R_CONN_P

SATA_HDD2_D2R_DF_N

SATA_HDD2_D2R_DF_P

SATA_HDD2_D2R_N

SATA_HDD2_D2R_P

SATA_HDD2_R2D_CONN_N

SATA_HDD2_R2D_CONN_P

SATA_HDD2_R2D_C_N

SATA_HDD2_R2D_C_P

SATA_HDD2_R2D_DF_N

SATA_HDD2_R2D_DF_P

SDCONNF_CLK

PP3V3_S5_SMC_AVCC

PP3V3_SW_S0_PWR

PP3V3_T29

PP3V3_T29_DP

PP3V3_T29_DPBIA5

PP3V3_T29_PLL

PP3V3_US3900

PP3V3_WLAN

PP3V3_WLAN_FET

PP4V5_AUDIO_ANALOG

PP5V_AUDIO_HPAMP

PP5V_G3H

PP5V_HDMI_DDC_CONN

PP5V_HDMI_DDC_FUSE

PP5V_S0

PP5V_S0_AUDIO

PP5V_S0_AUDIO_AMP

PP5V_S0_CPUVCCIO50_VCC

CC

PP5V_S0_CPU_VCORE_VCC

C

PP5V_S0_PCH_V5REF

PP5V_S0_VCCSA50_VCC

PP5V_S4

PP5V_S4_DORREG_V5IN

PP5V_S4_IR_R

PP5V_S5

PP5V_S5_PCH_V5REFSU

PP5V_S5_V5REFSU

PP5V_U7100_PVCC

PP5V_U7700_PVCC

PP5V_USB2_PORT0

PP5V_USB2_PORT0_F

PP5V_USB2_PORT1

PP5V_USB2_PORT1_F

PP5V_USB2_PORT2

PP5V_USB2_PORT2_F

PP5V_USB2_PORT3

PP12V_G3H

PP12V_G3H_CPUDDR

PP12V_G3H_DDRS3REG_FLT

ILT

PP12V_G3H_GPU

PP12V_G3H_PSU

PP12V_S0_AXG_FLTRD

PP12V_S0_CPU_FLTRD

PP12V_S0_FAN

PP12V_S0_FAN0_F

PPCPVUCCIO_S0_REG_R

PPDDRVRREF_S3

PP1RRRCVR_F

PP1RRRCVR_G3H_LED

PPUSB_HUB1_CRFILT

PPUSB_HUB1_P1_CRFILT

PPUSB_HUB2_CRFILT

PPUSB_HUB2_P1_CRFILT

PPVAXG_S0_CPU

PPVAXG_S0_CPU_RS

PPVBATT_G3_RTC

PPVBATT_G3_RTC_R

PPVCCSA_S0_CPU

PPVCCSA_S0_REG_R

PPVCORE_S0_CPU

PPVOUIT_G3_PCH_DCPRTC

PPVOUIT_S0_PCH_DCPSTS

PPVP_FW_PHY_CPS

PSU_TEMP

PU_FW_VP

PU_HDMI_LVDS_RS1V1

PU_SDCONN_OC_L

PU_USB1_PRT4N

PU_USB1_PRT4P

PU_USB2_PRT4N

PU_USB2_PRT4P

PVCCIO_S0_SMC_R

PVCCSA_EN

PVCCSA_P000D

RTC_RESET_L

RT_1V8S0

SGPDG_BJT_GND_R

SGPDG_C

SS_PWRG0

SATARDVRV_EN

SATA_HDD1_D2R_CONN_N

SATA_HDD1_D2R_CONN_P

SATA_HDD1_D2R_DF_N

SATA_HDD1_D2R_DF_P

SATA_HDD1_D2R_N

SATA_HDD1_D2R_P

SATA_HDD1_R2D_CONN_N

SATA_HDD1_R2D_CONN_P

SATA_HDD1_R2D_C_N

SATA_HDD1_R2D_C_P

SATA_HDD1_R2D_DF_N

SATA_HDD1_R2D_DF_P

SATA_HDD2_D2R_CONN_N

SATA_HDD2_D2R_CONN_P

SATA_HDD2_D2R_DF_N

SATA_HDD2_D2R_DF_P

SATA_HDD2_D2R_N

SATA_HDD2_D2R_P

SATA_HDD2_R2D_CONN_N

SATA_HDD2_R2D_CONN_P

SATA_HDD2_R2D_C_N

SATA_HDD2_R2D_C_P

SATA_HDD2_R2D_DF_N

SATA_HDD2_R2D_DF_P

SDCONNF_CLK

SDCONNF_CLK_FF

SDCONN_CLK

SDCONN_CLK_FF

SDCONN_CLK_R

SDCONN_CMD

SDCONN_CMD_R

SDCONN_DATA<0>

SDCONN_DATA<7..0>

SDCONN_DATA<1>

SDCONN_DATA<2>

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SDCONN_DATA<4>

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SDCONN_DATA<6>

SDCONN_DATA<7>

SDCONN_DATA_R<0>

SDCONN_DATA_R<7..0>

SDCONN_DATA_R<1>

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SDCONN_DATA_R<6>

SDCONN_DATA_R<7>

SDCONN_DETECT

SDCONN_DETECT_L

SDCONN_STATE_CHANGE

SDCONN_STATE_CHANGE_S1R

SDCONN_STATE_CHANGE_S2R

S2R

SDCONN_WP

SLG_ENET_RESET_L

SLG_ENET_RESET_R_L

SMBUS_PCH_CLK

SMBUS_PCH_DATA

SMBUS_SMC_0_S0_SCL

=I2C_T29THMSNS_SCL

SMBUS_SMC_0_S0_SDA

=I2C_T29THMSNS_SDA

SMBUS_SMC_A_S3_SCL

=SMBUS_REMOTE_THMSNS_SCL

SMBUS_SMC_A_S3_SDA

=SMBUS_REMOTE_THMSNS_SDA

SMBUS_SMC_B_S0_SCL

SMBUS_SMC_B_S0_SDA

SMBUS_SMC_MGMT_SCL

SMBUS_SMC_MGMT_SDA

SMB_PCH_DIMM_SCL

=I2C_S0DIMMB_SCL

=I2C_S0DIMMA_SCL

=I2C_S0DIMMB_SCL

=I2C_S0DIMMA_SCL

SMB_PCH_MKY_SCL

=I2C_MKIEY_SCL

SMB_PCH_MKY_SDA

=I2C_MKIEY_SDA

SMB_PCH_STR_SCL

SMB_PCH_STR_SDA

SMB_PCH_VRF_SCL

=I2C_PCA9557D_SCL

=I2C_VREFDAC5_SCL

=I2C_PCA9557D_SCL

SMB_PCH_VRF_SDA

=I2C_PCA9557D_SDA

=I2C_VREFDAC5_SDA

SMB_PCH_XDP_SCL

=I2C_PCA9557D_SDA

SMB_PCH_XDP_SDA

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8			7			6			5			4			3			2			1		
D	R6881	RES_402	m1b[62C3]	C	R7861	RES_402	m1b[71C5]	B	R9749	RES_402	m1b[79B5]	A	U5365	INA219_SC70	m1b[50A7]	A	U5365	INA219_SC70	m1b[50A7]	A	U5365	INA219_SC70	m1b[50A7]
	R6886	RES_402	m1b[62C2]		R7862	RES_402	m1b[71C5]		R9741	RES_402	m1b[79B5]		U5375	INA219_SC70	m1b[50C7]		U5375	INA219_SC70	m1b[50C7]		U5400	INA219_SC70	m1b[51C7]
	R6887	RES_402	m1b[62C3]		R7863	RES_402	m1b[71D5]		R9750	RES_402	m1b[79B7]		U5400	INA219_SC70	m1b[51C7]		U5403	INA214_SC70	m1b[51D7]		U5500	EMC1428_QFN	m1b[52D6]
	R6888	RES_402	m1b[62C2]		R7865	RES_402	m1b[71C7]		R9753	RES_402	m1b[79C7]		U5403	INA214_SC70	m1b[51D7]		U5500	EMC1428_QFN	m1b[52D6]		U5535	EMC1414_MSOP10_MSOP	m1b[52B6]
	R7101	RES_402	m1b[64C5]		R7870	RES_402	m1b[71B7]		R9754	RES_402	m1b[79B7]		U5500	EMC1428_QFN	m1b[52D6]		U5500	EMC1428_QFN	m1b[52D6]		U5600	CS4206B_QFN	m1b[53D5]
	R7102	RES_603	m1b[64C5]		R7920	RES_402	m1b[72C6]		R9755	RES_402	m1b[79C7]		U5535	EMC1414_MSOP10_MSOP	m1b[52B6]		U5600	CS4206B_QFN	m1b[53D5]		U5601	MAX9840_10QFN	m1b[53A5]
	R7103	RES_402	m1b[64B6]		R8000	RES_201	m1b[73B4]		R9756	RES_402	m1b[79B6]		U5600	CS4206B_QFN	m1b[53D5]		U5601	MAX9840_10QFN	m1b[53A5]		U5600	AMP_MAX97220_TQFN	m1b[55C4]
	R7104	RES_402	m1b[64C7]		R8001	RES_201	m1b[73B4]		R9757	RES_402	m1b[79C6]		U5601	MAX9840_10QFN	m1b[53A5]		U5600	AMP_MAX97220_TQFN	m1b[55C4]		U5900	AUD10_MAX98300_WLP	m1b[56C5]
	R7120	RES_603	m1b[64C3]		R9010	RES_402	m1b[74D5]		R9758	RES_402	m1b[79B6]		U5900	AUD10_MAX98300_WLP	m1b[56C5]		U6106	CD3282A1_WCSP	m1b[58D3]		U6400	FLASH_MX25L6406E_SOT	m1b[59C5]
	R7130	RES_402	m1b[64C4]		R9011	RES_402	m1b[74D5]		R9759	RES_402	m1b[79C6]		U6106	CD3282A1_WCSP	m1b[58D3]		U6400	FLASH_MX25L6406E_SOT	m1b[59C5]		U6701	SLG5AP021_TDFN	m1b[61B7]
C	R7140	RES_SENSE_0612	m1b[64C2]	B	R9021	RES_402	m1b[74C6]	A	R9761	RES_402	m1b[79C6]	A	U6701	SLG5AP021_TDFN	m1b[61B7]	A	U6800	ISL88042_TDFN	m1b[62A7]	A	U7100	ISL95870A_UTQFN	m1b[64C5]
	R7141	RES_402	m1b[64B3]		R9022	RES_402	m1b[74C6]		R9762	RES_402	m1b[79B3]		U6800	ISL88042_TDFN	m1b[62A7]		U7100	ISL95870A_UTQFN	m1b[64C5]		U7201	SC414_MLPQ	m1b[65D6]
	R7142	RES_402	m1b[64B3]		R9023	RES_402	m1b[74C6]		R9763	RES_402	m1b[79C3]		U7201	SC414_MLPQ	m1b[65D6]		U7300	TPS51916_QFN	m1b[66C6]		U7400	ISL6364_QFN	m1b[67C4]
	R7146	RES_402	m1b[64C6]		R9025	RES_402	m1b[74C6]		R9764	RES_402	m1b[79B3]		U7300	TPS51916_QFN	m1b[66C6]		U7400	ISL6364_QFN	m1b[67C4]		U7501	ISL6620A_QFN	m1b[68D7]
	R7147	RES_402	m1b[64B6]		R9029	RES_402	m1b[74B6]		R9765	RES_402	m1b[79C3]		U7501	ISL6620A_QFN	m1b[68D7]		U7521	ISL6609_QFN1	m1b[68C7]		U7601	ISL6620A_QFN	m1b[69C7]
	R7148	RES_402	m1b[64B6]		R9030	RES_402	m1b[74B6]		R9766	RES_402	m1b[79B3]		U7601	ISL6620A_QFN	m1b[69C7]		U7700	ISL95870_UTQFN	m1b[70C6]		U7800	TPS54218_QFN	m1b[71D7]
	R7160	RES_603	m1b[64B2]		R9031	RES_402	m1b[74A6]		R9767	RES_402	m1b[79B3]		U7800	TPS54218_QFN	m1b[71D7]		U7870	LREG_TPS720XX_DRV_SO	m1b[71B6]		U9800	T29_FC8GA	m1b[74D4]
	R7202	RES_402	m1b[65D6]		R9032	RES_402	m1b[74A7]		R9769	RES_402	m1b[79C3]		U9800	T29_FC8GA	m1b[75D5]		U9800	EEPROM_M95320_MLP8_M	m1b[74C7]		U9200	SLG4AP016_TDFN	m1b[76C7]
	R7203	RES_603	m1b[65D6]		R9040	RES_402	m1b[74A7]		R9769	RES_402	m1b[79C3]		U9210	TPS22924_CSP	m1b[76B7]		U9210	TPS22924_CSP	m1b[76B7]		U9215	TPS22924_CSP	m1b[76A7]
	R7204	RES_402	m1b[65D7]		R9041	RES_402	m1b[74A7]		R9769	RES_402	m1b[79C3]		U9215	TPS22924_CSP	m1b[76A7]		U9216	TPS22924_CSP	m1b[76A7]		U9216	TPS22924_CSP	m1b[76A7]
B	R7205	RES_402	m1b[65C6]	A	R9051	RES_402	m1b[74D2]	A	R9769	RES_402	m1b[79C3]	A	U9310	PS5801_A2_QFN	m1b[77C6]	A	U9310	PS5801_A2_QFN	m1b[77C6]	A	U9330	PC11112A_HVQFN25_HVQ	m1b[77A6]
	R7208	RES_402	m1b[65C5]		R9055	RES_603	m1b[74C3]		R9769	RES_402	m1b[79C3]		U9330	PC11112A_HVQFN25_HVQ	m1b[77A6]		U9359	SN74LVC1G04_SC70	m1b[77C3]		U9359	SN74LVC1G04_SC70	m1b[77C3]
	R7209	RES_402	m1b[65C5]		R9085	RES_402	m1b[74A3]		R9769	RES_402	m1b[79C3]		U9359	SN74LVC1G04_SC70	m1b[77C3]		U9390	CBTL04DP081_HVQFN	m1b[77B2]		U9401	ISL80101B_DFN	m1b[78D7]
	R7210	RES_402	m1b[65C4]		R9090	RES_402	m1b[74C8]		R9769	RES_402	m1b[79C3]		U9401	ISL80101B_DFN	m1b[78D7]		U9410	SN1801017_QFN	m1b[78D2]		U9420	MC74VHC1G08_SC70-HF	m1b[78C7]
	R7211	RES_402	m1b[65C4]		R9091	RES_402	m1b[74C8]		R9769	RES_402	m1b[79C3]		U9420	MC74VHC1G08_SC70-HF	m1b[78C7]		U9740	PS8171_QFN	m1b[79D4]		U9740	PS8171_QFN	m1b[79D4]
	R7214	RES_402	m1b[65C7]		R9092	RES_402	m1b[74C6]		R9769	RES_402	m1b[79C3]		U9740	PS8171_QFN	m1b[79D4]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7215	RES_402	m1b[65C5]		R9093	RES_402	m1b[74C6]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7251	RES_603	m1b[65D6]		R9095	RES_402	m1b[74C2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7254	RES_402	m1b[65B6]		R9096	RES_402	m1b[74C2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7255	RES_402	m1b[65A6]		R9098	RES_402	m1b[74C2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7256	RES_402	m1b[65A5]		R9099	RES_402	m1b[74B3]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
A	R7257	RES_402	m1b[65A4]	A	R9120	RES_402	m1b[75C7]	A	R9769	RES_402	m1b[79C3]	A	U9800	SWI_TPS2051B_SOT23	m1b[80D3]	A	U9800	SWI_TPS2051B_SOT23	m1b[80D3]	A	U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7258	RES_402	m1b[65A4]		R9150	RES_402	m1b[75D2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7259	RES_402	m1b[65A4]		R9160	RES_402	m1b[75C2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7260	RES_402	m1b[65A4]		R9203	RES_402	m1b[76C7]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7261	RES_402	m1b[65A3]		R9207	RES_402	m1b[76C6]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7301	RES_603	m1b[66C7]		R9210	RES_402	m1b[76A7]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7305	RES_402	m1b[66B7]		R9215	RES_402	m1b[76C7]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7310	RES_402	m1b[66B7]		R9221	RES_201	m1b[76B2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7320	RES_402	m1b[66B8]		R9222	RES_201	m1b[76B2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7321	RES_402	m1b[66B8]		R9223	RES_201	m1b[76B2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]
	R7330	RES_402	m1b[66C5]		R9224	RES_201	m1b[76B2]		R9769	RES_402	m1b[79C3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]		U9800	SWI_TPS2051B_SOT23	m1b[80D3]

